



ENVIRONMENTAL SOCIAL MONITORING FRAMEWORK



TECHNICAL ADVISORY SERVICES TO BANGLADESH INFRASTRUCTURE FINANCE FUND LIMITED (BIFFL)

Prepared By



Infrastructure Investment Facilitation Company

30 June 2014

BANGLADESH INFRASTRUCTURE FINANCE FUND LIMITED (BIFFL)

ENVIRONMENTAL SOCIAL MONITORING FRAMEWORK

TECHNICAL ADVISORY SERVICES TO BIFFL

Table of Contents

1	THE ESMF DOCUMENT	1
1.1	INTRODUCTION	1
1.2	STRUCTURE OF THE DOCUMENT	1
1.3	PURPOSE OF ESMF	2
1.4	USERS OF ESMF	2
2	DESCRIPTION OF BIFFL.....	4
2.1	BACKGROUND OF BIFFL.....	4
2.2	OBJECTIVES OF BIFFL	4
2.3	TYPES OF ACCEPTABLE PROJECTS.....	5
3	GENERAL ENVIRONMENTAL MANAGEMENT	7
3.1	OVERVIEW.....	7
3.2	TARGET AREAS OF ENVIRONMENTAL MONITORING.....	7
3.3	MAIN ENVIRONMENTAL CONCERNS.....	9
4	LEGAL AND INSTITUTIONAL FRAMEWORK (ENVIRONMENTAL)	11
4.1	OVERVIEW	11
4.2	ENVIRONMENT RELATED POLICIES IN BANGLADESH.....	11
4.3	RELEVANT LAWS AND REGULATIONS IN BANGLADESH.....	13
4.4	ENVIRONMENTAL GUIDELINES FOR PROJECTS IN BANGLADESH	16
4.5	INSTITUTIONAL ORGANIZATIONS	17
4.6	DOE STANDARDS ON ENVIRONMENTAL ASPECTS	18
4.7	DOE REQUIREMENTS ON ENVIRONMENTAL ASPECTS	18
4.8	BANGLADESH BANK’S ENVIRONMENTAL RISK MANAGEMENT (ERM).....	20
4.9	WORLD BANK SAFEGUARD POLICIES	21
4.10	LAND ACQUISITION IN BANGLADESH AND WORLD BANK SAFEGUARDS	21
4.11	ADB REQUIREMENTS ON ENVIRONMENTAL ASPECTS	24
5	OPERATIONAL MANUAL	26
5.1	GENERAL OPERATIONAL PROCEDURE	26
5.2	SEQUENCE OF TASKS OF ESMF IN DIFFERENT PHASES OF BIFFL PROJECT	34
5.3	INSTITUTIONAL FRAMEWORK FOR ESMF IMPLEMENTATION	46
5.4	ESMF IMPLEMENTATION BUDGET	52
5.5	ESMF UPDATE.....	53
5.6	MONITORING OF ESMF	53
5.7	ESMF AUDITING FRAMEWORK.....	54
5.8	PERIODIC REPORTING ON ESMF	55
6	ANNEXES	56

List of Tables

Table 4-1: Environment Related Laws and Regulations.....	13
Table 4-2: Validity Period of ECC.....	20
Table 4-3: Key comparisons between Bangladesh Law and ODs 4.30 and 4.20.....	24
Table 5-1: Environmental Key Requirements	28
Table 5-2: Project E&S Risk Rating Criteria in ESMF	29
Table 5-3: Summary Table for Document Requirement.....	36
Table 5-4: Steps in the Environmental and Social Assessment Process	45
Table 5-5: Allocation of Responsibilities	49
Table 5-6: ESMF Implementation Budget (sample)	52
Table 5-7: Monitoring Plan of ESMF	54
Table 6-1: World Bank Safeguard Policies	64
Table 6-2: Comparison between GOB and World Bank Guidelines.....	66
Table 6-3: Comparison: Bangladesh Law and World Bank OP 4.12.....	109
Table 6-4: Addressing World Bank Safeguard Policies.....	110
Table 6-5: Social Compliance Matrix.....	111
Table 6-6: Site Location and Sensitivity	125
Table 6-7: Major Environmental Impacts	128
Table 6-8: Mitigation Measures.....	131
Table 6-9: Relevant National Legislation	134
Table 6-10: Variables Matrix	142
Table 6-11: List of Probable Impacts during Various Phases of the Project	147
Table 6-12: Resettlement in Different Project Types.....	149
Table 6-13: Comparison of Bangladesh Law with World Bank OD 4.01.....	151
Table 6-14: Bangladesh Law Vs WB Policy on Land Acquisition and Resettlement	152

List of Figures

Figure 3-1: PDCA Cycle	7
Figure 5-1: Project Appraisal Activities by Key Stakeholders.....	33
Figure 5-2: DOE Clearance Process for Red Category Projects	39
Figure 5-3: Proposed ESC within BIFFL Organization Structure	47
Figure 6-1: Impact Structure	130
Figure 6-2: Basic Stages in Project Development	138
Figure 6-3: Flowchart for SMF.....	140
Figure 6-4: Steps in the Social Impact Assessment Process.....	146

List of Annexes

ANNEX 6-1: ECOLOGICALLY CRITICAL AREAS DECLARED BY DOE.....	57
ANNEX 6-2: STANDARDS SET BY DOE FOR EFFLUENT FROM INDUSTRIAL UNITS OR PROJECTS WASTE	59
ANNEX 6-3: STANDARDS SET BY DOE FOR GASEOUS EMISSION FROM INDUSTRIES OR PROJECTS	61
ANNEX 6-4: STANDARDS SET BY DOE FOR SOUND	62
ANNEX 6-5: STANDARDS SET BY DOE FOR ODOR	63
ANNEX 6-6: WORLD BANK SAFEGUARDS POLICIES (ENVIRONMENTAL)	64
ANNEX 6-7: OUTLINE OF AN ENVIRONMENTAL IMPACT ASSESSMENT REPORT	68
ANNEX 6-8: OUTLINE OF A RESETTLEMENT PLAN	71
ANNEX 6-9: IFC/MIGA EXCLUSION LIST OF PROJECTS.....	75
ANNEX 6-10: ADB PROHIBITED INVESTMENT ACTIVITIES LIST.....	77
ANNEX 6-11: SITE VISIT CHECKLIST – ENVIRONMENTAL SAFEGUARDS	78
ANNEX 6-12: FORMAT OF PRELIMINARY ENVIRONMENTAL APPRAISAL REPORT (PEAR).....	80
ANNEX 6-13: CONTENTS OF ANNUAL ENVIRONMENTAL MONITORING REPORT	81
ANNEX 6-14: ENVIRONMENTAL & SOCIAL DUE DILIGENCE (EDD) REPORT OUTLINE	83
ANNEX 6-15: FORMAT FOR CORRECTIVE ACTION PLAN	86
ANNEX 6-16: DOE PRESCRIBED PRO-FORMA FOR APPLYING FOR ECC.....	87
ANNEX 6-17: OUTLINE OF AN IEE REPORT	89
ANNEX 6-18: STRUCTURE FOR A SAMPLE EIA REPORT	92
Annex 6-19: Structure of EMP	102
ANNEX 6-20: EFFLUENT DISCHARGE REQUIREMENTS BY WORLD BANK.....	103
ANNEX 6-21: EMISSION DISCHARGE REQUIREMENTS BY WORLD BANK.....	105
ANNEX 6-22: NOISE LEVEL REQUIREMENTS BY WORLD BANK	106
ANNEX 6-23: WORLD BANK SAFEGUARDS POLICIES (SOCIAL)	107
ANNEX 6-24: RESETTLEMENT ACTION PLAN (RAP) PREPARATION	113
ANNEX 6-25: WORLD BANK ENVIRONMENTAL CLEARANCE PROCESS	120
ANNEX 6-26: ENVIRONMENTAL IMPACT AND MITIGATION MEASURES	125
ANNEX 6-27: GOB SOCIAL MANAGEMENT PROCESS	134
ANNEX 6-28: WB SOCIAL MANAGEMENT PROCESS	138
ANNEX 6-29: ADB SAFEGUARD POLICY STATEMENTS	153
ANNEX 6-30: GENERIC OUTLINE FOR AN INDIGENOUS PEOPLES DEVELOPMENT PLAN	160
ANNEX 6-31: GENERAL CHECK LIST TO EVALUATE IEE/ ESIA REPORTS.....	164
ANNEX 6-32: SECTOR SPECIFIC CHECKLISTS TO EVALUATE ESIA REPORTS.....	172
ANNEX 6-33: INVOLUNTARY RESETTLEMENT SCREENING AND CATEGORIZATION.....	194
ANNEX 6-34: ENVIRONMENTAL AND SOCIAL COVENANTS AND REQUIREMENTS FOR CONTRACTORS	196

Acronyms

APO	Annual Plan Operation
BIFFL	Bangladesh Infrastructure Finance Fund Limited
BMP	Best Management Practices
BRTA	Bangladesh Road Transport Authority
CO	Carbon Monoxide
CUL	Compensation under Law
DC	District Commissioner
DLAC	District Land Acquisition Committee
DoE	Department of Environment
EA	Environmental Assessment
ECA	Environment Conservation Act, 1995
ECC	Environmental Clearance Certificate
ECR	Environment Conservation Rules, 1997
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
EMMP	Environmental Mitigation and Management Plan
ERD	Economic Relations Division
ESA	Environmental and Social Assessment
E&S	Environmental and Social
ESDD	Environmental and Social Due Diligence
ESMF	Environmental and Social Monitoring Framework
ESMP	Environmental and Social Management Plan
FCD	Flood, Control and Drainage
FGD	Focus Group Discussion
FI	Financial Intermediary
FIL	Financial Intermediary Loan
GoB	Government of Bangladesh
ICT	Information and Communication Technology
IDA	International Development Association
IEE	Initial Environmental Examination
IIFC	Infrastructure Investment Facilitation Company
IPDP	Indigenous People Development plan
LAP	Land Acquisition Proposal
MoEF	Ministry of Environment & Forest
NEMAP	National Environmental Management Action Plan

NGO	Non-Government Organization
NOx	Oxides of Nitrogen
OD	Operational Directives
OP	Operational Policies
PAD	Project Appraisal Document
PAP	Project Affected Person
PI	Public Involvement
PPP	Public-Private Partnership
PRA	Participatory Rural Appraisal
RAP	Resettlement Action Plan
RDP	Regional Development Plan
SCC	Site Clearance Certificate
SIA	Social Impact Assessment
SMP	Social Management Plan
SOx	Oxides of Sulphur
TA	Technical Assistance
WB	World Bank
WQS	Water Quality Standard

1 THE ESMF DOCUMENT

1.1 Introduction

Public- Private Partnership (PPP) is a strategic mechanism for procuring, financing, and delivering public infrastructure and services. The PPP arrangement is meant to ensure sustainable investment in public infrastructure and service, contribute to soundness and accountability in public financial management and create an environment that facilitates a greater private sector role in financing and operating public infrastructure and services. As indicated in its Five Year Development Plan (FYP) 2010 – 2015, Government of Bangladesh decided to strengthen PPP in order to achieve its long-term vision as reflected in the Vision 2021. As part of PPP implementation plan, GOB published PPP Policy and Guidelines 2010, Guidelines and Scheme for PPPTAF 2012 and Guidelines for VGF for PPP Projects 2012.

Despite the above, one of the key challenges encountered in the PPP projects was with regard to compliance to the environment and social standards for effective planning, raising investment and for timely and smooth implementation of the projects.

Environmental and Social Monitoring Framework (ESMF) is a safeguard instrument and compliance policy covering GOB requirements. This establishes a mechanism to determine and assess future potential environmental and social impacts of different projects and other activities associated with the projects regardless of funding agency. The framework sets out mitigation, monitoring and institutional measures to be taken during design, implementation and operation of the project activities to eliminate adverse environmental and social impacts, offset them, or reduce them to acceptable levels. This instrument has been prepared as a separate and stand-alone document to be used for the BIFFL project. This framework of the BIFFL project is to carry out a due diligence on environment and social aspects of the project, after the investor makes an application for loan.

1.2 Structure of the Document

This document has been divided into eight chapters for the convenient use of this document. In the ESMF, the 'Environmental Monitoring Framework' and the 'Social Monitoring Framework' are dealt separately, so that users can easily go through both the frameworks without any difficulty.

The contents of the chapters are stated below.

- i) Chapter One gives an **overview of ESMF** including its purpose, users and environmental and social assessment process.
- ii) Chapter Two gives a **brief description of the BIFFL** including its objective, nature, components and the types of acceptable projects.
- iii) Chapter Three describes the relevant available target areas for **environmental management in Bangladesh**, which are important for environmental monitoring.

- iv) Chapter Four provides the **legal and institutional framework for establishing the environmental standards/guidelines** for BIFFL funded projects.
- v) Chapter Five is the **operational manual** including sub project implementation arrangements; and monitoring plans.
- vi) Chapter Six provides annexes with the **legal and institutional basis** for establishing the **social standards/guidelines**. It includes environmental and social impact assessment and mitigation measures with relevant **formats and checklists**.

1.3 Purpose of ESMF

The Environment and Social Monitoring Framework (ESMF) details the agreed policies, guidelines, and procedures to be integrated into the implementation of BIFFL projects. The main purposes of the ESMF are,

- To understand the process of undertaking ESAs for BIFFL sub-projects, which are usually PPP projects
- To establish clear procedures and methodologies for the Environmental and Social Due Diligence (ESDD) to be carried out before approval of loans to be financed by BIFFL;
- To specify appropriate roles and responsibilities, and outline the necessary reporting procedures, for managing and monitoring environmental and social concerns related to project investments;
- To provide practical information, resources for implementing the ESMF.

As a detailed guide on principles and procedures for incorporating environmental and social issues into project preparation, the ESMF would be a useful reference to potential project sponsors likely to participate in future stage of BIFFL projects.

1.4 Users of ESMF

As a core policy and procedural document for the BIFFL projects, the ESMF is designed for use by the following parties involved in a project:

- **BIFFL Project Staff:** As a document enshrining operating principles and guidelines to ensure that projects are adequately assessed to ensure compliance with environmental and social standards.
- **Project Sponsors:** As a document that spells out requirements with respect to environmental and social issues that need to be met, for obtaining finance for BIFFL.
- **Executing Agencies:** As a document for Executing Agencies to use a reference guide for ensuring E&S compliance prior to their award¹, for PPP projects that are likely to receive BIFFL funding.

¹ After the award of a sub-project to a project sponsor, the role of compliance primarily shifts to the project sponsor

- **Project Associates:** Stakeholders, consultants, lenders, etc.: As a document that explains the compliance requirements for environmental and social requirements for projects that receive financing for BIFFL projects.
- Other relevant government agencies, including DoE, for reference.

2 DESCRIPTION OF BIFFL

2.1 Background of BIFFL

In order to keep pace with sustainable development the Government of Bangladesh is keen to develop the infrastructure sector through private investment. As the government alone can hardly achieve such investment objective, it has prioritized Public-Private Partnership (PPP) as a key initiative for increasing the investment in infrastructure.

Government has taken up a supportive role to private sector investment by providing long term loan to private investors. In this respect, Bangladesh Infrastructure Finance Fund Limited (BIFFL) was incorporated on 21 March 2011 under Ministry of Finance (MoF) as a Special Purpose Vehicle (SPV) for financing infrastructure projects in Bangladesh. The main objective of BIFFL is to provide predominantly long-term financing for PPP projects through issuance of bonds and debt instruments and equity offerings. BIFFL envisages attracting private investments from local & foreign investors and to invest companies that are implementing infrastructure projects in Bangladesh.

Presently, the company has an eight-member Board of Directors. The overall Policy decisions of the Company are vested with the Board of Directors. The Board of Directors of BIFFL includes Secretary, Prime Minister's Office; Secretary, Power Division; Secretary, Energy & Mineral Resources Division; Secretary, Roads Division; Secretary, Bridges Division; and Additional Secretary of Finance Division. The Secretary Power Division is the Chairman of the Board.

2.2 Objectives of BIFFL

The primary objectives of BIFFL are:

- i) to receive and accept from GoB and/or any other foreign or local source including official or semi-official development sources, i.e.
 - o funds and moneys by way of loans, debts, debt backed by sovereign guarantees;
 - o bond specially designed for non-resident Bangladeshis, sub-sovereign debt, equity, preference shares or any other quasi-equity instruments;
 - o development of Limited Partner (LP) - for the development of infrastructure projects in Bangladesh;
- ii) to promote, encourage and finance private sector investment in all critical infrastructure sectors;
- iii) to create funds, sub-funds including Islamic funds and any other type of funds as deemed appropriate by the Company;
- iv) to engage Fund Managers, Investment Advisors, Management Consultants or any such other advisors or staff to manage the business of the Company;

2.3 Types of Acceptable Projects

BIFFL as a lender would seek participation in a) projects in which risks are most easily allocated to other parties; and b) projects for which risk mitigation measures are available and proven. Sectors by priority are:

2.3.1 Power and Energy

- i. generation of power, energy, nuclear energy, hydrocarbon exploration, hydropower, solar power, wind power;
- ii. production of biomass or biogas energy fueled by waste or process by-products in a manner that does not denigrate the environment within the territory of Bangladesh;
- iii. transmission of energy within the territory of Bangladesh or beyond for the purpose of economic development of the country;

2.3.2 Natural Resource Development

- i. exploration, production, transmission, distribution of natural gas & oil, petrochemicals, coal & other mineral resources, oil refinery & production of LPG;
- ii. production of fertilizer;
- iii. water supply and distribution, sewerage and drainage, effluent treatment plants (ETP);
- iv. land reclamation, dredging of rivers, canals, wetlands, lakes & other related facilities;

2.3.3 Transport

- i. roads and highways and expressways including mass rapid transport system, bridges, tunnels, flyovers, interchangers, city roads, bus terminals, commercial car parking;
- ii. airports, terminals and related aviation facilities;
- iii. port development (sea, river and land) including inland container terminals, inland container depot and other related services;
- iv. deep sea port development;
- v. railway systems, rolling stock, equipment and facilities;

2.3.4 Telecommunication

- i. telecommunication systems (cellular, broadband, satellite, cable and data transmission);
- ii. networks and services including information and communication technology;
- iii. e-service delivery to citizens;

2.3.5 Social Infrastructure

- i. health, education, human resource development, research and development, and cultural facilities;
- ii. poverty alleviation projects, e.g. pourashava and village water power supply systems, rural gas supply, rural internet projects, terminals/landing stations, rural health services and hospital; and irrigation and other agricultural services;

2.3.6 Other Infrastructure Projects

- i. basic and all necessary infrastructure related to tourism industry;
- ii. environmental, industrial and solid waste management projects;
- iii. economic zone, industrial estates and parks, city and property development;
- iv. services to support commercial and non-commercial activities.

3 GENERAL ENVIRONMENTAL MANAGEMENT

3.1 Overview

An Environmental Management Plan (EMP) is an implementation plan for mitigation, protection and/or enhancement measures, which are recommended in the EIA. It details:

- (a) The measures to be taken during the implementation and operation of a project to eliminate or offset adverse environmental impacts, or to reduce them to acceptable level.
- (b) The actions needed to implement these measures.

EMP documents are key elements of environmental management including the environmental policy, responsibilities, applicable standard operating procedures and best management practices (BMP), record keeping, reports, communication, training, monitoring, and corrective action. It features the “PLAN, DO, CHECK, ACT” model for future and ongoing improvement of environment.

Figure 3-1: PDCA Cycle



Before preparing Environmental Management Plan, it is necessary to get a brief idea on available environmental resources of Bangladesh.

3.2 Target Areas of Environmental Monitoring

The BIFFL sub projects are expected to be located all over Bangladesh. Therefore, it is necessary to keep in mind the general variation in the environmental conditions expected in different parts of the country. A brief description of the important environmental resources and issues are discussed in this section.

3.2.1 Geography

Bangladesh is located between 20°34' and 26°38' N, and 88°01' and 92°41' E. The area of the country is 1, 47,570 square km with a coastline of about 700 km in length. The land is flat to the extent of about 80% intersected by numerous rivers and their distributaries. The rest 20% of the area consists of uplands (~8%) and hill areas (~12%). The hill areas consist of Chittagong Hill Tracts, hill ranges of northeastern Sylhet and hills along the narrow strip of Sylhet and Mymensingh Districts.

Most of the land is used for agriculture. Land use is generally classified into five categories as agricultural, forest, cultivable waste, current fallow and as 'not available for cultivation'. The category of 'not available for cultivation' consists of mainly of urban, rural settlements and industrial land.

3.2.2 Hydrology

Bangladesh is characterized by an exceptional hydro-geographical setting. It is one of the largest deltas in the world and is criss-crossed by numerous rivers, their tributaries and distributaries. There are about 700 rivers, canals, and streams in Bangladesh, with a total length of approximately 22,155 km, which occupy a riverine area of about 9,384 sq km. A vast amount of water flows through Bangladesh mostly during June-October. The rivers of Bangladesh also carry huge amounts of sediment, an estimated 24 billion tons/year. The ultimate result is additional new land in some places due to accretion, forming islands called chars, and loss of land in some other places due to erosion.

Hydrology of a given area is extremely important for any sub-project, which may have impact upon water bodies. Impact on water flow by any subproject should be carefully considered in subproject planning.

3.2.3 Water Resources

Groundwater is a vital natural resource for the reliable and economic provision of potable water supply in both the urban and rural environment. Presently about 80% of the people in rural Bangladesh depend on groundwater for drinking. Urban water supply is also largely dependent on groundwater. Until the emergence of the Arsenic problem, groundwater was hailed for providing the access to clean drinking water in Bangladesh.

Groundwater in Bangladesh occurs at a very willow depth where the recent river-borne sediments form prolific aquifers in the floodplains. The main component of discharge is the withdrawal of groundwater by different types of tube wells. A minor component of discharge is natural flow towards lower gradient. The groundwater level is at or very close to the surface during the monsoon whereas it is at maximum depth during the months of April and May. Because of uneven distribution of water resources, dry season demand for water in Bangladesh is mainly met by groundwater.

3.2.4 Ecology

The most important terrestrial ecosystem in Bangladesh is forest. The total land under forest in Bangladesh is about 2.56 million ha. A significant part of the existing forest area is designated as State Forest; most of this land is actually barren of tree

vegetation. Again, Bangladesh has an enormous area of seasonal wetland. In fact, half of the country could be delineated as such. The area under perennial wetlands is much smaller and is principally permanent rivers and streams, willow freshwater lakes and marshes, fish ponds and estuarine systems in the extensive mangrove swamps.

Wetland resources are crucial to the environment of Bangladesh. Both perennial and seasonal wetlands provide habitats to a large variety of flora and fauna. Wetlands also provide subsistence for a significant proportion of the population through their fishery resources. But the wetlands of Bangladesh are being drastically affected by the impacts of the increasing human population. The wetlands are being lost to flood control, drainage and irrigation development. Severe erosion in the catchments areas is causing increased siltation, and having major impacts on the key wetland areas.

3.2.5 Climate

Bangladesh has a subtropical monsoon climate characterized by wide seasonal variations in rainfall, moderately warm temperatures, and high humidity. Regional climatic differences in this flat country are minor. Three seasons are generally recognized: a hot, humid summer from March to June; a cool, rainy monsoon season from June to October; and a cool, dry winter from October to March. Heavy rainfall is characteristic of Bangladesh. About 80% rainfall occurs during the monsoon season. The monsoons result from the contrasts between low and high air pressure areas that result from differential heating of land and water.

3.2.6 Environmentally and Ecologically Sensitive Areas

Environmentally sensitive areas are those having significant value in their natural state, or having socio-cultural significance or sensitivity. Cultural or historical sites and densely populated urban centers are examples of the latter category. Ecologically Sensitive Areas can be defined as areas that may contain unique features, maintain key natural processes, support endangered, endemic or threatened plants or animals and their habitats, or provide important breeding areas for wildlife. In sensitive areas, environmental and ecological impacts tend to be more severe than elsewhere, and therefore, extra precautions must be taken to avoid significant environmental impacts. In many cases, this will mean extra investments in mitigation measures, while in some cases these areas will simply have to be avoided, resulting, for example, in the abandonment of a subproject or there-alignment of a section of road. The list of ecologically critical areas declared by DoE is attached in **ANNEX 6-1**.

3.3 Main Environmental Concerns

The key environmental concerns in Bangladesh include the following and special care needs to be taken for preparing an EMP:

- Liquid effluents
- Ambient Air Pollution
- Surface water quality
- Solid and Hazardous Waste management
- Seasonal fluctuations in ground water table

- Noise pollution
- Groundwater quality
- Arsenic Contamination of Aquifers
- Traffic management
- Saline intrusion (coastal areas)
- Drainage
- River bank erosion
- Flooding
- Wetland deterioration
- Land degradation
- Indoor Air Pollution
- Forestry management
- Biodiversity conservation
- Fish and fisheries resource management

4 LEGAL AND INSTITUTIONAL FRAMEWORK (ENVIRONMENTAL)

4.1 Overview

The institutional and legal setting in Bangladesh provides the context within which the environmental assessment procedure for BIFFL projects has been developed. It is imperative that the project sponsors for BIFFL projects are aware of and comply with the legal and institutional requirements of the GOB as applicable.

4.2 Environment Related Policies in Bangladesh

This section will discuss the relevant policies, which regulate environmental management in Bangladesh and are also relevant to projects that may be financed by BIFFL. The relevant policies are,

1. National Environmental Policy, 1992
2. National Conservation Strategy, 1992
3. National Environmental Management Action Plan, 1995
4. National Water Policy, 2000
5. National Water Management Plan, 2001

4.2.1 National Environmental Policy, 1992

The National Environmental Policy sets out the basic framework for environmental action together with a set of broad sectoral action guidelines. Key elements of the policy are:

- Maintenance of the ecological balance and overall progress and development of the country through protection and improvement of the environment;
- Protection of the country's assets, properties and resources against natural disasters;
- Identification and regulation of all types of activities which pollute and degrade the environment;
- Ensuring sustainable utilization of all natural resources;
- Promoting active association with all environment related international initiatives.

The Environmental Policy requires the following specific actions with respect to the 'Industrial' sector:

- To phase in corrective measures in polluting industries;
- To conduct Environmental Impact Assessment (EIA) for all new public and private industrial developments;
- To ban, or find environmentally sound alternatives for, the production of goods that cause environmental pollution; and
- To minimize waste and ensure sustainable use of resources by industry.

Under the National Environmental Policy, Department of Environment is directed to review and approve all Environmental Impact Assessments.

4.2.2 *National Conservation Strategy, 1992*

The National conservation strategy provides recommendations for sustainable development in the industrial sector as follows:

- All industries will be subject to EIA and adoption of pollution prevention/control technologies will be enforced;
- Hazardous or toxic materials/wastes will not be imported as raw materials for industry;
- Import of appropriate and environmentally sound technology will be ensured; and
- Dependence on imported technology and machinery should gradually be reduced in favor of sustainable local skills and resources.

4.2.3 *National Environmental Management Action Plan, 1995*

The National Environmental Management Action plan (NEMAP) is a wide-ranging and multi-faceted plan, which builds on and extends the statements set out in the National Environmental Policy. NEMAP was developed to address issues and management requirements and set out the framework within which the recommendations of the National Conservation Strategy are to be implemented. NEMAP has the following broad objectives:

- Identification of key environmental issues affecting Bangladesh;
- Identification of actions necessary to halt or reduce the rate of environmental degradation;
- Improvement of the natural environment;
- Conservation of habitats and bio-diversity;
- Promotion of sustainable development; and
- Improvement of the quality of life of the people.

4.2.4 *National Water Policy, 2000*

The National water policy recognizes that continued development and management of the nation's water resources is essential and includes protection, restoration and preservation of the environment and biodiversity including wetlands, mangrove and other natural forests, endangered species and water quality. It also states objectives for all agencies and departments entrusted with water management in regards to their responsibilities for regulation, planning, construction, operation and maintenance. Pollution of surface and ground water around various industrial centers from untreated effluent discharge into water courses is a critical water management issue. The Policy of the Government in this regard is that:

- Zoning regulations will be established for location of new industries in consideration of safe water availability and suitable effluent discharge possibilities;
- Effluent disposal will be monitored by relevant government agencies to prevent water pollution;
- Standards of effluent disposal into common watercourses will be set by Water Resources Planning Organization (WARPO) of the Ministry of Water Resources in consultation with DoE;
- Industrial polluters will be required by law to pay for remedial clean up of water bodies polluted by them.

4.2.5 National Water Management Plan, 2001

The National Water Management Plan addresses options for water quality, considerations behind measures to clean up industrial pollution, where effluent discharge monitoring and zoning regulations for new industries are emphasized.

4.3 Relevant Laws and Regulations in Bangladesh

There are a large number of laws related to environmental issues, some dating back to 19 century exist in Bangladesh. The most important of these are the Environmental Conservation Act, 1995 (ECA 1995) and the Environmental Conservation Rules (under the ECA, 1995) 1997, (ECR 1997). Many of the other laws are cross-sectoral and are only partially related to environmental issues. Following table presents an outline of the National legal instruments along with a list of key governmental institutions that have regulatory power over environmental aspects associated with the projects financed by BIFFL.

Table 4-1: Environment Related Laws and Regulations

	Laws/ Regulations	Enforcing Agencies/Ministry/Authorities
1	The Environment Conservation Act, 1995 and subsequent amendments in 2000 and 2002	Department of Environment, Ministry of Environment and Forest
	<i>Regulated/ Enforced Items</i> • Declaration of Ecologically Critical Areas; • Obtaining Environmental Clearance Certificate; • Regulation with respect to vehicles emitting smoke harmful for the environment; • Regulation of development activities from environmental perspective; • Promulgation of standards for quality of air, water, noise, and soils for different areas and for different purposes; • Promulgation of acceptable limits for discharging and emitting waste; • Formulation of environmental guidelines relating to control and mitigation of environmental pollution, conservation and improvement of environment.	

	Laws/ Regulations	Enforcing Agencies/Ministry/Authorities
2	Environment Conservation Rules, 1997 and subsequent amendments in 2002 and 2003	Department of Environment, Ministry of Environment and Forest
	<i>Regulated/ Enforced Items</i> - Declaration of ecologically critical Area; - Requirement of Environmental Clearance Certificate for various categories of projects; Requirement for IEE/EIA according to the appropriate category of the project; - Renewal of the environmental clearance certificate within 30 days after the expiry; Provision of standards for quality of air, water & sound and acceptable limits for emission/discharges from vehicles and other sources.	
3	Environment Court Act, 2000 and subsequent amendments in 2002	Judiciary Division, Ministry of Environment & Forest
	<i>Regulated/ Enforced Items</i> GOB has given highest priority to environment pollution and passed 'Environment Court Act, 2000' for completing environment related legal proceedings effectively.	
4	The Vehicle Act, 1927 The Motor Vehicles Ordinance, 1983 The Bengal Motor Vehicle Rules, 1940	Bangladesh Road Transport Authority (BRTA)
	<i>Regulated/ Enforced Items</i> - Exhaust emission; - Vehicular air and noise; - Road safety; - Licensing.	
5	The Brick Burning (Control) Act, 1989 The Brick Burning (Control) Amendment Act, 1992	Ministry of Environment & Forest
	<i>Regulated/ Enforced Items</i> - Control of brick burning requires a license from the MoEF; - Restricts brick burning with fuel wood.	
6	The Removal of Wrecks and Obstructions in inland Navigable Water Ways Rules 1973	Bangladesh Inland Water Transport Authority
	<i>Regulated/ Enforced Items</i> Removal of wrecks and obstructions in inland Navigable Waterways.	
7	Water Supply and Sanitation Act, 1996	Ministry of Local Government, Rural Development and Cooperatives
	<i>Regulated/ Enforced Items</i> Management and Control of water supply and sanitation in urban areas.	
8	The Ground Water Management Ordinance 1985	Upazila Parishad
	<i>Regulated/ Enforced Items</i> Management of ground water resources; Tube well will not be installed in any place without the license granted by Upazila Parishad.	
9	The Forest Act, 1927 and subsequent amendments in 1982 and 1989	Ministry of Environment and Forest

	Laws/ Regulations	Enforcing Agencies/Ministry/Authorities
	<i>Regulated/ Enforced Items</i> Reserve Forests; Protected Forests; Village Forests.	
10	The Private Forests Ordinance Act, 1959	Regional Forest Officer, Forest Department
	<i>Regulated/ Enforced Items</i> Conservation of private forests and for the afforestation on wastelands.	
11	Bangladesh Wild Life (Preservation) Act, 1974	Ministry of Environment and Forest Bangladesh Wild Life Advisory Board
	<i>Regulated/ Enforced Items</i> Preservation of Wildlife Sanctuaries, parks, reserves.	
12	The Protection and Conservation of Fish Act 1950 subsequent amendments in 1982	Ministry of Fishery
	<i>Regulated/ Enforced Items</i> Protection and Conservation of fishes in Government owned water bodies.	
13	Natural Water Bodies Protection Act 2000	RAJUK/Town Development Authority/ Municipalities
	<i>Regulated/ Enforced Items</i> According to this Act, the character of water bodies i.e. rivers, canals, tanks, or floodplains identified as water bodies in the master plans or in the master plans formulated under the laws establishing municipalities in division and district towns will not be changed without approval of the concerned ministry.	
14	The Embankment and Drainage Act 1952	Ministry of Water Resources and FCD
	<i>Regulated/ Enforced Items</i> An Act to consolidate the laws relating to embankment and drainage and to make better provision for the construction, maintenance, management, removal and control of embankments and water courses for the better drainage of lands and for their protection from floods, erosion and other damage by water.	
15	Noise Pollution Regulations, 2006	Department of Environment.
	<i>Regulated/ Enforced Items</i> Regulations to address the problem of noise being above the acceptable limits as set in the regulations.	

4.4 Environmental Guidelines for Projects in Bangladesh

As pointed out earlier, the most important laws/rules in Table 4.1 are the ECA 1995 and the ECR 1997. The ECA 1995 is primarily an instrument of the Department of Environment (DoE) for controlling industrial pollution. The ECR, 1997 was promulgated to operationalize the enforcement of the Act. Depending on the extent of impact on the environment, it classifies all the projects in four categories. These are:

- i) Green;
- ii) Orange - A;
- iii) Orange- B; and
- iv) Red

4.4.1 Green Category

Projects, which do not have any negative impact on the environment, belong to Green category. For this category of projects, no Initial Environmental Examination (IEE) or Environmental Impact Assessment (EIA) is required. However, the project proponent will have to submit an application in a prescribed format to DoE for Environmental Clearance Certificate (ECC).

Examples

- Bamboo and cane goods;
- Candle, watches etc. assembling and manufacturing; and
- Cork (excluding metallic item).

4.4.2 Orange A and B Category

Orange category includes those projects that produce such wastes that can produce moderate or significant impacts on environment but the impacts could be mitigated easily if proper action is undertaken. Depending on the nature and extent of impacts the projects under Orange category has been sub-divided into two sub-categories- Orange A and Orange B. The projects/industries likely to produce some wastes but those are not harmful for surrounding environment and can be managed easily are categorized under “Orange-A”. Orange A & B projects require the two stages of clearances.

Example

- Small hotel/restaurant business;
- Weaving factory;
- Handloom industry;

The “Orange-B” category projects/industries are those likely to produce adverse environmental impacts but not to any significant level and that the impacts can be easily mitigated with no residual adverse impacts. Orange-B projects require an IEE.

Examples

- Mineral water, soft drink, etc., manufacturing and bottling;
- Processing of fish;
- Meat and other food items;

4.4.3 Red Category

This category includes industries, first requiring IEE for the purpose of obtaining site clearance, and then EIA, for obtaining environmental clearance (ECC). In this case also an application has to be made in a prescribed format along with an IEE report, on the basis of which site clearance may be granted with suitable conditions or the project may be rejected, on grounds of unsuitable location. If the site clearance is granted, the project proponent can go ahead with implementation of the project subject to the conditions as may be stipulated while granting the site clearance.

Examples

- Port development including container terminals, island container depot etc.;
- Telecommunication systems, networks and services including information
- Communication technology (ICT);
- Power generation, transmission, distribution and services.

4.5 Institutional Organizations

4.5.1 Department of Environment (DoE)

The primary institution for environmental regulation in Bangladesh is the Department of Environment (DoE), working under the Ministry of Environment and Forest (MoEF). Its mission is to assist in securing a clean and healthy environment through the fair and consistent application of environmental rules and regulations. The Department was created in 1989 to ensure sustainable development and to conserve and manage the environment of Bangladesh. The following Policy, Acts and Rules facilitate the activities of the Department

- Environment Policy, 1992;
- Environmental Conservation Act, 1995 and subsequent amendments in 2000 and 2002;
- Environmental Conservation Rules, 1997 and subsequent amendments in 2002 and 2003;
- Environment Court Act, 2000 and subsequent amendments in 2002

The principal activities of DoE are:

- Defining Environmental Impact Assessment (EIA) procedures and issuing environmental clearance permits - the latter being legal requirements before proposed projects can proceed to implementation stage;

- Providing advice or taking direct action to prevent degradation of the environment;
- Pollution control, including the monitoring of effluent sources and ensuring mitigation of environmental pollution;
- Setting the Water Quality Standard (WQS) for particular uses of water and for discharges to water bodies; and
- Declaring Ecologically Critical Areas (ECAs) where the ecosystem has been degraded to a critical state.

4.5.2 Department of Forest

The Department of Forest is responsible for sensitive area protection in four types of legally protected areas- wildlife sanctuaries, game reserves, reserved forests, and natural forests.

4.6 DOE Standards on Environmental Aspects

Following are the standards set by DoE regarding relevant environmental aspects:

- i) Standards Set by DoE for Effluent from Industrial Units (ref: **Annex 6-2**)
- ii) Standards Set by DoE for Gaseous Emission from Industries or Projects (ref: **Annex 6-3**)
- iii) Standards Set by DoE for Sound (ref: **Annex 6-4**)
- iv) Standards Set by DoE for Odor (ref: **Annex 6-5**)

4.7 DOE Requirements on Environmental Aspects

There are two stages of clearance required depending on the project category, these stages are:

- a. Initial Stage: Site Clearance Certificate (SCC)
- b. Advanced Stage: Environmental Clearance Certificate (ECC)

4.7.1 Initial Stage

For obtaining the SCC, the following are the requirements according to the categories of projects:

Green

- No Site Clearance Certificate required.

Orange A

- General information on the industry
- Description of raw materials and finished products
- Process flow diagram
- Layout plan showing ETP
- Waste discharge arrangement
- Relocation plan, if any
- NOC

Orange B

- Feasibility Report (for new industries only)
- Initial Environmental Examination (IEE) report including process flow diagram, layout plan and details of ETP
- Environment Management Plan (EMP) (Applicable for existing industries)
- Pollution effect abatement plan and emergency plan
- Outline of relocation plan
- A NOC

Red

- Feasibility Report (for new industries only)
- Initial Environmental Examination (IEE) report including TOR for EIA
- Process flow diagram
- Layout plan and details of ETP
- Time frame
- EMP with process flow diagram, layout plan with location of ETP and information on ETP (for existing industries only)
- Pollution effect abatement plan and emergency plan
- NOC from local authority
- Relocation or Rehabilitation plan (if any)

Without the SCC, the sponsor cannot apply for approval of the civil design of the project to the local authorities. After the SCC is obtained, the sponsor can proceed for land development and other primary civil works.

4.7.2 Advanced Stage

After receiving the SCC, the sponsors need to proceed for ECC. The EIA has to be submitted to DOE for clearance. After EIA approval, construction of super-structure and erection of capital machinery can be done. ECC needs to be obtained before commercial operation. The activities (document requirements in the case of Green category projects) that the project sponsors needs to carry out before applying for ECC are:

Green

- General information
- Description of raw materials and finished products
- A NOC

Orange A

The development of land and infrastructure may be carried out with letter of credit opened for machinery equipment and ETP. Installation of machinery including ETP may also take place. The next step is to apply for ECC. After ECC is issued, the project sponsor may apply for gas connection and commence trial and commercial production.

Orange B

The development of land and infrastructure may be carried out with letter of credit opened for machinery equipment and ETP. Installation of machinery including ETP may also take place. The next step is to apply for ECC. After ECC is issued, the project sponsor may apply for gas connection and commence trial and commercial production.

Red

The project sponsor may start land and infrastructure development. The sponsor then applies for approval of EIA and ETP design. On obtaining the approval, the ETP is installed and application is made for ECC. Once the ECC is obtained, application may be made for gas connection and trial and commercial production may be commenced.

4.7.3 Validity Period of Environmental Clearance Certificate

The table below gives the respective validity periods for the ECC for the various project categories:

Table 4-2: Validity Period of ECC

Categories of Project	Validity	Renewal Period
Green	3 years	30 days before expiry of the validity period
Orange A & B	1 year	
Red	1 year	

4.8 Bangladesh Bank's Environmental Risk Management (ERM)

Bangladesh Bank (BB), the central bank and apex regulatory body for the country's monetary and financial system has prepared Environmental Risk Management (ERM) guidelines for banks and financial institutions in January 2011. BB recognizes that environmental risk affects credit risk to a great extent; and hence has developed a risk based approach to account for environmental risk when assessing financing opportunities.

Principles: All Banks and FIs should make an "in principle" commitment to ERM in general and to the following in specific:

- Integration of ERM in their credit policies and procedures.
- Building awareness and providing Constant training and capacity building of their staff relevant to ERM.
- Adoption of a value adding approach to ERM with their potential borrower and aim to facilitate the borrower in addressing environmental issues that could lead to risks.

Requirements: BB recommends that all Banks and FIs should pass a Resolution of the Board or appropriate top and senior management committee on the adoption of these Guidelines and acceptance of the above principles. On an annual basis, all Banks/FIs should undertake a top management review to determine whether ERM is being effectively practiced in its operations.

Procedures: The procedures to be followed by FIs as per BB's ERM guidelines are as follows:

- Relationship banking / marketing: Identifying and planning for Environmental risks
- Credit risk management: Integrating Environmental risks

- iii. Credit processing and approval process: Incorporating Environmental risk covenants
- iv. Credit Administration: Verifying Environmental risk considerations
- v. Credit Monitoring: Carrying out Environmental risk monitoring
- vi. Database on Non-Performing Loans (NPLs) due to Environmental risks
- vii. Reporting system

The risk based approach recommended in BB's ERM will be applicable to FIs and will be incorporated in the operational procedures of BIFFL's ESMF

4.9 World Bank Safeguard Policies

Safeguard policies, incorporating environmental principles, are designed to protect the interests of third parties from adverse impacts of World Bank-assisted projects. The objective of these policies is to prevent and mitigate undue harm to people and their environment in the development process. BIFFL will need to adhere to the policies in case of handling WB funded projects. The Safeguards are designed to protect the environment from possible adverse effects of the projects. The World Bank's Safeguard Policies are described in **Annex 6-6**.

4.10 Land Acquisition in Bangladesh and World Bank Safeguards

4.10.1 Legal Context of Land Acquisition in Bangladesh

Currently the only legal framework that governs land acquisition in Bangladesh is the **Acquisition and Requisition of Immovable Property Ordinance, 1982**. However, its provisions are not adequate to address adverse impacts associated with land acquisition and involuntary displacement and do not fully satisfy the requirements of the Bank's Operational Policy (OP 4.12) on Involuntary Resettlement or that of the international practices. In essence, the law is largely indifferent to the landowners' present socio-economic conditions, or the long-term adverse impacts on incomes and livelihood that the acquisition and displacement may cause on the affected people. Also, there are no other policies that complement the acquisition ordinance in ways to assess, mitigate and monitor adverse impacts that the affected people may suffer. Some of the salient gaps in the existing legal framework are summarized below:

- **Avoiding/Minimizing Land Acquisition:** The law only implicitly discourages unnecessary acquisition, as lands acquired for one purpose cannot be used for a different purpose, and lands that remain unused be returned to the original owners. However, there are no mechanisms to monitor if these conditions are actually adhered to.
- **Eligibility for Compensation:** The law stipulates compensation only for the persons who appears in the land administration records as the owners. It does not recognize the rights of those, such as squatters, who do not possess legal title to the lands they live in or make a living from. There is thus no provision to mitigate the adverse impacts they suffer.
- **People who are impacted through loss of income are not recognized.** The Land Acquisition Act provides for compensation for lands and other fixed assets built and

grown on them (structures, trees and orchards, crops and any other developments like ponds, built amenities, etc.). However there is no provision to assess the impacts on peoples' incomes, livelihood, loss of employment and businesses for mitigation measures to restore loss of incomes and livelihood.

- **Compensation Standards:** Although the law stipulates payment of compensation at 'market prices' for acquired lands as the just compensation, the legal assessment procedures used almost always results in prices that are far below the actual market prices.
- **Relocation of Displaced Persons.** There is no provision in the existing laws for relocation of displaced families who are affected by the loss of their assets: land and/or structures.
- **Ensuring Payment/Receipt of Compensation.** The legal process to determine entitlements are too cumbersome and time consuming and do not ensure payment of compensation prior to their displacement. Lands are legally acquired and handed over to the project execution agency as soon as the authority identifies the owners (or 'awardees'), by examining the records, and sends a legal notice advising them to claim the compensation (or 'awards'). The onus is left on the affected land owners to prove, by producing an array of documents, that the acquired lands legally belong to them. As gathering these documents is a long, expensive and cumbersome process, many landowners may be unable to claim their awards. The project has meanwhile started to use the lands.
- **Socio-economic Rehabilitation.** Existing legal framework does not have any provisions to mitigate long-term impacts on peoples' livelihood caused by their displacement. Except for the compensation at the 'market price' for the loss of land, there are no other provisions, in the acquisition or other laws that require the government to mitigate the resultant adverse impacts caused by the acquisition. Socioeconomic rehabilitation of the involuntarily displaced persons is totally absent in the legal regime of the country.

4.10.2 World Bank's OP 4.12 Requirements

The primary objective of the World Bank policy on 'Involuntary Resettlement' is to explore all alternatives to avoid or at least minimize involuntary resettlement. Where resettlement is unavoidable, the resettlement activities should be conceived and executed as sustainable development programs, providing sufficient resources to enable affected persons to share in project benefits and assisted in their efforts to improve their livelihood and standard of living, or at least to restore them to pre-project level. The policy also requires that affected people are meaningfully consulted and should have opportunities to participate in planning and implementing resettlement programs. The policy applies to the taking of land and other assets when involuntary resettlement results in the loss of shelter, the loss of all or part of productive assets, or access to them, and the loss of income sources or means of livelihood, with or without physical displacement.

Measures required to ensure that resettlement has a positive outcome include:

- Providing Project-affected persons with options;
- Permitting their participation in planning and selecting these options;
- Prompt compensation at full replacement cost for losses;
- Choosing relocation sites that provide, at a minimum, the same benefits as the sites they replace;
- Providing allowances and other assistance to make a smooth transition after displacement;
- Identifying vulnerable groups and providing and special assistance to these groups; and,
- Implementing an institutional structure that supports the process to a successful end.

The World Bank's Policy on Involuntary Resettlement requires payment of compensation and other assistance to project affected people before they are displaced from their existing locations. Further the policy requires income rehabilitation assistance to those affected severely due to the loss of their productive assets or loss of incomes and livelihood. The Outline of a Resettlement Plan and the Resettlement Action Plan Preparation are shown in **Annex 6-8** and **Annex 6-24** respectively.

Absence of legal title does not exclude individuals from the eligibility to receive compensation and/or other assistance. The displaced or affected population eligible for compensation for losses include: those who have formal legal rights to land or other assets, and those who initially do not have formal legal rights to land or other assets but have a claim to legal rights based upon the laws of the country. The Policy also requires that those without legal title should be given assistance to meet the objectives of the policy. The genesis of these rights may come from continued possession of public land where the government has not sought their eviction. Bank's policy also recognizes that stakeholders who illegally occupy project-affected areas after established cut-off-date for any components are not eligible for compensation and other assistance provided that adequate measures are taken for information dissemination to people.

The gaps in the existing legal framework of Bangladesh and the objectives and requirements of the World Bank and other multilateral agencies are well recognized. Therefore, institutional or project specific policies are prepared to address these gaps and to meet the requirements of multilateral agencies in projects funded by them. The following table shows Comparison of Bangladesh Law with World Bank Groups Operational Directives on Involuntary Resettlement and Indigenous Peoples (OD 430, OD 4.20)

Table 4-3: Key comparisons between Bangladesh Law and ODs 4.30 and 4.20

Criteria	Requirement as per Bangladesh law	Requirement as per ODs 4.30 and 4.20
1. Scope/Applicability	Country Specific	Guiding principles
2. Schedule of Social Assessment	Not specified	Defined linkages between social assessment and project cycle
3. Preparation of Social Assessment	Not specified	Mandatory
4. Social Assessment Outputs	Not specified	Mandatory RAP and IPDP
5. Definition of PAP	Not specified	Inclusive
6. Indigenous Peoples	No special provisions	Special status
7. Extent of compensation	Provision in principle, but no clear guidelines	Replacement value and benefits
8. Nature of compensation	Limited	Comprehensive
9. Public consultation	Provisions, no clear guidelines	Participatory, entire project cycle
10. Disclosure	Not mandatory	Mandatory

4.11 ADB Requirements on Environmental Aspects

ADB Strategy 2020 emphasizes assisting Developing Member Countries (DMCs) to pursue environmentally sustainable and inclusive economic growth. ADB is committed to ensuring the social and environmental sustainability of the projects. In this context, the objective of ADB Safeguard Policy Statement (SPS) is to promote the sustainability of project outcomes by protecting the environment and people from potential adverse impacts.

The objectives of ADB safeguards are to:

- i. avoid adverse impacts of projects on the environment and affected people, where possible;
- ii. minimize, mitigate, and/or compensate for adverse project impacts on the environment and affected people when avoidance is not possible; and
- iii. assist borrowers/clients to strengthen their safeguard systems and develop the capacity to manage environmental and social risks.

Requirements of ADB Safeguard Policy Statement for Financial Intermediaries (FIs) are as follows:

- i. FIs will be required to establish an appropriate Environmental and Social Management System (ESMS) commensurate with the nature and risks of the FI's likely future portfolio to be maintained as part of the FI's overall management system;

- ii. FIs investments will be in compliance with applicable national laws and regulations and will apply the prohibited investment activities list (**Annex 6-10**) to subprojects financed by ADB.
- iii. Where the FI's investments have minimal or no adverse environmental or social risks, the FI project will be treated as category C project and need not apply any other specific requirements;
- iv. Where the FI's subprojects to be funded by ADB, have potential for significant environmental or social impacts, the FI will be required to ensure that such subprojects meet ADB's safeguard policy requirements, including those specified in Safeguard Requirements. Refer **Annex 6-29**.

The Environmental and Social Management System (ESMS) for the FIs will incorporate the following elements:

- i. environmental and social policies;
- ii. screening, categorization, and review procedure;
- iii. organizational structure and staffing including skills and competencies in environmental and social areas;
- iv. training requirements; and
- v. monitoring and reporting.

i). Environmental Safeguards

The ADB Safeguard Policy Statement includes explicit policy principles and requirements on:

- a. environmental assessment process
- b. biodiversity protection and natural resources management
- c. pollution prevention and abatement
- d. occupational and community health and safety
- e. physical cultural resources

More information about GOB and the World Bank related environmental and social aspects are described in the following Annexes:

- a) **Annex 6-26:** Environmental Impact and Mitigation Measures
- b) **Annex 6-27:** GOB Social Management Process
- c) **Annex 6-28:** WB Social Management Process

5 OPERATIONAL MANUAL

5.1 General Operational Procedure

General operational procedures of project appraisal in terms of environmental and social compliance described in this section are applicable to the projects seeking BIFFL loan and will be followed by Environmental and Social Cell (ESC) of BIFFL in coordination with the sponsors.

i) **Meeting with the Project Proponent**

The project proponent will make a loan application for the consideration of BIFFL financing. An initial query meeting will be conducted between the project proponent and BIFFL. The project proponent at this stage will also be referred to the ESC of BIFFL for understanding environmental social requirements and procedures of for the project. The ESC will guide the project proponent for filling the environmental social related aspects of the loan application form. The meeting will discuss the following:

- i) BIFFL's business process with ESMF integration
- ii) Environmental and social risk rating
- iii) Appraisal report preparation by ESC
- iv) Terms of Reference for EIA studies to follow DOE requirements or Terms of Reference for Initial Environmental Examination (IEE) studies as per ESMF
- v) Site visit requirement
- vi) Environmental and social monitoring protocol
- vii) Conditions precedent and subsequent to disbursement

ii) **Identify the Investment Project**

The primary objective of BIFFL is to provide long term financing for infrastructure projects. BIFFL will seek loan application to encourage and finance private sector funding in the infrastructure sectors as mentioned in section 2.3. The ESC of BIFFL will verify project against the prohibited investment activities as identified by IFC and ADB. Any project featuring in the exclusion lists will not be financed by BIFFL. The Exclusion lists of IFC and ADB Prohibited Investment Activities Lists are attached as **Annex 6-9** and **Annex 6-10** respectively.

iii) **Assess Environmental and Social Risk**

The main determinants of environmental risk are the nature of a borrower's business activity and the vulnerability of the location of the business activity. Environmental risks will vary according to different forms of transactions, e.g. risks associated to a long-term financing will be comparatively more than the risk associated with short term financing.

The overall purpose of environmental risk assessment is to understand and manage risks that arise from environmental concerns. The specific purposes are to:

- examine the environmental issues and concerns associated with potential business
- activities proposed for financing;
- identify, evaluate and manage the environmental risks and the associated financial implications arising from these issues and concerns;
- enhance the credit risk appraisal process.

iv) Follow Risk Rating System

This the risk rating criteria to be followed in the ESMF for projects that are acceptable to BIFFL financing. Refer table 5.2 below.

The risk rating criteria is applicable to new (Greenfield) as well as existing projects (where refinancing is required).

The risk rating criteria for ESMF harmonizes requirements of:

- a) GOB 'ECR' 97 - Green, Orange-A, Orange-B and Red categories;
- b) BB's ERM risk rating approach;
- c) Criteria for classifying projects as Category A/B/C/F1 according to the World Bank Operation Policies (OP 4.01);
- d) Criteria for classifying projects as Category A/B/C/F1 according to ADB's SPS;
- e) Capacity of the project proponent to manage E&S risks.

Review of the project documents is recommended before evaluating the risk rating of the project.

The project proponent will provide ESC required information related to Environmental and Social aspects of the project in the Loan Application Form.

Harmonization of Environmental safeguards

The requirements of GOB environmental safeguards, ADB's, Safeguard Policy Statement, BB's Environmental Risk Management (ERM) and the World Bank ESMF are shown in the following Table.

Table 5-1: Environmental Key Requirements

GOB Environmental Safeguards	ADB Safeguard Policy Statement
• Requirements for Industrial Units and Projects • Location Clearance • Categorization: Green, Orange-A, Orange-B, Red • Category Orange-B/Red: ▪ Feasibility Report, Initial Environmental Examination Report ▪ Environmental Management Plan (EMP) Report ▪ No objection certificate from the Local Authority ▪ Emergency Plan. ▪ Outline of the relocation, rehabilitation plan ▪ Labour Code 2006 for health and safety	• Environmental and Social Management System (ESMS) for Financial Intermediaries • Prohibited Investment Activities List • Categorization: ▪ Category A – EIA ▪ Category B – IEE ▪ Category C – Environmental Review • Public Consultation • Information Disclosure • Monitoring and Reporting • Grievance Redress Mechanism
Bangladesh Bank ERM	World Bank's ESMF
• Environmental Risk Management (ERM) in credit risk for Banks and Financial Institutions • ERM Policy • Identifying and categorizing environmental risk – High, moderate, Low (Sector Specific) • Incorporating risk monitoring • Environmental risk monitoring • Database on Non-Performing Loans (NPLs) due to Environmental risks • Reporting system	• General Principles for Environmental Safeguard Management ▪ All projects classified as Category B ▪ Bank policy OP 4.10 will not be triggered ▪ Monitoring ▪ Educate the communities ▪ GOB Labour Code 2006 to be followed by manufacturers/suppliers ▪ Monitoring (new battery) and collection (expired battery) • EIA will follow GOB procedure for ECC • EMP and mitigation measures • Capacity-building and monitoring of safeguard framework implementation • Consultation and Disclosure

Based on the above environmental key requirements the criterion of risk rating may be set as per list of queries described in table below.

Table 5-2: Project E&S Risk Rating Criteria in ESMF

Sl. No	Environmental and Social Risk Rating Criteria	Yes	No
1.	For new projects i. Does the project have any pending compliance such as Location and Environmental Clearance based on its category (Red, Orange-A, Orange-B and Green), from the DOE? For refinancing projects ii. Does the project have a pending DOE Clearance or pending necessary operating licenses and permits on EHS from respective regulatory authorities?		
2.	Is the project located in the immediate vicinity (likely to cause adverse impact) of environmentally critical areas (national parks, wetlands, wildlife habitats, important bird areas, and protected areas)? Ref: Environmentally Critical Areas Rules, 2010		
3.	Does the project construction and/or operation lead to environmental impacts that are diverse, irreversible and / or Unprecedented in nature? Refer to IEE/EIA reports if available or Environmental Due Diligence (EDD) during site visit, to answer this question		
4.	Does the project require involuntary resettlement that results in loss of land or livelihoods or physically displaces more than 200 persons?		
5.	Is the project site on or in immediate vicinity of socially vulnerable or Indigenous People owned or occupied land and has the potential to cause an adverse impact on their culture and identity?		
6.	Is the project vulnerable to climate change related impacts?		
	<i>E&S Capacity of the Borrower</i>		
7.	Does the Borrower have a documented Policy on Environmental and Social Performance?		
8.	Does the Borrower have dedicated human resources to address Environmental and Social performance?		
9.	Has the Borrower established and implemented Environmental, Health & Safety Management Systems and Social Accountability Systems for the Project SPV or in the parent company?		

Diverse, Irreversible and Unprecedented Environmental Impacts

- **Diverse Impacts** – impacts resulting on multiple E&S components or receptors over a varying time and spatial scale (e.g. activities that can cause large scale adverse impacts on local air quality, noise levels, generation of hazardous wastes as well as nuisance to community)
- **Irreversible Impacts**– impacts on E&S components that, in all practical terms are permanent in nature and cannot be reversed in spite of the removal of the causal stress factor (e.g. construction or change in land use that permanently alters the natural drainage or destroys habitats used by migratory birds)
- **Unprecedented Impacts** – are impacts that are first of its kind in terms of available knowledge of their potential to cause harm to the E&S components and their effective mitigation (e.g. impact of noise pollution on an endangered faunal species in a geographical region where no prior studies are available on impact tolerance and response of the species)

v) Determine the Risk Rating for the Project

- a) Based on the Table above, if any question numbered 1 to 6 is answered as Yes, the Project E&S Risk is rated **“High”**.
- b) If all the questions numbered 1 to 6 are a No and at least one of the questions numbered 7 to 9 is a No, then the Project E&S Risk is rated as **“Moderate”**.
- c) In case of refinancing of existing projects in advanced stage, if all the questions numbered 1 to 6 are a No and any two of the questions numbered 7 to 9 is a **No**, then the Project E&S Risk is rated as **“High”**.
- d) If all questions numbered 1 to 6 are a **No** and all questions rated 7 to 9 are a **Yes** then the Project E&S Risk is rated as **“Low”**.

A. New /Greenfield Projects

High Risk Projects (applicable if risk rating is High on account of questions 1,2,3 or 6) of the above table: For new projects categorized as “High “risk, the following procedures will be followed:

- i. The ESC will visit the project area. A site visit checklist is attached in **Annex 6-11**
- ii. The project proponent will submit the IEE checklist to BIFFL that was submitted to DOE while applying for Site Clearance
- iii. The ESC will prepare a preliminary environmental appraisal report (PEAR) based on the risk rating, site visit and IEE checklist. Refer **Annex 6-12** for the format of PEAR. The ESC will discuss with the project proponent before sending the report for BIFFL Board for approval.
- iv. Once BIFFL issues the preliminary letter of support and letter of engagement, ESC will conduct a detailed environmental appraisal based on the EIA and EMP report submitted by the project proponent.

- v. The project proponent will submit half yearly monitoring reports on implementation of EMP, the scope of the report is given in **Annex 6-13**

Moderate Risk Projects: For new projects categorized as “Moderate” risk, the following procedures will be followed:

- i. The ESC will conduct a desk review of the environmental section of the loan application form;
- ii. The project proponent will submit the IEE checklist to BIFFL that was submitted to DOE while applying for Site Clearance;
- vi. The ESC will prepare a PEAR based on the risk rating, desk review and ESC will discuss with the project proponent before sending the report for BIFFL Board for approval;
- vii. Once BIFFL issues the preliminary letter of support and letter of engagement, ESC will conduct a detailed environmental appraisal based on the EIA and EMP report submitted by the project proponent;
- viii. The project proponent will submit half yearly monitoring reports on implementation of EMP, the scope of the report is given in **Annex 6-13**.

Low Risk Projects: For new projects categorized as “Low” risk, the following procedures will be followed:

- i. The ESC will prepare a PEAR (refer **Annex 6-12** for format) based on review of the loan application form and risk rating;
- ii. The project proponent will submit the Site and/or Environmental Clearance Certificate from DOE to BIFFL before loan disbursement.
- iii. The requirements of BIFFL’ ESMF on GOB legal compliance, implementation of EMP, monitoring, reporting, information disclosure and grievance redress mechanism will be inserted in the loan documentation as covenants, if any.

B. Refinancing/Existing Projects in advanced Stage

High Risk Projects: For refinancing/existing projects categorized as „High“ risk, the following procedures will be followed:

- i. The project proponent will submit the EIA and EMP reports as well as the SCC and ECC Certificate granted by DOE to BIFFL;
- ii. The ESC will in addition visit the project area to check the status of works and compliance with the EMP.
- iii. The ESC will prepare a PEAR based on the risk rating, site visit and review of EIA and EMP reports for approval. Refer **Annex 6-12** for the format of PEAR.
- iv. Once BIFFL issues the preliminary letter of support and letter of engagement, ESC will conduct a detailed environmental appraisal based on the EIA and EMP report submitted by the project proponent.

- v. Environmental Due Diligence and Corrective Action Plan (CAP) as per **Annex 6-14** and **Annex 6-15** will be conducted preferably by the external consultant and submitted for approval.
- vi. The project proponent will submit annual monitoring reports on implementation of CAP.
- vii. The requirements of ESMF on GOB legal compliance, implementation of CAP, monitoring, reporting, information disclosure, grievance redress mechanism shall be inserted in the loan documentation as covenants.

Moderate Risk Projects: For refinancing/existing projects categorized as “Moderate risk, the following procedures will be followed:

- i. The project proponent will submit the EIA and EMP reports as well as the SCC and ECC Certificate granted by DOE to BIFFL;
- ii. The ESC will conduct a desk review of the environmental section of the loan application form;
- iii. The ESC will in addition visit the project area to check the status of works and compliance with the EMP;
- iv. The ESC will prepare a PEAR based on the risk rating, site visit and review of EIA and EMP reports. **Refer Annex 6-12** for the format of PEAR.
- v. Once BIFFL issues the preliminary letter of support and letter of engagement, ESC will conduct a detailed environmental appraisal based on the EIA and EMP report submitted by the project proponent;
- vi. Environmental Due Diligence and Corrective Action Plan (CAP) will be conducted preferably by the external consultant and submitted for approval.
- vii. The project proponent will submit annual monitoring reports on implementation of CAP.
- viii. The requirements of ESMF on GOB legal compliance, implementation of CAP, monitoring, reporting, information disclosure, grievance redress mechanism shall be inserted in the loan documentation as covenants.

Low Risk Projects: For refinancing/existing projects categorized as “Low” risk, the following procedures will be followed:

- i. The ESC will prepare a PEAR (**refer Annex 6-12** for the format) based on review of the loan application form and risk rating.
- ii. The project proponent will submit the Site and/or Environmental Clearance Certificate from DOE to BIFFL before loan disbursement.
- iii. The requirements of ESMF on GOB legal compliance, implementation of EMP monitoring, reporting, information disclosure, grievance redress mechanism shall be inserted in the loan documentation as covenants, if any

PSC of BIFFL will evaluate the IEE and EIA reports submitted by the project proponent using prescribed checklists as per

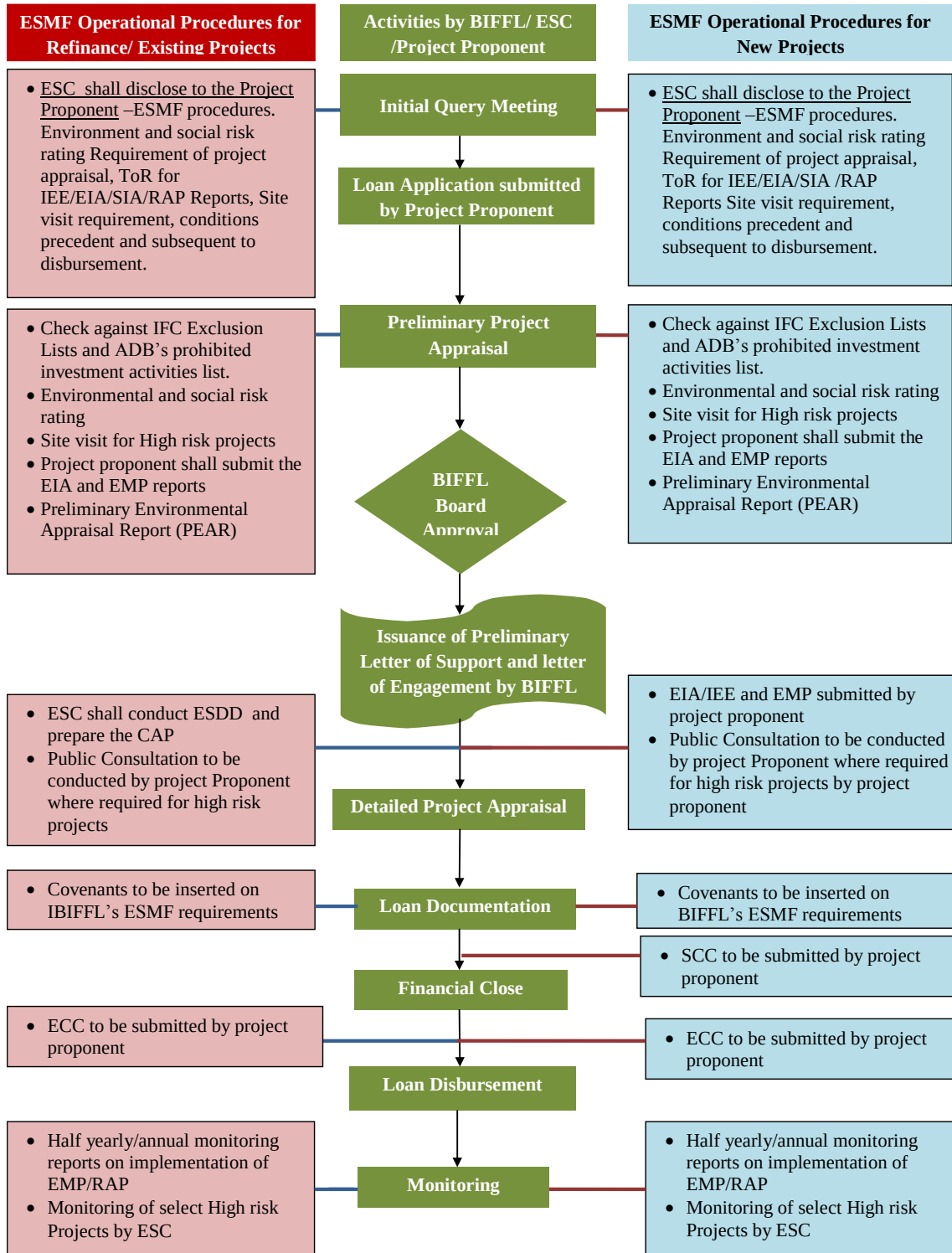
a) **Annex 6-31:** General Checklist to Evaluate IEE/EIA

b) **Annex 6-32:** Sector Specific Checklists for to Evaluate ESIA Reports

After evaluation ESC needs to include covenants and requirements in the agreements of contractors (like EPC contractor). Environmental and Social covenants and requirements are included in **Annex 6-34**.

Project appraisal activities pertaining to environmental and social aspects to be carried out by PSC and Project proponent are summarized in the following figure.

Figure 5-1: Project Appraisal Activities by Key Stakeholders



5.2 Sequence of Tasks of ESMF in Different Phases of BIFFL Project

The ESMF tasks have been grouped in five phases, as mentioned below. The activities, by a project sponsor, in different phases are described in detail in the subsequent sections. The phases are:

- i. Pre-Construction Phase
- ii. Land Acquisition Phase
- iii. Financing Phase
- iv. Construction Phase
- v. Post-Construction Operation Phase

i) **Pre-Construction Phase**

- a) Environmental and Social Screening, report on Impact Identification and project alternatives (prepared by the sponsor)
- b) Initial Environmental Examination IEE
Site Clearance Certificate (SCC) obtained for minor civil works.
- c) Social Impact Assessment (SIA), after base line data collection, will include²:
 - Social Management Plan (SMP), with Mitigation measures
 - Rehabilitation Plan (RP)
- d) Environmental Impact Assessment EIA, will include:
 - Environmental Management Plan EMP

Environmental Clearance Certificate (ECC) obtained and project construction may start, subject to availability of fund.

DOE prescribed Pro-forma for applying for ECC is provided as **Annex 6-16**.

ii) **Land Acquisition Phase**

Case A: Land Purchase by Sponsor (Through Negotiation)

Land is purchased directly by the Sponsor, through negotiation with the landowners. The usual policy is to pay the market price or the replacement value of the land and the structures. A 'Land Acquisition Specialist' engaged by the project sponsor (or requiring body, RB) and its representative works jointly, in collaboration with a Legal Advisor.

No RAP is required in such cases and development may start immediately.

Case B: Land Acquisition By GoB

- a) Land Acquisition Plan (LAP), prepared by requiring body, RB. Includes:
 - Land plot schedule of the project foot-print area, along with Mouza maps

² Applicable only in case of financing WB funded projects

- Entitlement list and Compensation package prepared by joint verification survey, by representatives of RB and Deputy Commissioner(DC)'s office
- b) Estimated land compensation money deposited by RB and Compensation payment made by DC to the affected persons.

Land acquired and handed over to RB and development starts

Case C: Land Acquisition by GoB (for availing World Bank finances)

1. Disclosure (of Project information) and Consultations (with potential affected persons), by project sponsor and/or engaged consultants.
2. Socio-Economic (sample) Survey, PRA and FGD in Project and Surrounding Area, the methodology depending on pattern of livelihood and other parameters.
3. Resettlement Action Plan (RAP) preparation, after Socio-economic (census) Survey in the Project footprint area. The RAP will include:
 - Project Affected People (PAP) list
 - Compensation policy
 - Compensation Matrix and Entitlement Plan
 - Relocation Plan (RP), if any
 - Indigenous People Development Plan, IPDP (if required)

Generic Outline for Indigenous Peoples Development Plan is shown in **Annex 6-30**

[For projects not requiring specific resettlement and LA studies, a 'Social Performance Framework' will be sufficient]

4. Engagement of RAP Implementation Agency (Consulting Company or NGO)
5. Disclosure of RAP information and Consultations with PAP
6. Grievances and Redress Mechanism management
7. RAP approved by WB
8. Compensation Payment to PAP
 - Compensation as per GoB policy by DC
 - Additional Entitlement to PAP, by the project

iii) Financing Phase

Loan Application is submitted to BIFFL, by the private sector sponsor, at the stage when the project is awarded by the public sector and agreement signed between them. By this time, most of the studies may already have been completed and the land acquired.

- a) Review of and due diligence through an ESDD on already completed EIA and SIA Reports. Assess if all issues have been adequately addressed. Decide if further alternatives are to be examined or there are any existing performance gaps and corrective actions needed.

iv) Construction Phase

- a) Implementation of Environmental Management Plan (EMP) and its monitoring
- b) Monitoring of Social Management Plan (SMP), which includes RAP, Relocation, Livelihood restoration, Training activities, etc.)

v) Post-Construction Operation Phase

- a) Environmental and Social Audit, as per recommendation of EIA and SIA.
- b) Regular monitoring of environmental issues as per EMP, and keep records.

5.2.1 Pre-Construction Phase Activities

After identification of an infrastructure project in PPP, the project sponsor will carry out the Environmental and Social screening of the project. After screening, sponsor will apply for site clearance certificate for stating the project at site. National Environment Conservation Rules (ECR) 1997 provides a category-wise list of industries, projects and development activities. Projects having insignificant or no impact on environment belong to Green category list. Environmental screening of a Green or Orange-A category project, on the basis of the category list for DoE clearance, will however be reviewed if more adverse environmental impact of the project is subsequently identified and Initial Environmental Examination (IEE) will need to be conducted. More adverse environmental impact of the project may be identified at the stage of construction, implementation and operation of the project and may occur in case of Red category project as well.

a) Initial Screening

The sponsor thus conducts a preliminary screening, for DoE compliance whether EIA process and, if so, the type IEE or EIA is needed to be applied to the specific case and also reviews impacts of green and orange-A category projects in subsequent stages. All the applications are required to be submitted in DoE prescribed Form-3. Category wise DOE Clearance and documents required for the clearance are given in table below.

Table 5-3: Summary Table for Document Requirement

Project Category	DoE Certificate	Documents to be submitted
Green	ECC	• Application • General information on the industry, Raw materials & finished products • Registration certificate from BOI (if applicable), • NOC from local authority, • Land ownership/ rental deed etc (if applicable),

Project Category	DoE Certificate	Documents to be submitted
		Resettlement Action Plan (RAP) for affected people, if any, Indigenous People's Development Plan (IPP), if applicable etc.
Orange-A	SCC,ECC	- Application Same documents, as required for Green Category projects. Additional documents: - documents showing, process flow diagram, layout plan, waste discharge, treatment and disposal arrangement etc.
Orange-B	SCC,ECC	- Application Same documents, as required for Green Category projects. Additional documents: - Initial Environmental Examination (IEE), - Feasibility Report - Information / report as specified in the checklist given in the IEE e.g, map, process flow diagram, layout plan showing waste discharge, treatment and disposal arrangement.
Red	SCC, Approval of EIA, ECC	- Application (in Form-3), Same documents, as required for Green Category projects. Additional documents: - Initial Environmental Examination (IEE), - Feasibility Report - Information / report as specified in the checklist given in the IEE e.g, map, process flow diagram, layout plan showing waste discharge, treatment and disposal arrangement etc, - design for Effluent Treatment Plant (ETP) and other applicable environmental pollution control devices with Environmental Management Plan (EMP), Pollution effect abatement & emergency plans etc.

Application has to be submitted to the concerned Divisional Office of the DoE in prescribed form (Form-3) for obtaining ECC for Green category projects, SCC & ECC for Orange-A & Orange-B category projects and SCC, EIA approval & ECC for Red category projects. No separate application for SCC, EIA approval and ECC are needed as all the clearances are issued against the original application for ECC and the submitted documents.

Issuing ECC and SCC

DoE does not ask for any separate initial examination or assessment of social impact of the project and hence no separate Social Clearance Certificate, other than ECC, is issued. Documents for assessment, compensation/mitigation and management of social impact of the project are submitted along with the application (Form-3) for environmental clearance from DoE. The prescribed pro forma for applying for ECC is annexed in **Error! Reference source not found..**

Green category projects: Sponsors of Green category projects do not need SCC to proceed for land development and primary civil works. But, without SCC, sponsors of Orange-A, Orange-B and Red category projects can not apply for approval of the civil design of the project to the local authorities. After having the SCC, the sponsor can proceed for land development and primary civil works.

For ECC for Green category projects, the followings are to be attached to the application (in Form-3) to the DoE:

- Necessary treasury challan,
- General information on the industry, Raw materials & finished products etc,
- Registration certificate from BOI (if applicable),
- NOC from local authority,
- Land ownership/ rental deed etc (if applicable),

After having the ECC, the sponsors can commence commercial production.

The application and documents required for Green Category projects are common for all category. Additional documents for Orange-A category project are needed depending on the extent and impact of the project on the environment and, for Orange-B and Red Category projects, on the environmental risks identified in the Initial Environmental Examination (IEE) report.

Orange-A category projects: For SCC & ECC, for Orange-A category projects, the followings documents, in addition to those required for *Green Category projects*, are to be attached to the application (in Form-3) to the DoE:

- process flow diagram,
- layout plan,
- waste discharge,
- treatment and disposal arrangement.

After having SCC, the sponsors can develop land & infra-structure, build super structure and install machinery. After having the ECC, the sponsors can install gas connection (if needed) and commence trial & commercial productions.

Orange-B category projects: For SCC & ECC, for Orange-B category projects, the following documents, in addition to those required for *Green Category Projects*, are to be attached to the application (in Form-3) to the DoE:

- Initial Environmental Examination (IEE),
- Feasibility Report
- Information / report as specified in the checklist given in the IEE e.g, map, process flow diagram, layout plan showing waste discharge, treatment and disposal arrangement.

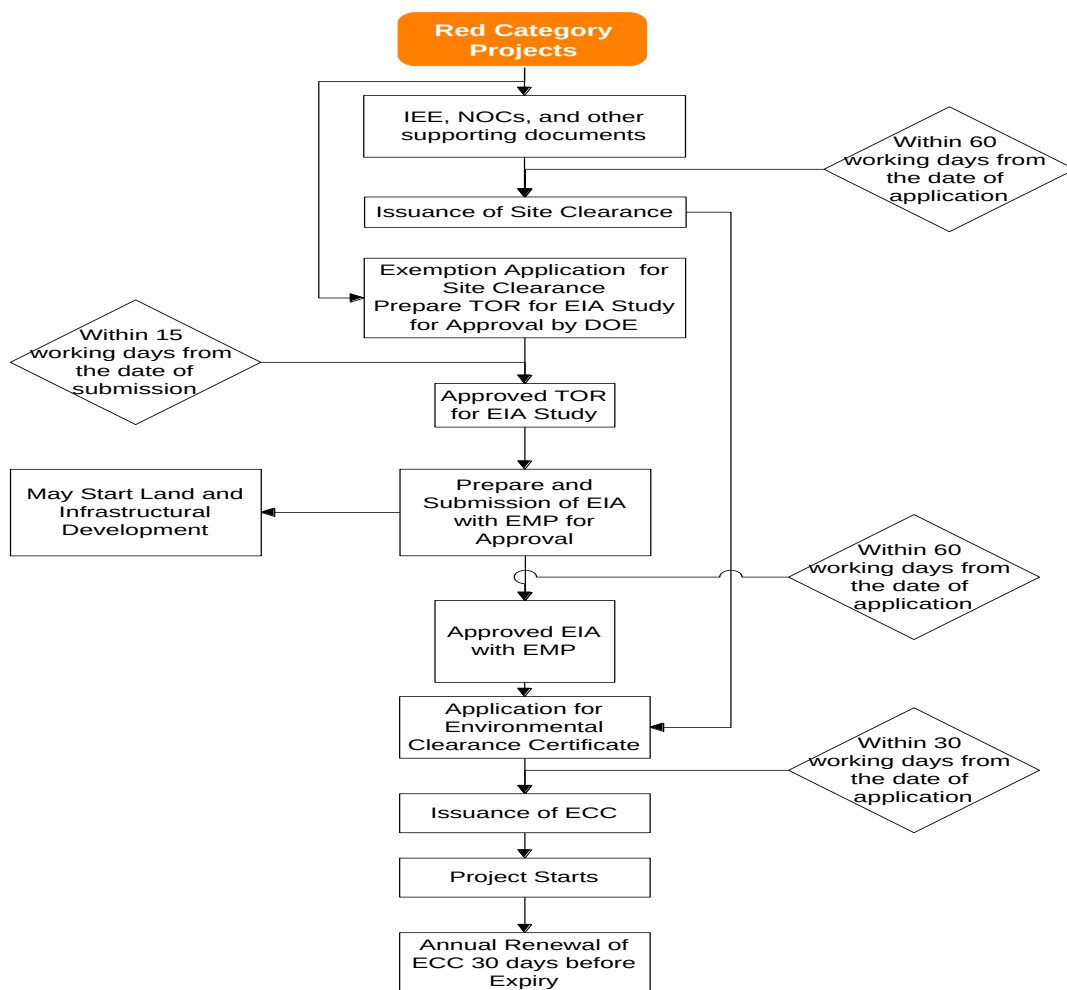
After having SCC, the sponsors can develop land & infrastructure, build super structure and install machinery. DoE will issue ECC to the sponsors after they have

installed the waste treatment device, if any. After having the ECC, the sponsors can install gas connection (if needed) and commence trial & commercial productions.

Red category projects : For SCC, approval of EIA & ECC for Red category projects, the following documents, in addition to those required for *Green Category projects*, are to be attached to the application (in Form-3) to the DoE:

- Initial Environmental Examination (IEE),
- Feasibility Report
- Information / report as specified in the checklist given in the IEE e.g, map, process flow diagram, layout plan showing waste discharge, treatment and disposal arrangement etc,
- design for Effluent Treatment Plant (ETP) and other applicable environmental pollution control devices with Environmental Management Plan (EMP), Pollution effect abatement & emergency plans.

Figure 5-2: DOE Clearance Process for Red Category Projects



After having SCC, the sponsors can develop land & infrastructure. The sponsor will next submit EIA and ETP design along with Environment Management Plan (EMP), Pollution effect abatement & emergency plans etc for approval. After having EIA

approved they will build super structure and procure & install machinery. DoE will issue ECC to the sponsors after they have installed the ETP as required. After having the ECC, the sponsors can install gas connection (if needed) and commence trial & commercial production. .

The Content of Initial Environmental Examination (IEE) Report

IEE is an important tool which should be carried out as early as the project planning stage as part of feasibility thus it can assure that the project will be environmentally feasible. The outline of IEE report is shown in **Annex 6-17**

b) The EIA Process

Guidelines set forth by DOE define EIA as a comprehensive process involving study of the anticipated impacts due to a proposed project upon the physical, biological and socio-economic environment of the project command area and formulation of a suitable Environment Management Plan (EMP) toward offsetting the adverse impacts and enhancing positive impacts as well as the overall environmental quality. The objectives of EIA, as stipulated in the above guidelines are as under:

- identifying sources of impacts that the proposed project could have upon the various components of the environment and recognizing the critical sources of such impacts;
- predicting the impacts using qualitative and, if possible, quantitative methods;
- recommending an Environment Management Plan (EMP) to reduce or mitigate adverse impacts and enhance positive contributions of the proposed project by exploring changes to be brought about in technology, use of raw materials, fuels, designs, operational practices, or even by evaluating alternative sites for the project;
- presenting results of impact identification, prediction and assessment with suggested mitigation measures;
- setting the basis of continuous monitoring of the key activities of the proposed project and that of the environmental conditions.

The EIA process needs to pass through the following three tiers in order to optimize the resources required for conducting EIA studies:

- Screening: to decide whether the EIA process needs to be applied to the specific case and, if so, the type (IEE or EIA);
- Initial Environmental Examination (IEE): to help delineate potential adverse impacts upon the environmental components and to find ways to mitigate the impacts or enhance the quality of those environmental components;
- EIA(Full-scale EIA): to carry out detailed examination of the impacts through conducting surveys and monitoring with the application of

rigorous impact prediction tools, where necessary, and ensuring effectiveness of the mitigation and enhancement measures through formulation of EMP.

Adequacy of mitigation measures and EMP needs to be ascertained through a public consultation process involving interaction among DOE, concerned other departments/authorities, NGO community and people likely to be affected by the proposed project. Views and recommendations emanating through such a public consultation process should form the basis of preparing further proposals for quality enhancement of the project and development of adequate and pragmatic environmental planning. To be more specific, the public disclosure and consultation should form the core of the EIA process.

The outline of an EIA report and structure of a sample EIA Report are annexed as **Annex 6-7** and **Annex 6-18** respectively.

Environment Management (and Monitoring) Plan (EMP)

Environment Management (and Monitoring) Plan (EMP) is a component of the EIA process. EMP could either be incorporated in the EIA report or prepared as a separate document and could form a basis of chalking out elaborate and time-bound program of activities to be undertaken by the entrepreneur during pre-construction and construction stages of the project. Aspects to be addressed in EMP need to be derived from the EIA findings and recommendations. A simple structure of EMP is given in **Annex 6-19**.

c) Environmental and Social Screening of the project for World Bank Clearance

World Bank requirements for Environmental clearance are given in the World Bank Environmental Safeguard Policies OP 4.01 and OP 4.04.

d) Criteria for Site Clearance of Projects/Industries

As stated in the EIA Guidelines for Industries, Department of Environment, June 1997, for establishing industrial plant, especially with significant potential for environmental impacts, only the EIA will indicate the suitability of a site. Selection of sites for orange and red category industries should be done keeping in view the following criteria:

1. The location of an industry should be in the area, which is designated or likely to be designated as an industrial zone under the town and country planning regulations.
2. The sites for industries of the Orange and Red categories should consider the following features:
 - Environmentally or other sensitive areas
 - High tide line in coastal areas, or natural or modified flood plain boundary
 - Boundary of the nearest human settlement
 - Highway/ Railway boundary

3. No “forest land” or prime agricultural land will be used as industrial sites.
4. The industrial premises should be adequately large to provide for;
 - (i) Storage of solid wastes;
 - (ii) Treatment of waste water;
 - (iii) Reuse of treated waste water, if feasible; and
 - (iv) Green belt of reasonable width around the site perimeter.
5. Industrial plants with high potential for pollution should not be located in or near;
 - (v) Catchments areas of public water supply reservoir;
 - (vi) Recharge area of aquifer of present or possible water supply use;
 - (vii) Areas known to suffer from frequent inversions;
 - (viii) A habit of endangered species;
 - (ix) Areas prone to flooding or earthquake.

Other factors, which need to be taken into consideration from the point of view of potential impacts include,

- (a) Assimilative capacity of receiving body of water;
- (b) Effect on availability of existing infrastructure facilities to the local population like water supply, housing, power;
- (c) Likely induced growth around the industrial site; and
- (d) Views of the local people likely to be affected.

e) **Land Acquisition Phase**

The land procurement for the project may happen in three different ways;

1. Land purchase by the sponsor through negotiation
2. Land acquisition by GoB
3. Land acquisition by GoB for availing World Bank finances

In case 1, no RAP will be required as the purchase is done from willing seller, by directly paying the cost of land and structures.

In case 2, the land will be acquired under the Acquisition and Requisition of Immovable Property Ordinance 1982 (Ordinance 11 of 1982).

In case 3, the land acquisition process must be in conformity with World Bank safeguard policies, as WB fund will be utilized for land acquisition or development of the project. However, if the land acquisition took place at a time reasonably long

before applying for WB finance, it may obtain waiver from the WB from WB's safeguard policies.

All these have been described in relevant chapters of this framework.

f) Financing Phase

Loan Application is received by BIFFL from the private sector sponsor, at the stage when the project is awarded by the public sector and agreement signed between them. At this stage, most of the studies may have been completed and the land acquired. The BIFFL project will review already completed EIA and SIA Reports. BIFFL project will assess if all issues have been adequately addressed and decide if further alternatives are to be examined.

g) Environmental and Social Due Diligence (ESDD)

The level of E&S due diligence will be based on the sub-project's E&S risk profile and potential impacts, for example, whether they are major (e.g., irreversible, unprecedented, involving large resettlement) and moderate (generally site-specific, largely reversible and mitigable). In most circumstances, the ESDD will be contracted to an external consultant or consultant team, for which BIFFL should prepare the Terms of References (ToR) to ensure that the necessary due diligences is conducted. The E&S due diligence will typically consist of:

- (i) review of all relevant documents and information provided by the sponsor and other sources; and
- (ii) site reconnaissance comprising visual observations of relevant areas and meetings and interviews (with relevant stakeholders sponsor personnel, governmental officials, affected communities, etc.).

Upon completion of the due diligence, the findings, conclusions, and recommendations will be presented in the ESDD report. The recommendations should include the necessary actions which must be implemented for the proposed investment to proceed to financial closure. At a minimum, these will consist of a set of mitigation, management, monitoring, and institutional measures to be taken during project implementation and operation to address any gaps with the Fund's Social and Environmental Policy. Any mitigating actions addressed in the ESDD report should clearly indicate the (achievable) level of environmental compliance with the Policy, the existing performance gaps, and the corrective actions that need to be taken to close these gaps along with reasonable timelines (collectively, the action plan), and should be thoroughly discussed and agreed with the project sponsor.

BIFFL should remain engaged in the due diligence process, and involve other investors or stakeholders in reviewing the due diligence report and determining the soundness of the action plan. Additional reviews or due diligence work may be triggered as a result of such stakeholder engagement. Finally, the results of the E&S due diligence should be presented to the Fund's investment committee in

concert with the results of the other due diligence conducted on the prospective investment. The ESDD report outline is described in **Annex 6-14**

E&S Conditions of Investment

BIFFL should negotiate and agree with sponsor regarding the E&S provisions and investment conditions in the term sheet and investment agreements. These typically consist of standard E&S terms applicable to all IPFF investments, and project-specific conditions identified during the E&S due diligence of a proposed investment. They are represented in legal documents as general E&S definitions, representations and warranties, disbursement conditions, and/or covenants regarding compliance with the requirements as defined by the Fund's Social and Environmental Policy. Critical mitigating actions that address gaps with the Fund's Social and Environmental Policy must be included in the investment agreement as conditions of investment. These actions may be represented in an action plan, as appropriate. Post-investment reporting requirements for the sponsor, particularly on the status of the implementation of the mitigating measures (or action plan, as appropriate) should be included as covenants in the investment agreement as well. The sponsor should be provided with a format for the Annual Environmental and Social Performance Monitoring Report (AMR) to be submitted to the Bangladesh Bank, annually.

h) Monitoring

BIFFL should actively supervise and monitor the E&S performances of each of its portfolio projects. As noted above, the investment agreements should require the sponsor to report periodically (at least once a year) on its E&S performance according to a pre-agreed AMR form. BIFFL should also require timely reports from the sponsor regarding the implementation of the critical mitigating actions and any significant E&S incidents. Based on this reporting, BIFFL should discuss with the sponsor any outstanding or potential E&S performance issues and if needed, agree to additional measures to address them. During construction and at project completion BIFFL should make supervision site visits to confirm that all related E&S requirements are being implemented. From time to time during the operation phase, BIFFL should also make supervision site visits and consider making independent E&S audits (involving external consultants if necessary) as part of its overall supervision and monitoring process.

A Supervision Report should be prepared after each of such visits or audits. In certain cases, a pre-divestment environmental and social audit may be needed to determine whether there are any potential issues requiring attention prior to disposal.

i) Environmental and Social Assessment Process

The ESMF will establish a unified process for addressing all the environmental and social safeguard issues. It will help to systematically identify, predict, and evaluate beneficial and adverse environmental and social impacts of development

activities, designing enhancement measures for beneficial impacts, and implement mitigation measures for the adverse impacts.

Table below describes the steps in environmental and social assessment process at different stages, against which the ESDD will be carried out for compliance.

Table 5-4: Steps in the Environmental and Social Assessment Process

	EIA and SIA Steps in Different Stages of Project Cycle
A	<i>Project Identification & Pre-Feasibility Studies (Pre-Construction) Phase</i>
	1. Environmental and Social screening 2. Social performance frameworks for projects not requiring specific resettlement and land acquisition studies
B	<i>Project Design Phase</i>
	3. Initial Environmental Examination (IEE): assess environmental impacts to determine if EIA is required; its recommendations are incorporated into the project design. 4. Scoping: identify significant potential impacts and project alternatives, and propose terms of reference for the EIA and SIA. 5. Baseline Data Collection, as part of SIA: identify current environmental and socio-economic conditions without the project and update it during the project in accordance with changes in the project. 6. Public consultation and participation of stakeholders of different level at various stages in the assessment process to ensure quality, comprehensiveness and effectiveness, and that stakeholders' views are adequately addressed. 7. Prepare EIA Report and SIA Report: summarizes all information obtained, analyzed and interpreted in a report form; should contain a non-technical summary including methods used, results, interpretations and conclusions. The report should also include recommendations for mitigation of negative impacts, enhanced opportunities and relevant policy and regulatory actions. The report should be shared with stakeholders participating in the consultation process and affected by the recommendations and time for feedback should be allowed. 8. Prepare Environmental and Social Management Plan (ESMP) of the project to determine specific actions to be taken during the designing of the project that includes plans for land acquisition, engineering design and construction stages to minimize or mitigate adverse environmental and social impacts. 9. Preparation of Resettlement Action Plan (RAP) and Indigenous People's Development Plan (IPDP) on the basis of SIA and update them in accordance with changes in the project requiring land acquisition. 10. Design mitigation measures: to avoid, reduce and minimize adverse environmental and social impacts and enhance beneficial impacts.
C	<i>Project Appraisal/ Approval (Financing) Phase</i>
	11. Review and Approval of EIA and SIA Report: review report to assess if all issues have been adequately addressed and to facilitate the decision-making process; decide if project should proceed, or if further alternatives must be examined.

EIA and SIA Steps in Different Stages of Project Cycle	
D	Construction Phase
	12. Implementation of Environmental Management Plan (EMP) and RAP to address adverse environmental and social impacts and update it in accordance with changes in the project 13. Environmental and Social Monitoring: determines compliance with ESMP Mid- term independent evaluation to assess the continued relevance of the mitigation plans and need for any alterations based on actual developments during the construction phase
E	Post-Construction Phase
	14. Environmental and social Audit: As per the recommendations of EIA and SIA study. 15. Regular monitoring arrangements to record and evaluate progress against initial plans and potential new challenges and opportunities.

j) **Construction Phase**

During the construction phase, BIFFL project will monitor the implementation of Environmental Management Plan (EMP) and Social Management Plan (SMP). This will include monitoring of RAP, Relocation, Livelihood restoration, Training activities, etc.

k) **Post-Construction Operation Phase**

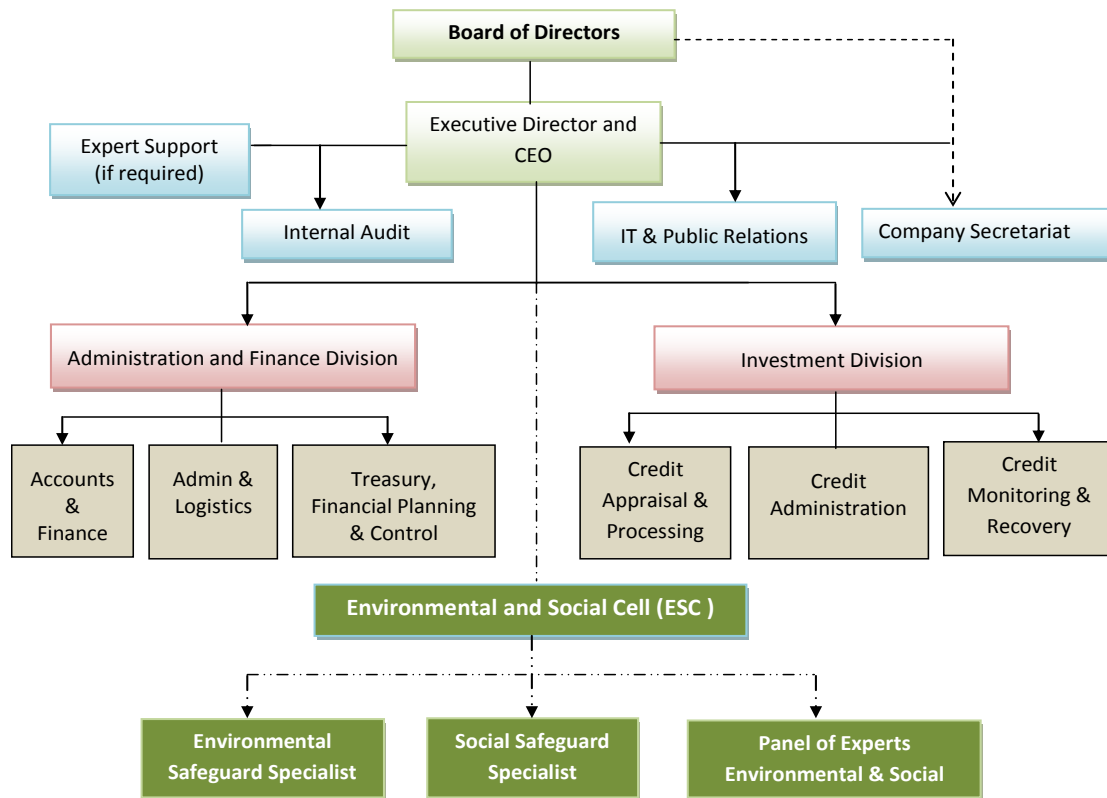
In the operation phase BIFFL project will not have much activity with regards to ESMF. However, an Environmental and Social Audit is recommended as per EIA and SIA.

5.3 Institutional Framework for ESMF Implementation

5.3.1 Environment and Social Cell (ESC) of BIFFL

To coordinate the environmental and social assessment processes and to ensure the effective implementation of the Environmental and Social Management Plan, the fund manager (the BIFFL project) will establish an Environment and Social Cell (BIFFL ESC). BIFFL may also hire a consulting company for the purpose, who will work on behalf of the BIFFL project. The proposed ESC is depicted in the figure below.

Figure 5-3: Proposed ESC within BIFFL Organization Structure



This ESC (or the hired company) should have an ‘E&S Manager’ from the fund manager’s senior management who will assume responsibility for overseeing all environmental and social matters, ensure resources are made available for the E&S management. The E&S Manager may be supported by an Environment Safeguard Specialist and Social Safeguard Specialist who should be technically qualified, to implement and supervision activities. The E&S Manager and his supporting staffs should ensure the investment teams and legal counsels are all trained on the E&S policy and procedures and can call for meetings to discuss any policy or project related issues.

The fund manager should build and maintain a roster of qualified E&S consultants who can be called upon to assist for E&S reviews. It is strongly recommended that E&S responsibilities and outcomes be included in fund employees’ performance measurement system.

Responsibilities of ESC

The environmental safeguards specialist shall deliver the following services:

- Participate in the meetings with client
- Advise project proponent on completing the Environmental section of the Loan Application Form
- Review of Loan Application Form to draw indicative term sheet
- Categorization of project
- Prepare PEAR

- (f) Issue Terms of Reference (ToR) for conduct of EIA/IEE/EDD/CAP
- (g) Review EIA/ IEE/ EDD/ CAP submitted for a project
- (h) Coordinate (including hiring consultants) conduct of monitoring of implementation of EMP/ CAP of select High risk projects
- (i) Review half-yearly monitoring reports of the compliance and effectiveness of the EMP/ CAP
- (j) Report progress of projects annually
- (k) Submit EIA of High risk projects for approval
- (l) Forward EIA/ IEE/ EDD/ CAP/ monitoring reports for disclosure and information

The social safeguards specialist will deliver the following services:

- (a) Participate in the meetings with client
- (b) Advise project proponent on completing the social section of Loan Application Form
- (c) Conduct review of Loan Application Form to draw indicative term sheet
- (d) Categorization of project
- (e) Issue ToR for conduct of SIA and RP/IPP
- (f) Review SIA and RP/ADP submitted for a project
- (g) Coordinate (including hiring consultants) conduct of monitoring of implementation of RP/IPP of select high risk projects
- (h) Review half-yearly monitoring reports of the compliance and effectiveness of the RP/IPP
- (i) Report on the progress of projects annually to ADB
- (j) Forward SIA and RP/ADP monitoring reports to ADB for disclosure
- (k) Submit RP/ADP of High risk projects to ADB for approval
- (l) Ensure financial commitment by the project proponent for RP/ADP implementation prior to loan disbursement
- (m) Review monitoring report submitted by PP/IA

Allocation of Responsibilities

Project proponent shall be responsible for regular monitoring and periodic evaluation of safeguard compliance of the project with agreed milestones and performance levels particularly for purposes of disbursement of BIFFL funds. It shall send periodic safeguard progress reports in such form and, at such times, as may be prescribed by ESMF. The project proponent will therefore be responsible to report E&S compliance of the project to ESC in the formats as prescribed by the ESMF. **Table 5-5** summarizes allocation of responsibilities for implementation of E&S safeguards of BIFFL.

Table 5-5: Allocation of Responsibilities

Sl. No	Tasks	Primary Responsibility	Secondary Responsibility
1.	Submit all documentation related to environmental clearances, NoCs etc. and EIA/EMP reports	Project Proponent	Review and Check by ESC
2.	Submit Involuntary Resettlement and Indigenous Peoples (tribal) Screening Checklist along with project concept note	Project Proponent	Review and Check by ESC
3.	Check each project for completion of application and list of IFC/MIGA exclusion list of prohibited activities by ADB	ESC/BIFFL	-
4.	Categorization and Issue ToR for conduct of EIA/IEE/ SIA & Resettlement Plan	ESC/BIFFL	-
5.	Identify and submit all relevant environmental information of high risk projects	ESC/BIFFL	-
6.	Check environmental compliance	ESC/BIFFL	Project Proponent for providing additional information as required
7.	Check adequacy and effectiveness of EMP	ESC/BIFFL	Project Proponent for providing additional information as required
8.	Conduct EIA/ IEE/ EDD/ SIA	Project Proponent	ESC/BIFFL – support on ToR preparation and identifying agency
9.	Prepare and Implement EMP/ RP	Project Proponent	ESC/BIFFL – support on ToR preparation and identifying agency
10.	Implement CAP	Project Proponent	-
11.	Review EIA/ IEE/ EMP/ CAP/ RP prepared by Project Proponent	ESC/BIFFL	-
12.	Approval for Moderate and Low risk projects	ESC/BIFFL	-
13.	Submit EIA/RP of High risk projects to ADB for approval	ESC/BIFFL	-
14.	Ensure financial commitment of the	ESC/BIFFL	Project Proponent

Sl. No	Tasks	Primary Responsibility	Secondary Responsibility
	Project Proponent to implement RP /EMP		
15.	Conduct monitoring of implementation of EMP/ RP	Project Proponent	ESC/BIFFL
16.	Prepare and submit to ESC half-yearly/annual E&S monitoring report	Project Proponent	-
17.	Review half-yearly/annual E&S monitoring report	ESC/BIFFL	-
18.	Meet environmental compliance	Project Proponent	-
19.	Attend to public grievances	Project Proponent	-
20.	Conduct E&S monitoring and evaluation (select High risk projects)	ESC/BIFFL	Project Proponent
21.	Conduct annual E&S audit and disclose the annual E&S performance report and submit the final report to ADB	ESC/BIFFL	-
22.	Submit EIA/ IEE/ EDD/ CAP/ RP for disclosure	ESC/BIFFL	Project Proponent
23.	Conduct awareness and training of project proponents and other stakeholders	ESC/BIFFL	Project Proponent
24.	Update ESMF based on FI experience	ESC/BIFFL	-

5.3.2 The Project Management Unit (PMU) in Sub-project

Each sub-project sponsor will establish a Project Management Unit (PMU). Implementation of the ESMF will be the responsibility of this PMU and its monitoring would be the responsibility of the ESC of BIFFL.

Key responsibilities of PMU will include the following:

- To monitor and prepare reports on physical and financial progress of implementation
- To maintain financial accounts
- To conduct technical, financial and procurement audit.
- To ensure implementation of Environmental and Social Management Plan (ESMP) as set out in the Environmental Assessment (EA).

- To expedite the Land Acquisition process and provide compensation to the Project Affected Persons (PAPs) and implementation of Resettlement Action Plan (RAP)
- To liaise with the ministries / planning commissions and other institutions / agencies / ministries etc., if required.
- To make timely and efficient disbursements to all the parties working on the project.

5.3.3 Project Environmental and Social Cell (PESC) under sub-project

Each PMU will also establish a Project Environmental and Social Cell (PESC)

The Functions of PESC will be:

- Ensure implementation of the ESMP throughout the project implementation period,
- Ensure integration of the ESA & resulting ESMP into the sub-project design and implementation plans (contract documents)
- Monitor implementation of the mitigation measures by the Contractors,
- Assist the Engineers at site by providing appropriate environmental and social advice, and developing appropriate environmental and social mitigation measures for the sub-projects
- Assist PMU to carry out participatory consultation during planning, design and implementation of the sub-projects
- Prepare regular periodic progress reports on the implementation of the ESMP throughout the project implementation period.

5.3.4 Resettlement Unit (RU) under PESC

Every Project Environment and Social Cell (PESC) will set up a Resettlement Unit (RU) to organize, manage and coordinate works related to preparation and implementation of the Resettlement Plan. It should be mentioned that this RU will be formed only for implementation of the resettlement plan. The RU will be responsible for coordination, management and day to day monitoring of the resettlement work and all works related to RAP and to determine entitlement package for each affected person.

5.3.5 Grievance Redress Mechanism

The borrower/client will establish a mechanism to receive and facilitate the resolution of affected person's concerns and grievances about physical and economic displacement and other project impacts, paying particular attention to the impacts on vulnerable groups. The grievance redress mechanism should be scaled to the risks and adverse impacts of the project. It should address affected person's concerns and complaints promptly, using an understandable and transparent process that is gender responsive, culturally appropriate, and readily accessible to the affected persons at no costs and without retribution. The mechanism should not impede access to the country's judicial or administrative remedies. The borrower/client will inform affected persons about the mechanism.

5.3.6 Monitoring and Reporting

The borrower/client will monitor and measure the progress of implementation of the resettlement plan. The extent of monitoring activities will be commensurate with the project's risks and impacts. In addition to recording the progress in compensation payment and other resettlement activities, the borrower/client will prepare monitoring reports to ensure that the implementation of the resettlement plan has produced the desired outcomes. For projects with significant involuntary resettlement impacts, the borrower/client will retain qualified and experienced external experts or qualified NGOs to verify the borrower's/client's monitoring information. The external experts engaged by the borrower/client will advise on safeguard compliance issues, and if any significant involuntary resettlement issues are identified, a corrective action plan will be prepared to address such issues. Until such planning documents are formulated, disclosed and approved, the borrower/client will not proceed with implementing the specific project components for which involuntary resettlement impacts are identified.

The borrower/client will prepare semiannual monitoring reports that describe the progress of the implementation of resettlement activities and any compliance issues and corrective actions. These reports will closely follow the involuntary resettlement monitoring indicators agreed at the time of resettlement plan approval. The costs of internal and external resettlement monitoring requirements will be included in the project budget.

5.4 ESMF Implementation Budget

In view of the environmental and social monitoring measures suggested above for implementation, necessary budgetary provisions should be made for different components of individual projects. Tentative budget for each of the projects should include the environmental monitoring costs other than the good engineering practices, cost of environmental and resettlement monitoring. It is assumed that each of the sub-projects involving civil works will involve environmental and resettlement costs. These costs do not involve administration costs of environment and social staff at PMU.

A sample table for details of costs is provided in table below.

Table 5-6: ESMF Implementation Budget (sample)

Project Name: _____

Components, with Sub-Component	Environmental Monitoring	R&R Monitoring	Environmental Monitoring	RAP Budget	Training/Capacity Building Budget	Total Cost

5.5 ESMF Update

The ESMF will be monitored regularly and reviewed each year after the results of annual environmental and social audit. Based on the experience gained and considering further changes, if any, in GOB's environmental policy and regulatory framework or any environmental and social safeguard policy of ADB and the World Bank the ESMF document will be appropriately updated.

5.6 Monitoring of ESMF

The objectives of ESMF monitoring are

- To provide timely information about the success or otherwise of the Environmental and Social Monitoring process outlined in the ESMF in such a manner that changes can be made as required to ensure continuous improvement to the process; and
- To evaluate the performance of the ESMF by determining whether the mitigation measures designed in the project have been successful in such a way that the pre-program environmental condition has been restored, improved upon or worse than before and to determine what further mitigation measures may be required.

It is suggested that environmental monitoring studies should be conducted separately because different activities during construction and operation period of the Project will result in different possible environmental impacts. Monitoring plan of the project is presented in construction and operation phases, respectively.

5.6.1 Construction Phase Monitoring

Monitoring every environmental and social parameter continuously is not practical and necessary. Environmental and social monitoring functions according to the characteristics of existing environmental and social conditions and identifies unacceptable changes during construction period. In this context, monitoring plan is designed to commit that "project activities have no permanent negative effect on environmental resources".

For construction period, different components of Environment and social issues relevant to the project will be monitored and the results will be compared to legal limiting values.

5.6.2 Operational Phase Monitoring

In the operational phase of the project, fewer environmental and social impacts are expected than construction period. In addition, it is also recommended to carry out social survey prior to the operational phases in order to determine the needs of the local society and to develop social responsibility projects. Furthermore, in the operational phase of the project, there should be a routine social monitoring to follow up the development of the social responsibility projects and to record and resolve any complaint about the operations of project.

5.6.3 Monitoring Plan

The table below gives the general format of planning and implementation for a monitoring plan.

Table 5-7: Monitoring Plan of ESMF

Phase	What is to be monitored	Where is the parameter be monitored	How is the parameter to be monitored	When is the parameter to be monitored frequency of measurement	Monitoring cost	Responsibility
Construction						
Operation						

5.7 ESMF Auditing Framework

Audit is the process to review the effectiveness of environmental and social monitoring. Its objectives are to:

- Determine whether a project complies with all regulatory and environmental and social performance standards or other monitoring requirements for their operations.
- Ensure conformity with environmental and social assessment requirements, and test the accuracy of the assessment. It entails a systematic, documented and periodic review of project implementation. Therefore, its results could be very useful to improve project management performance.

The principal focus of the audit will be to assess the extent to which the ESMP has led to the implementation of the environmental codes of practices, appropriate mitigation measures in accordance with the ESMPs/ Specifications in the sub-projects.

5.8 Periodic Reporting on ESMF

The ESMF is not a static document. It may be improved with passage of time, based on experience gathered in execution of different sub-projects.

The Fund Manager should require its environmental manager/ environmental coordinator along with the investment transaction leader/officer to prepare periodic reports on ESMF.

The Fund Manager should also provide proper and periodic ESMF reports to its investors and other stakeholders. These internal and external reports can also form the basis of the reports, which the fund manager is required to prepare annually and periodically to the competent authority. Internally and in addition to the annual external investor/stakeholder report, the fund manager should use all these opportunities to review and adjust its E&S policies and procedures and identify areas for improving their implementation.

6 ANNEXES

ANNEX 6-1: ECOLOGICALLY CRITICAL AREAS DECLARED BY DOE.....	57
ANNEX 6-2: STANDARDS SET BY DOE FOR EFFLUENT FROM INDUSTRIAL UNITS OR PROJECTS WASTE	59
ANNEX 6-3: STANDARDS SET BY DOE FOR GASEOUS EMISSION FROM INDUSTRIES OR PROJECTS	61
ANNEX 6-4: STANDARDS SET BY DOE FOR SOUND	62
ANNEX 6-5: STANDARDS SET BY DOE FOR ODOR	63
ANNEX 6-6: WORLD BANK SAFEGUARDS POLICIES (ENVIRONMENTAL)	64
ANNEX 6-7: OUTLINE OF AN ENVIRONMENTAL IMPACT ASSESSMENT REPORT.....	68
ANNEX 6-8: OUTLINE OF A RESETTLEMENT PLAN	71
ANNEX 6-9: IFC/MIGA EXCLUSION LIST OF PROJECTS.....	75
ANNEX 6-10: ADB PROHIBITED INVESTMENT ACTIVITIES LIST.....	77
ANNEX 6-11: SITE VISIT CHECKLIST – ENVIRONMENTAL SAFEGUARDS	78
ANNEX 6-12: FORMAT OF PRELIMINARY ENVIRONMENTAL APPRAISAL REPORT (PEAR).....	80
ANNEX 6-13: CONTENTS OF ANNUAL ENVIRONMENTAL MONITORING REPORT	81
ANNEX 6-14: ENVIRONMENTAL & SOCIAL DUE DILIGENCE (EDD) REPORT OUTLINE.....	83
ANNEX 6-15: FORMAT FOR CORRECTIVE ACTION PLAN	86
ANNEX 6-16: DOE PRESCRIBED PRO-FORMA FOR APPLYING FOR ECC.....	87
ANNEX 6-17: OUTLINE OF AN IEE REPORT	89
ANNEX 6-18: STRUCTURE FOR A SAMPLE EIA REPORT	92
ANNEX 6-19: Structure of EMP	102
ANNEX 6-20: EFFLUENT DISCHARGE REQUIREMENTS BY WORLD BANK.....	103
ANNEX 6-21: EMISSION DISCHARGE REQUIREMENTS BY WORLD BANK.....	105
ANNEX 6-22: NOISE LEVEL REQUIREMENTS BY WORLD BANK	106
ANNEX 6-23: WORLD BANK SAFEGUARDS POLICIES (SOCIAL)	107
ANNEX 6-24: RESETTLEMENT ACTION PLAN (RAP) PREPARATION	113
ANNEX 6-25: WORLD BANK ENVIRONMENTAL CLEARANCE PROCESS.....	120
ANNEX 6-26: ENVIRONMENTAL IMPACT AND MITIGATION MEASURES	125
ANNEX 6-27: GOB SOCIAL MANAGEMENT PROCESS	134
ANNEX 6-28: WB SOCIAL MANAGEMENT PROCESS	138
ANNEX 6-29: ADB SAFEGUARD POLICY STATEMENTS	153
ANNEX 6-30: GENERIC OUTLINE FOR AN INDIGENOUS PEOPLES DEVELOPMENT PLAN	160
ANNEX 6-31: GENERAL CHECK LIST TO EVALUATE IEE/ ESIA REPORTS.....	164
ANNEX 6-32: SECTOR SPECIFIC CHECKLISTS TO EVALUATE ESIA REPORTS.....	172
ANNEX 6-33: Involuntary Resettlement Screening and Categorization.....	194
ANNEX 6-34: ENVIRONMENTAL AND SOCIAL COVENANTS AND REQUIREMENTS FOR CONTRACTORS	196

ANNEX 6-1: ECOLOGICALLY CRITICAL AREAS DECLARED BY DOE

Ecologically Critical Areas (ECAs)	Total Area, (Ha)	Name of District	Name of Upazila	Name of Union/ Pourashava	Name of Mouza
Sundarban	7,62,034	Bagerhat, Khulna and Satkhira	Upazilas within 10 km peripheral distance of the identified reserve forest area.	Unions/Pourashavas under the Upazilas listed in the previous column.	Mouzas under the Unions/Pourashavas listed in the previous column.
Cox's Bazar-Teknaf sea beach	10,465	Cox's Bazar	Cox's Bazar	Cox's Bazar, Zilonja, and Khuruskul	Sea beach, sand rim, estuary, forest, wetland, etc. as recorded in the register of Land Revenue Dept., Cox'sBazar.
			Ramu	Khunia Palong	Jungle Khunia Palong, Jungle Dhoa Palong, Pecher Deep and Jungle Gorasia Palong.
			Ukhia	Ukhia and Zalira Palong	Zalira Palong and Inani
			Teknaf	Teknaf, Bahar Chara and Sabrang	Teknaf (excluding Bazar and Border Check post), Silkhali, Sabrang, Shah Porar Deep (excluding border check post) and Bordayle
Saint Martin Deep (island)	590	Cox's Bazar	Teknaf	Saint Martin Deep (island)	Narikel Jinjira
Sonadia Deep (island)	4,916	Cox's Bazar	Moheshkhali	Kutubjum	Sonadia, Ghoti Bhanga (part)
Hakaluki Haor	18,383	Moulvi Bazar and Sylhet	Borolekha, Kulaura, Fenchuganj and Golapganj	Sujanagar, Barni, Talikpur, Poschimjuri, Jafarnagar, Boromchol, Boksimali, Vatera, Gilachhara, Uttar Bade Pasa and Sharifganj	Water bodies, recorded as Beel in the register of Land Revenue Department, located in all Mouzas/part of Mouzas under the jurisdiction of Unions listed in the previous column.

Ecologically Critical Areas (ECAs)	Total Area, (Ha)	Name of District	Name of Upazila	Name of Union/ Pourashava	Name of Mouza
Tanguar Haor	9,727	Sunamganj	Taherpur and Dharmopasa	Uttar Srepur, Dokkhin Srepur, Uttar Bonsikundi and Dokkhin Bonshikundi	Water bodies, recorded as Beel in the register of Land Revenue Department, located in all Mouzas/part of Mouzas under the jurisdiction of Unions listed in the previous column.
Marjat Baor	200	Jhenaidah	Kaliganj	Water bodies, recorded as Beel in the register of Land Revenue Department, located in all Mouzas/part of Mouzas under the jurisdiction of Unions listed in the previous column.	Water bodies, recorded as Beel in the register of Land Revenue Department, located in all Mouzas/part of Mouzas under the jurisdiction of Unions listed in the previous column.

ANNEX 6-2: STANDARDS SET BY DOE FOR EFFLUENT FROM INDUSTRIAL UNITS OR PROJECTS WASTE

Sl. No	Parameter	Unit	Inland surface water	Public sewerage system connected to treatment at second stage	Irrigated land
1	Ammonical nitrogen as elementary N	mg/L	50	75	75
2	Ammonia as free ammonia	"	5	5	15
3	Arsenic	"	0.2	0.05	0.2
4	BOD at 20°C	"	50	250	100
5	Boron	"	2	2	2
6	Cadmium	"	0.5	0.05	0.05
7	Chloride	"	600	600	600
8	Chromium as total Cr	"	0.5	1	1
9	COD	"	200	400	400
10	Chromium as hexavalent Cr	"	0.1	1	1
11	Copper as Cu	"	0.5	3	3
12	Dissolved oxygen (DO)	"	4.5-8.0	4.5-8.0	4.5-8.0
13	Electrical conductivity (EC)	micro mho/cm	1,200	1,200	1,200
14	Total dissolved solids	mg/L	2,100	2,100	2,100
15	Fluoride as F	"	2	15	10
16	Sulfide as S	"	1	2	2
17	Iron as Fe	"	2	2	2
18	Total nitrogen as N	"	100	100	100
19	Lead as Pb	"	0.1	1	0.1
20	Manganese as Mn	"	5	5	5
21	Mercury as Hg	"	0.01	0.01	0.01
22	Nickel as Ni	"	1	2	1
23	Nitrate as elementary N	"	10	Not yet fixed	10
24	Oil and grease	"	10	20	10

Sl. No	Parameter	Unit	Inland surface water	Public sewerage system connected to treatment at second stage	Irrigated land
25	Phenolic compounds as C ₆ H ₅ OH	"	1	5	1
26	Dissolved phosphorus as P	"	8	8	10
27	Radioactive substance	"	To be specified by Bangladesh Atomic Energy Commission		
28	p ^H		6-9	6-9	
29	Selenium as Se	mg/L	0.05	0.05	0.05
30	Zinc as Zn	"	5	10	10
31	Total dissolved solids	"	2,100	2,100	2,100
32	Temperature	°C			
	Summer		40	40	40
	Winter		45	45	45
33	Total Suspended solids (TSS)	mg/L	150	500	200
34	Cyanide as CN	"	0.1	2	0.2

Note 1: The above standards are applicable to all industries or projects other than those specified for sector-wise standards.

Note 2: Compliance with these standards to be ensured from the moment an industrial unit starts trial production, and in other cases, from the moment a project starts operation.

Note3: Inland surface water means drains/ ponds/tanks/water bodies/ ditches/canals/rivers/ springs and estuaries.

Note 4: Public sewerage system means treatment facilities of the first and second stage and also combined and complete treatment facilities.

Note 5: Irrigated land means land area, which is sufficiently irrigated by waste water for cultivation of selected crops.

ANNEX 6-3: STANDARDS SET BY DOE FOR GASEOUS EMISSION FROM INDUSTRIES OR PROJECTS

SL No.	Parameter	Standard present in a unit of mg/Nm ³
1.	Particulate	
	a) Power plant with capacity of 200 Megawatt or above	150
	b) Power plant with capacity less than 200 Megawatt	350
2.	Chlorine	150
3.	Hydrochloric acid vapour and mist	350
4.	Total fluoride F	25
5.	Sulphuric acid mist	50
6.	Lead particulate	10
7.	Mercury particulate	0.2
8.	Sulphur dioxide	
	a) Sulphuric acid production (double conversion, double absorption process)	4 kg/ton
	b) Sulphuric acid production (single conversion, single absorption process)	10 kg/ton
	Lowest height of stack for dispersion of sulphuric acid (in meter)	
	a) Coal based power plant	
	1) 500 Megawatt or above	275 kg/ton
	2) 200 to 500 Megawatt	220 kg/ton
	3) Less than 200 Megawatt	14 (Q) ^{0.3} kg/hour
	b) Boiler	
	1) Steam per hour up to 15 tons	11 kg/ton
	2) Steam per hour more than 15 tons	14 (Q) ^{0.3} kg/hour
	Note : Q = Emission of sulphur dioxide , kg/hour	
9	Oxides of nitrogen	
	a) Nitric acid production	3 kg/ton
	b) Gas fuel based power plant	
	1) 500 Megawatt or above	50 ppm
	2) 200 to 500 Megawatt	40 ppm
	3) Below 200 Megawatt	30 ppm
	c) Metallurgical oven	200 ppm
10	Soot and dust particles	
	a) Air ventilated furnace	500 mg/Nm ³
	b) Brick-field	1000 mg/Nm ³
	c) Coke oven	500 mg/Nm ³
	d) Lime kiln	250 mg/Nm ³

ANNEX 6-4: STANDARDS SET BY DOE FOR SOUND

SL. No.	Category of areas	Standards determined at dBa unit	
		Day	Night
1	Silent zone	45	35
2	Residential area	50	40
3	Mixed area (mainly residential area, and also simultaneously used for commercial and industrial purposes)	60	50
4	Commercial area	70	60
5	Industrial area	75	70

Note 1: The time from 6 a.m to 9 p.m is counted as day time.

Note 2: The time from 9 p.m to 6 a.m is counted as night time.

Note 3: Area up to a radius of 100 meters around hospitals or educational institutions or special institutions/establishments identified/to be identified by the Government is designated as Silent Zones where use of horns of vehicles or other audio signals, and loud speakers are prohibited.

ANNEX 6-5: STANDARDS SET BY DOE FOR ODOR

Parameter	Standards Limit (ppm)
Acetaldehyde	0.5-5
Ammonia	1-5
Hydrazine sulphide	0.02-0.2
Methyl disulphide	0.009-0.1
Methyl mercaptan	0.02-0.2
Methyl sulphide	0.01-0.2
Styrene	0.4-2.0
Tri methylamine	0.005-0.07

Note 1: Following regulatory limit will be generally applicable to emission/exhaust outlet pipe of above 5 meter height:
 $Q = 0.108 \times H_e^2 C_m$ (Where Q = Gas Emission rate Nm^3 /hour)
 Here,
 Q = gas exhaust rate in Nm^3 /hour
 H_e = height of exhaust pipe in meter (m)
 C_m = standard limit mentioned above (ppm)

Note 2: In case where limits of quality standard have been specified, the lower limit for purpose of warning and the higher for legal proceedings or penal action will be used.

ANNEX 6-6: WORLD BANK SAFEGUARDS POLICIES (ENVIRONMENTAL)

World Bank strict adherence of its policies by all the parties related to infrastructure projects. This section will describe World Bank environmental safeguard policies, project categories and comparison of WB and GoB policies.

Table 6-1: World Bank Safeguard Policies

Policy	Summary of core requirements	Applicability to lending instruments	Public Consultation	Explicit provision for exceptions
OP/BP 4.01, Environmental Assessment	Screen early for potential impacts and select appropriate instrument to assess, minimize, and mitigate potentially adverse impacts.	All investment projects. Applicable to sector adjustment loans.	Consult affected groups and NGOs as early as possible (for Category A and B projects).	Yes. For emergency recovery operations with approval of the Regional VP, ENV Chair, and LEG.
OP/BP 4.04, Natural Habitats	Do not finance projects that degrade or convert critical habitats. Support projects that affect noncritical habitats only if no alternatives are available and if acceptable mitigation measures are in place.	All investment projects.	Consult local people in planning, designing, and monitoring projects.	NO
OP/4.09, Pest Management	Support integrated approaches to pest management. Identify pesticides that may be financed under the project and develop appropriate pest management plan to address risks.	All investment projects and adjustment operations.	Consult local people in planning, designing, and monitoring projects	NO
OP 4.36, Forestry	Support sustainable and conservation-oriented forestry. Do not support commercial logging in primary moist tropical forests.	All investment projects.	Consult local people, the private sector, and interest groups in forest area	NO
OP/BP 4.37, Safety of Dams	For large dams, carry out technical review and periodic safety inspections by independent dam safety	All investment projects.	No public consultation	NO

Policy	Summary of core requirements	Applicability to lending instruments	Public Consultation	Explicit provision for exceptions
	professionals.			
OP/BP 7.50, Projects in International Waterways	Ascertain whether riparian agreements are in place, and ensure that riparian states are informed of and do not object to project interventions.	All investment projects.	No Public consultations. Riparian notification required	NO
OP/BP 7.60, Projects in Disputed Areas	Ensure that claimants to disputed areas have no objection to proposed project.	All investment projects.	No Public consultations. Claimants informed	NO

1. Categories of Projects According to the World Bank

Based on project type and scale, project site, sensitivity of issues, nature of impacts and magnitude of impacts, World Bank has classified all projects into three categories, viz: Category A, Category B and Category C. These are described in the following sections.

a) Category A:

A project is classified as Category A if it is likely to have significant adverse environmental impacts that are sensitive, diverse, or unprecedented. These impacts may affect an area broader than the sites or facilities subject to physical works. EA for a Category A project examines the project's potential negative and positive environmental impacts, compares them with those of feasible alternatives (including the "without project" situation), and recommends any measures needed to prevent, minimize, mitigate, or compensate for adverse impacts and improve environmental performance. For a Category A project, the borrower is responsible for preparing a report, normally an EIA (or a suitably comprehensive regional or sectoral EA)

b) Category B:

A project is classified as Category B if its potential adverse environmental impacts on human population or environmentally important areas--including wetlands, forests, grasslands, and other natural habitats--are less adverse than those of Category A projects. These impacts are site-specific; few if any of them are irreversible; and in most cases mitigating measures can be designed more readily than for Category A projects. The scope of EA for a Category B project may vary from project to project, but it is narrower than that of Category A EA. Like Category A EA, it examines the project's potential negative and positive environmental impacts and recommends any measures needed to prevent,

minimize, mitigate, or compensate for adverse impacts and improve environmental performance.

c) Category C

A project is classified as Category C if it is likely to have minimal or no adverse environmental impacts. Beyond screening, no further EA action is required for a Category C project.

Majority of projects requiring BIFFL finance are expected to fall under environmental category 'A'.

2. COMPARISON BETWEEN GOB AND WB GUIDELINES

After reviewing the laws of GoB and World Bank guidelines, it is necessary to identify the similarities and differences between those so that the more stringent requirements can be applied for the sub projects. In general OP 4.01 requirements are more comprehensive when compared with the requirements of Bangladesh legal system. Following are some key comparisons between GoB and World Bank guidelines.

Table 6-2: Comparison between GOB and World Bank Guidelines

SL. No.	Criteria	Requirements as per GoB law	Requirements as per OP 4.01
	Type of Environmental Analysis	Project specific	Project specific, regional and sectoral
1.	Basis for Categorization	Currently, screening criteria available only for industrial projects, where assessment is done based on: • Level of pollution emission • Type of project and location • Scale of project • Operational activities Non-industrial projects are reviewed on a case by case basis by DoE and addressed in ECR, 1997	Detailed screening criteria for all projects based on • Sensitivity • Nature and magnitude of potential impacts
2.	EA Outputs	Since detailed rules and regulations for EA have not been prescribed, EA outputs are not specified. However, the industrial sector guidelines, the water sector guidelines and the road sector guidelines have specific EA output requirements, such as: • Baseline survey • IEE/EIA Report	• EA Report • Analysis of alternatives • Environmental Management Plan

SL. No.	Criteria	Requirements as per GoB law	Requirements as per OP 4.01
		• Site clearance • Risk analysis and management • Analysis of alternatives	
3.	Public Consultation	No special mention is made for public consultation in BECA. Sectoral guidelines mentioned above have prescribed consultation.	Mandatory at the stage of • Preparation of EA • Project appraisal and • Project design • Project implementation and monitoring
4.	Disclosure of Information	BECA makes no reference to disclosure. The Sectoral guidelines prescribe some provisions for disclosure	Mandatory at • Summary of project description and potential adverse impact • Summary of EA report and conclusion EA report

3. WB Standards of Environmental Aspects:

In any WB funded projects, the WB standards are mandatory for the project company to comply with. Following are the WB requirements on different environmental aspects:

- I. Effluent Discharge Requirements by the World Bank (ref: Annex 6-20)
- II. Emission Discharge Requirement by the World Bank (ref: Annex 6-21)
- III. Noise level Requirement by the World Bank (ref: Annex 6-22)

ANNEX 6-7: OUTLINE OF AN ENVIRONMENTAL IMPACT ASSESSMENT REPORT

This outline is part of the Safeguard Requirements. An environmental assessment report is required for all environment category Red and Orange B projects. Its level of detail and comprehensiveness is commensurate with the significance of potential environmental impacts and risks. A typical EIA report contains the following major elements, and an IEE may have a narrower scope depending on the nature of the project. The substantive aspects of this outline will guide the preparation of environmental impact assessment reports, although not necessarily in the order shown.

1. Executive Summary

This section describes concisely the critical facts, significant findings, and recommended actions.

2. Policy, Legal, and Administrative Framework

This section discusses the national and local legal and institutional framework within which the environmental assessment is carried out. It also identifies project-relevant international environmental agreements to which the country is a party.

3. Description of the Project

This section describes the proposed project; its major components; and its geographic, ecological, social, and temporal context, including any associated facility required by and for the project (for example, access roads, power plants, water supply, quarries and borrow pits, and spoil disposal). It normally includes drawings and maps showing the project's layout and components, the project site, and the project's area of influence.

4. Description of the Environment (Baseline Data)

This section describes relevant physical, biological, and socioeconomic conditions within the study area. It also looks at current and proposed development activities within the project's area of influence, including those not directly connected to the project. It indicates the accuracy, reliability, and sources of the data.

5. Anticipated Environmental Impacts and Mitigation Measures

This section predicts and assesses the project's likely positive and negative direct and indirect impacts to physical, biological, socioeconomic (including occupational health and safety, community health and safety, vulnerable groups and gender issues, and impacts on livelihoods through environmental media, and physical cultural resources in the project's area of influence, in quantitative terms to the extent possible; identifies mitigation measures and any residual negative impacts that cannot be mitigated; explores opportunities for enhancement; identifies and estimates the extent and quality of available data, key data gaps, and uncertainties associated with predictions and specifies topics that do not require further attention; and examines global, trans-boundary, and cumulative impacts as appropriate.

6. Analysis of Alternatives

This section examines alternatives to the proposed project site, technology, design, and operation—including the no project alternative—in terms of their potential environmental impacts; the feasibility of mitigating these impacts; their capital and recurrent costs; their suitability under local conditions; and their institutional, training, and monitoring requirements. It also states the basis for selecting the particular project design proposed and, justifies recommended emission levels and approaches to pollution prevention and abatement.

7. Information Disclosure, Consultation, and Participation

This section:

- i. describes the process undertaken during project design and preparation for engaging stakeholders, including information disclosure and consultation with affected people and other stakeholders;
- ii. summarizes comments and concerns received from affected people and other stakeholders and how these comments have been addressed in project design and mitigation measures, with special attention paid to the needs and concerns of vulnerable groups, including women, the poor, and Indigenous Peoples; and
- iii. describes the planned information disclosure measures (including the type of information to be disseminated and the method of dissemination) and the process for carrying out consultation with affected people and facilitating their participation during project implementation.

8. Grievance Redress Mechanism

This section describes the grievance redress framework (both informal and formal channels), setting out the time frame and mechanisms for resolving complaints about environmental performance.

9. Environmental Management Plan

This section deals with the set of mitigation and management measures to be taken during project implementation to avoid, reduce, mitigate, or compensate for adverse environmental impacts (in that order of priority). It may include multiple management plans and actions. It includes the following key components (with the level of detail commensurate with the project's impacts and risks):

- (i) Mitigation:
 - a. identifies and summarizes anticipated significant adverse environmental impacts and risks;
 - b. describes each mitigation measure with technical details, including the type of impact to which it relates and the conditions under which it is required (for instance, continuously or in the event of contingencies), together with designs, equipment descriptions, and operating procedures, as appropriate; and

- c. provides links to any other mitigation plans (for example, for involuntary resettlement, Indigenous Peoples, or emergency response) required for the project.
- (ii) Monitoring:
 - a. describes monitoring measures with technical details, including parameters to be measured, methods to be used, sampling locations, frequency of measurements, detection limits and definition of thresholds that will signal the need for corrective actions; and
 - b. describes monitoring and reporting procedures to ensure early detection of conditions that necessitate particular mitigation measures and document the progress and results of mitigation.
- (iii) Implementation arrangements:
 - a. specifies the implementation schedule showing phasing and coordination with overall project implementation;
 - b. describes institutional or organizational arrangements, namely, who is responsible for carrying out the mitigation and monitoring measures, which may include one or more of the following additional topics to strengthen environmental management capability: technical assistance programs, training programs, procurement of equipment and supplies related to environmental management and monitoring, and organizational changes; and
 - c. estimates capital and recurrent costs and describes sources of funds for implementing the environmental management plan.
- (iv) Performance indicators: describes the desired outcomes as measurable events to the extent possible, such as performance indicators, targets, or acceptance criteria that can be tracked over defined time periods.

10. Conclusion and Recommendation

This section provides the conclusions drawn from the assessment and provides Recommendations

ANNEX 6-8: OUTLINE OF A RESETTLEMENT PLAN

This outline is part of the Safeguard Requirements. A resettlement plan is required for all projects with involuntary resettlement impacts. Its level of detail and comprehensiveness is commensurate with the significance of potential involuntary resettlement impacts and risks. The substantive aspects of the outline will guide the preparation of the resettlement plans, although not necessarily in the order shown.

1. Executive Summary

This section provides a concise statement of project scope, key survey findings, entitlements and recommended actions.

2. Project Description

This section provides a general description of the project, discusses project components that result in land acquisition, involuntary resettlement, or both and identify the project area. It also describes the alternatives considered to avoid or minimize resettlement. Include a table with quantified data and provide a rationale for the final decision.

3. Scope of Land Acquisition and Resettlement

This section:

- I. discusses the project's potential impacts, and includes maps of the areas or zone of impact of project components or activities;
- II. describes the scope of land acquisition (provide maps) and explains why it is necessary for the main investment project;
- III. summarizes the key effects in terms of assets acquired and displaced persons; and
- IV. provides details of any common property resources that will be acquired.

4. Socioeconomic Information and Profile

This section outlines the results of the social impact assessment, the census survey, and other studies, with information and/or data disaggregated by gender, vulnerability, and other social groupings, including:

- I. define, identify, and enumerate the people and communities to be affected;
- II. describe the likely impacts of land and asset acquisition on the people and communities affected taking social, cultural, and economic parameters into account;
- III. discuss the project's impacts on the poor, indigenous and/or ethnic minorities, and other vulnerable groups; and
- IV. identify gender and resettlement impacts, and the socioeconomic situation, impacts, needs, and priorities of women.

5. Information Disclosure, Consultation, and Participation

This section:

- I. identifies project stakeholders, especially primary stakeholders;
- II. describes the consultation and participation mechanisms to be used during the different stages of the project cycle;
- III. describes the activities undertaken to disseminate project and resettlement information during project design and preparation for engaging stakeholders;
- IV. summarizes the results of consultations with affected persons (including host communities), and discusses how concerns raised and recommendations made were addressed in the resettlement plan;
- V. confirms disclosure of the draft resettlement plan to affected persons and includes arrangements to disclose any subsequent plans; and
- VI. describes the planned information disclosure measures (including the type of information to be disseminated and the method of dissemination) and the process for consultation with affected persons during project implementation

6. Grievance Redress Mechanisms

This section describes mechanisms to receive and facilitate the resolution of affected persons concerns and grievances. It explains how the procedures are accessible to affected persons and gender sensitive.

7. Legal Framework

This section:

- I. describes national and local laws and regulations that apply to the project and identify gaps between local laws and ADB's policy requirements; and discuss how any gaps will be addressed.
- II. describes the legal and policy commitments from the executing agency for all types of displaced persons;
- III. outlines the principles and methodologies used for determining valuations and compensation rates at replacement cost for assets, incomes, and livelihoods; and set out the compensation and assistance eligibility criteria and how and when compensation and assistance will be provided.
- IV. describes the land acquisition process and prepare a schedule for meeting
- V. key procedural requirements.

8. Entitlements, Assistance and Benefits

This section:

- I. defines displaced persons' entitlements and eligibility, and describes all resettlement assistance measures (includes an entitlement matrix);
- II. specifies all assistance to vulnerable groups, including women, and other special

groups; and.

- III. outlines opportunities for affected persons to derive appropriate development benefits from the project.

9. Relocation of Housing and Settlements

This section:

- I. describes options for relocating housing and other structures, including replacement housing, replacement cash compensation, and/or self-selection (ensure that gender concerns and support to vulnerable groups are identified);
- II. describes alternative relocation sites considered; community consultations conducted; and justification for selected sites, including details about location, environmental assessment of sites, and development needs;
- III. provides timetables for site preparation and transfer;
- IV. describes the legal arrangements to regularize tenure and transfer titles to resettled persons;
- V. outlines measures to assist displaced persons with their transfer and establishment at new sites;
- VI. describes plans to provide civic infrastructure; and
- VII. explains how integration with host populations will be carried out.

10. Income Restoration and Rehabilitation

This section:

- I. identifies livelihood risks and prepare disaggregated tables based on demographic data and livelihood sources;
- II. describes income restoration programs, including multiple options for restoring all types of livelihoods (examples include project benefit sharing, revenue sharing arrangements, joint stock for equity contributions such as land, discuss sustainability and safety nets);
- III. outlines measures to provide social safety net through social insurance and/or project special funds;
- IV. describes special measures to support vulnerable groups;
- V. explains gender considerations; and
- VI. describes training programs.

11. Resettlement Budget and Financing Plan

This section:

- I. provides an itemized budget for all resettlement activities, including for the resettlement unit, staff training, monitoring and evaluation, and preparation of resettlement plans during loan implementation.
- II. describes the flow of funds (the annual resettlement budget should show the budget

scheduled expenditure for key items).

- III. includes a justification for all assumptions made in calculating compensation rates and other cost estimates (taking into account both physical and cost contingencies), plus replacement costs.
- IV. includes information about the source of funding for the resettlement plan budget.

12. Institutional Arrangements

This section:

- I. describes institutional arrangement responsibilities and mechanisms for carrying out the measures of the resettlement plan;
- II. includes institutional capacity building program, including technical assistance, if required;
- III. describes role of NGOs, if involved, and organizations of affected persons in resettlement planning and management; and
- IV. describes how women's groups will be involved in resettlement planning and management,

13. Implementation Schedule

This section includes a detailed, time bound, implementation schedule for all key resettlement and rehabilitation activities. The implementation schedule should cover all aspects of resettlement activities synchronized with the project schedule of civil works construction, and provide land acquisition process and timeline.

14. Monitoring and Reporting

This section describes the mechanisms and benchmarks appropriate to the project for monitoring and evaluating the implementation of the resettlement plan. It specifies arrangements for participation of affected persons in the monitoring process. This section will also describe reporting procedures.

ANNEX 6-9: IFC/MIGA EXCLUSION LIST OF PROJECTS

(A) IFC Exclusion List

The IFC Exclusion List defines the types of projects that IFC does not finance. The exclusion list is also applicable for projects to be financed under IPFF.

The IFC's exclusion list of projects is as follows:

1. Production or trade in any product or activity deemed illegal under host country laws or regulations or international conventions and agreements, or subject to international bans, such as pharmaceuticals, pesticides/herbicides, ozone depleting substances, PCB, wildlife or products regulated under CITES.
2. Production or trade in weapons and munitions.¹
3. Production or trade in alcoholic beverages (excluding beer and wine).¹
4. Production or trade in tobacco.¹
5. Gambling, casinos and equivalent enterprises.¹
6. Production or trade in radioactive materials. This does not apply to the purchase of medical equipment, quality control (measurement) equipment and any equipment where IFC considers the radioactive source to be trivial and/or adequately shielded.
7. Production or trade in un-bonded asbestos fibers. This does not apply to purchase and use of bonded asbestos cement sheeting where the asbestos content is less than 20%.
8. Drift net fishing in the marine environment using nets in excess of 2.5 km. in length.

A reasonableness test will be applied when the activities of the project company would have a significant development impact but circumstances of the country require adjustment to the Exclusion List.

All financial intermediaries (FIs), except those engaged in activities specified below*, must apply the following exclusions, in addition to IFC's Exclusion List:

1. Production or activities involving harmful or exploitative forms of forced labor²/harmful child labor.³
2. Commercial logging operations for use in primary tropical moist forest.
3. Production or trade in wood or other forestry products other than from sustainably managed forests.

*When investing in microfinance activities, FIs will apply the following items in addition to the IFC Exclusion List:

1. Production or activities involving harmful or exploitative forms of forced labor²/harmful child labor.
2. Production, trade, storage, or transport of significant volumes of hazardous chemicals, or commercial scale usage of hazardous chemicals. Hazardous chemicals include gasoline, kerosene, and other petroleum products.

3. Production or activities that impinge on the lands owned, or claimed under adjudication, by Indigenous Peoples, without full documented consent of such peoples.

*Trade finance projects, given the nature of the transactions, FIs will apply the following items in addition to the IFC Exclusion List:

- Production or activities involving harmful or exploitative forms of forced labor²/harmful child labor.

(B) MIGA Exclusion List of Activities

The type of projects for which MIGA will not provide guarantees are as follows:

1. Products or activities deemed illegal by the host country or international law.
2. Drift net fishing using nets in excess of 2.5 km. in length.
3. Activities prohibited by MIGA's environmental and social safeguard policies (e.g., natural habitats, forestry, pest management, etc.)
4. Manufacture and trade in weapons and munitions, distilled spirit for human consumption, tobacco, wildlife and wildlife products that are regulated under CITES, radioactive materials for non-medical uses, unbonded asbestos fibers, products containing PCBs, pharmaceuticals subject to international phase out or ban, pesticides subject to international phase out or ban, and ozone depleting substances subject to international phase out and
5. Enterprises whose primary sources of income are gambling, prostitution, or involved forced labour or child labour inconsistent with internationally recognized norms.

Endnotes

- 1 This does not apply to project sponsors who are not substantially involved in these activities. "Not substantially involved" means that the activity concerned is ancillary to a project sponsor's primary operations.
- 2 Forced labor means all work or service, not voluntarily performed, that is extracted from an individual under threat of force or penalty.
- 3 Harmful child labor means the employment of children that is economically exploitive, or is likely to be hazardous to, or to interfere with, the child's education, or to be harmful to the child's health, or physical, mental, spiritual, moral, or social development.

ANNEX 6-10: ADB PROHIBITED INVESTMENT ACTIVITIES LIST

The following do not qualify for Asian Development Bank financing:

- (i) production or activities involving harmful or exploitative forms of forced labor or child labor;
- (ii) production of or trade in any product or activity deemed illegal under host country laws or regulations or international conventions and agreements or subject to international phase outs or bans, such as (a) pharmaceuticals, pesticides and herbicides, (b) ozone-depleting substances, (c) polychlorinated biphenyls and other hazardous chemicals, (d) wildlife or wildlife products regulated under the Convention on International Trade in Endangered Species of Wild Fauna and Flora, and (e) trans-boundary trade in waste or waste products;
- (iii) production of or trade in weapons and munitions, including paramilitary materials;
- (iv) production of or trade in alcoholic beverages, excluding beer and wine;
- (v) production of or trade in tobacco;
- (vi) gambling, casinos, and equivalent enterprises;
- (vii) production of or trade in radioactive materials, including nuclear reactors and components thereof;
- (viii) production of, trade in, or use of unbonded asbestos fibers;
- (ix) commercial logging operations or the purchase of logging equipment for use in primary tropical moist forests or old-growth forests; and
- (x) marine and coastal fishing practices, such as large-scale pelagic drift net fishing and fine mesh net fishing, harmful to vulnerable and protected species in large numbers and damaging to marine biodiversity and habitats.

ANNEX 6-11: SITE VISIT CHECKLIST – ENVIRONMENTAL SAFEGUARDS

A. For Greenfield Projects

A reconnaissance visit should be conducted to observe the:

1. Current land use of project site
2. Land use of adjoining areas
3. Topography of the site & average levels of the site
4. Traces of pollution (land, air, water)

An interview should be organized with the local authority to identify significant issues related to the project site as listed below:

5. Distance from designated Protected Areas/ National Parks/ Wildlife Sanctuaries
6. Distance from designated Ecologically Critical Areas (as declared by DoE)
7. Whether the site/adjoining properties has any history of industrial pollution? Any action has been initiated against the present/past land owner due to the pollution?
8. Is the site located near any river/ stream/ perennial water body? Do the local people located downstream from the site depend on this for their domestic & agricultural water needs?
9. Do local fishermen use this stream/ river for their livelihood?
10. Is the site prone to flooding? Has there been any major flooding in the last 10 years?
11. Will the construction of the project change the natural drainage pattern of the site and affect any river / stream/ perennial nallah?
12. Any chances of major severance (approach road, water body, access road to farm lands, school, healthcare, mosque etc.)
13. Does the site have any major trees? Approximately how many of these are to be cut due to project?
14. Is the area known for having rare/endangered/ migratory species?
15. Is the site & surrounding area known for having unique fragile ecosystem?
16. Whether there are any structures of cultural/historical/religious importance near the site or any designated archaeological sites/ remnants?
17. The main economic activities in the nearby town/city/villages that may be altered by the construction and /or the operation of the project?
18. Whether there are any tourist attractions near the site, which may be affected due to this project?

B. Brownfield Projects

The checklist of questions for Greenfield projects shall also be applicable for Brownfield projects. In additions the questions listed below shall also be applicable.

1. Nos. & types of structures at site (detailed description & possible age of structure)
2. Adjacent land use and structures on them
3. What was the previous land use? If industrial land use, what was the type of industry? What was being manufactured?
4. Does the present land use involve hazardous waste generation & storage and /or use of petroleum products (barring any natural gas)?
5. Does the past land use involve hazardous waste generation & storage and /or use of petroleum products (barring any natural gas)?
6. Source of utilities to site (gas, electricity, water supply and sewage)?
7. Are there any pits/ponds/ lagoons in the existing structure?
8. Are there any storage tanks (above and/or underground) in the property? if yes, please note the details of their capacity, content and age)
9. Are there any visible stains, marks or corrosions (of spillage) in the existing structure or on the ground/ soil/pavement?
10. Are vegetation in and/or near the project appear to be stressed (discoloration and/or necrosis of leaves, droopiness of the structure or death of plant)?
11. Presence of apparently filled/graded with non natural material or solid waste
12. Is there any presence of friable asbestos?
13. Is there any presence of lead based paints?
14. Are there any transformer oil stored within site that contains Poly-chlorinated Biphenyls (PCB)?
15. Presence of any of the following within or in the close vicinity of the site:
 - a) Bore well
 - b) Injection well / recharge well
 - c) Septic tank
 - d) Sewers / Drains / sumps
 - e) Effluent Treatment Plant (ETP)

(Signature of the visiting Officer, BIFFL)

(Signature of ESC in charge BIFFL)

Name:

Name:

Date:

Date:

ANNEX 6-12: FORMAT OF PRELIMINARY ENVIRONMENTAL APPRAISAL REPORT (PEAR)

Project Categorization as per BIFFL's ESMF	
Environmental Category of the Project as per WB, ADB and DOE	
Project Description: (location, components, cost)	
Application of GOB environmental regulations/ Permits required, if any	
Status of clearances and permits: • Site Clearance Certificate • ToR for EIA • Environmental Clearance Certificate • Factories Inspectorate license	
Environmental Implications (summary of sector specific checklist, permits required under GOB regulations)	
Positive impacts:	
Recommended E&S covenants by ESC for loan documentation	
Attach filled in formats listed below: 1. E&S Risk Rating checklist 2. Sector specific checklist for the project	

(Signature of the Relevant Officer, BIFFL)

(Signature of ESC in charge BIFFL)

Name:

Name:

Date:

Date:

ANNEX 6-13: CONTENTS OF ANNUAL ENVIRONMENTAL MONITORING REPORT

Chapter No.	Chapter & Description
	Executive Summary
Chapter 1	Project Background
i.	Location
ii.	Capacity
iii.	Study Period
iv.	Present Status of the Subprojects
v.	Key Developments and Required Changes
Chapter 2	Environmental Permits and Compliance Certificates
	Issuance Certificates by government agency(ies)
	Issuance dates and duration of validity
	Permit conditions and renewal requirements
	Other standards and/or certificates
	Systems and Pollution Control Standards
Chapter 3	Scope & Methodology
	i. Scope of Services
	ii. Methodology
	iii. Data Collection and Analysis
Chapter 4	Grievance Redress Mechanism
	i. Date established and description of organizational arrangements
	ii. Complaints, grievance, or protests received from local communities, recorded dates and organizations involved, actions taken to resolve grievances; any
	iii. Outstanding issues and proposed measures for resolution
Chapter 5	Occupational and Community Health and Safety
	i. Training programs carried out
	ii. Incidents recorded, including date, scale of damage and injury ⁴⁰ , if any; authorities in charge of investigation/recording, and media or community reactions, if any; action taken to respond to the incident; and any outstanding issues and proposed measures

	iii. Other health and safety events or out-reach activities
Chapter 6	Environmental Management Capacity
	i. Number of staff, qualifications and experience in environmental management
	ii. Awareness of: (i) environmental issues, (ii) health and safety, (iii) national environmental laws and regulations, and (iv) applicable ADB's environmental safeguard requirements
	iii. Training programs carried out
	iv. Assessment of environmental management capacity needs (including planned Capacity building programs)
Chapter 7	Stakeholder Consultation
	i. Details of information disclosure and consultations, if any, with affected people, local communities, civil society groups, and other stakeholders
	ii. Details of approach/methodology on addressing the concerns and issues raised at consultations
Chapter 8	Compliance with Applicable Environmental Safeguard Requirements as Defined in BIFFL's ESMF
	Implementation of EMP (citing changes in the EMP, where warranted) at construction or operation phase, focusing on mitigation measures and monitoring program. Attach copies of monitoring reports. If standards or thresholds are exceeded, specify the corrective actions implemented or to be implemented.
Chapter 9	Other Information and Feedback
	Please list any other information that the Bank should be aware of. This may relate to broader environmental matters, community initiatives within your company, positive media or NGO attention, training activities, management system initiatives or cost savings through process efficiency, environment-friendly processes.
Chapter 10	Summary of Conclusion and Recommendation
Chapter 11	Follow-up Actions

Name and Signature of monitoring officer

Signature of ESC in charge BIFFL)

Position
Date

Name:
Date:

ANNEX 6-14: ENVIRONMENTAL & SOCIAL DUE DILIGENCE (EDD) REPORT OUTLINE

Chapter No.	Chapter & Description
	Table of Content
	Executive Summary
	Abbreviations
Chapter 1	Introduction
1.1	Project Background.
1.2	E&S categorization Rationale
1.3	Applicable Performance Standard
Chapter 2	Scope of Review and Technology
Chapter 3	Environmental and Social Issues and Mitigation (By relevant Performance Standards: examine environmental and social issues and compliance with respective performance standards, identify performance gaps and corrective action plans)
3.1	<i>Environmental and Social Assessment and Management System</i>
	i. Environmental and Social Assessment
	ii. Management Program
	iii. Organization
	iv. Training
	v. Community Engagement
	vi. Monitoring
	vii. Reporting
3.2	<i>Labor and Working Conditions</i>
	i. Human Resources Policy and Management
	ii. Worker's Organization
	iii. Non-Discrimination and Equal Opportunity
	iv. Retrenchment
	v. Protecting the Work Force
	vi. Occupational Health and Safety

Chapter No.	Chapter & Description
3.3	<i>Pollution Prevention and Abatement</i>
	i. Pollution Prevention, Resource Conservation and Energy Efficiency
	ii. Wastes
	iii. Hazardous Materials
	iv. Emergency Preparedness and Response
	v. Ambient Considerations
	vi. Green House Gas Emissions
	vii. Pesticide Use and Management
	viii. Pollution Prevention, Resource Conservation and Energy Efficiency
3.4	<i>Community Health, Safety and Security</i>
	i. Community Health and Safety
	ii. Emergency Preparedness and Response
	iii. Security Personnel Requirements
3.5	<i>Land Acquisition and Involuntary Resettlement</i>
	i. Project Design
	ii. Compensation and Benefits for Displaced Persons
	iii. Consultation and Grievance Mechanism
	iv. Resettlement Planning and Implementation
	v. Physical Displacement
	vi. Economic Displacement
	vii. Private Sector Responsibilities under Government-Managed Resettlement
3.6	<i>Biodiversity Conservation and Sustainable Natural Resource Management</i>
	i. Protection and Conservation of Biodiversity
	ii. Management and Use of Renewable Natural Resources
3.7	<i>Indigenous Peoples</i>
	i. Avoidance of Adverse Impacts
	ii. Consultation and Informed Participation
	iii. Impacts on Traditional or Customary Lands under Use
	iv. Relocation of Indigenous Peoples (IPs) from Traditional or Customary Lands
	v. Cultural Resources

Chapter No.	Chapter & Description
3.8	<i>Cultural Heritage</i>
	i. Protection of Cultural heritage in Project Design and Execution
	ii. Project use of Cultural Heritage
Chapter 4	Details of project consultation and local disclosure undertaken
Chapter 5	Other project specific issues
Chapter 6	Environment Management /Corrective Action Plan, if any
Chapter 7	Summary and Recommendations
Chapter 8	Financing Covenants/Conditions

ANNEX 6-15: FORMAT FOR CORRECTIVE ACTION PLAN

Name of Sub-Project					Date of Audit:		
Documents reviewed:							
Sl. No	Identified Non-Compliances	Recommended Corrective Actions	Time frame	Action completion indicator	Means of Evaluation	Name of Responsible person	Remarks
(Signature of Environmental Safeguards Specialist)							
Name:							

ANNEX 6-16: DOE PRESCRIBED PRO-FORMA FOR APPLYING FOR ECC

Director/Deputy Director

Department of Environment

Dhaka/ Chittagong/ Khulna/ Rajshahi (Bogra)/ Barisal/ Sylhet Division

Sir,

I do hereby apply for environmental clearance certificate (ECC) for my proposed industrial unit or project, or for the existing industrial unit or project, and enclose the required papers and furnish all information as follows:

1.(a)	Name of the industrial unit or project:	
(b)	Location of the industrial unit or project:	
(c)	Address of present office:	
2.(a)	Proposed industrial unit or project:	
(b)	Expected starting date of construction:	
)	Expected completion date of construction:	
	Expected trial production/operation starting date of industrial unit/project:	
2.(b)	Existing industrial unit or project:	
(b.a)	Expected trial production/operation starting date of industrial unit/project:	
3	Name of the product and quantity to be produced (daily/monthly/yearly):	
4.(a)	Name of raw materials and quantity required (daily/monthly/yearly):	
(b)	Source of raw materials:	
5(a)	Quantity of water to be used daily:	
(b)	Source of water :	
6.(a)	Name of fuel and quantity required (daily/monthly/yearly) :	
(b)	Source of fuel :	
7. (a)	Probable quantity of daily liquid waste :	
(b)	Location of waste discharge :	

(c)	Probable quantity of daily emission of gaseous substance :	
(d)	Mode of emission of gaseous substance:	
8	Mouza (village) map indicating "Daag" (plot) number and "Khatiyan" number :	
9	Approval of Rajdhani Unnayan Katripakkhya/Chittagong Development Authority/Khulna Development Authority/ Rajshahi City Corporation/Barisal City Corporation/Sylhet City Corporation/Concern Pourashava/ Union Parishad, etc :	
10.(a)	Design and time schedule of proposed effluent treatment plant (ETP) :	
(b)	Fund allocated:	
(c)	Area :	
11	Process flow diagram :	
(a)	Location map of industrial unit or project :	
(b)	Layout plan with location of ETP :	
(a)	Initial environmental examination (IEE)/ environmental impact assessment (EIA) report, if applicable:	
(b)	Environmental management plan (EMP), if applicable:	
(c)	Feasibility report, if applicable:	

Signature of the entrepreneur

Seal

Name :

Address

Phone :

Date :

Declaration

I do hereby declare that all information provided by me in this application are true to the best of my knowledge and no information has been concealed or distorted herein.

(Name and signature of the entrepreneur)

ANNEX 6-17: OUTLINE OF AN IEE REPORT

The contents of an IEE report are stated below;

- a) Introduction
- b) Description of the Project
- c) Description of the Environment
- d) Screening of Potential Environmental Impacts and Mitigation Measures
- e) Institutional Requirements and Environmental Monitoring Plan
- f) Public Consultation and Information Disclosure
- g) Findings and Recommendation
- h) Conclusions

The detail of the contents is stated below.

1. Introduction

This section usually includes the following:

- Purpose of the report, including (a) identification of the project and project proponent; (b) brief description of the nature, size, and location of the project and of its importance to the country; and (c) any other pertinent background information; and
- Extent of the IEE study: scope of study, magnitude of effort, person or agency performing the study, and acknowledgement.

2. Description of the Project

Furnish sufficient details to give a brief but clear picture of the following (include only applicable items):

- type of project;
- category of Project;
- need for project;
- location (use maps showing general location, specific location, and project site);
- size or magnitude of operation;
- proposed schedule for implementation; and
- descriptions of the project, including drawings showing project layout, and project components. This information should be of the same type and extent as is included in feasibility reports for proposed projects to give a clear picture of the project and its operations.

3. Description of the Environment

Furnish sufficient information to give a brief but clear picture of the existing environmental resources in the area affected by the project, including the following (to the extent applicable):

- i. Physical Resources: (e.g.)
 - a. atmosphere (e.g. air quality and climate)
 - b. topography and soils,
 - c. surface water
 - d. groundwater
 - e. geology/ seismology.
- ii. Ecological Resources: (e.g.)
 - a. fisheries
 - b. aquatic biology
 - c. wildlife
 - d. forests
 - e. rare or endangered species
 - f. protected areas
 - g. coastal resources
- iii. Economic Development: (e.g.)
 - a. industries
 - b. infrastructure facilities (e.g. water supply, sewerage, flood control)
 - c. transportation (roads, harbors, airports, and navigation)
 - d. land use (e.g. dedicated area uses)
 - e. power sources and transmission
 - f. agricultural development, mineral development, and tourism facilities
- iv. Social and Cultural Resources: (e.g.)
 - a. population and communities (e.g. numbers, locations, composition, employment)
 - b. health facilities
 - c. education facilities
 - d. socio-economic conditions (e.g. community structure, family structure, social well being)
 - e. physical or cultural heritage
 - f. current use of lands and resources for traditional purposes by Indigenous Peoples
 - g. Structures or sites that are of historical, archaeological, pale ontological or architectural significance.

4. Screening of Potential Environmental Impacts and Mitigation Measures

Using the checklist of environmental parameters for projects, this section will screen out “no significant impacts” from those with significant adverse impact by reviewing each relevant parameter according to the following factors or operational stages. Mitigation measures, where appropriate, will also be recommended environmental problems due to project location, and related to project design, construction, and operations. Potential environmental enhancement measures and additional considerations will also be covered.

5. Institutional Requirements and Environmental Monitoring Plan

This section should state the impacts to be mitigated, and activities to implement the mitigation measures, including how, when, and where they will be implemented. Institutional arrangements for implementation should be described. The environmental monitoring plan will describe the impacts to be monitored, and when and where monitoring activities will be carried out, and who will carry them out. The environmental management and monitoring costs should also be described.

6. Public Consultation and Information Disclosure

This section will describe the process undertaken to involve the public in project design and recommended measures for continuing public participation; summarize major comments received from beneficiaries, local officials, community leaders, NGOs, and others, and describe how these comments were addressed; list milestones in public involvement such as dates, attendance, and topics of public meetings; list recipients of this document and other project related documents; describe compliance with relevant regulatory requirements for public participation; and summarize other related materials or activities, such as press releases and notifications. This section will provide of summary of information disclosed to date and procedures for future disclosure.

7. Findings and Recommendations

This section will include an evaluation of the screening process and recommendation will be provided whether significant environmental impacts need further detailed study or EIA. If there is no need for further study, the IEE itself, which at times may need to be supplemented by a special study in view of limited but significant impacts, becomes the completed environmental assessment for the project and no follow-up EIA will be needed. If an EIA is needed, then this section will include a brief terms of reference (TOR) for the needed follow-up EIA, including approximate descriptions of work tasks, professional skills required, time required, and estimated costs. The Bank's Environment Guidelines provides a guide for preparing the TOR for different projects.

8. Conclusions

This section will discuss the result of the IEE and justification, if any, of the need for additional study or EIA. If an IEE, or an IEE supplemented by a special study, is sufficient for the project, then the IEE with the recommended institutional and monitoring program becomes the completed EIA.

ANNEX 6-18: STRUCTURE FOR A SAMPLE EIA REPORT

A structure for sample EIA report proposed for BIFFL is given below:

1. Executive Summary

2. Introduction

This chapter should cover:

- a. Project Background.
- b. Purpose of the Study, Project, Project Proponent, brief description of the project- name, nature, size.
- c. Need for the Project.
- d. Location of the project- land description, extent of the land, plot/ survey nos./ village, upazilla & district.
- e. Importance of the Project to the Region & Country.
- f. Scope of the EIA Study
- g. Disclosure of EIA Team.

Essential Maps to be Provided

- h. A map specifying location of the site, district and the Country (Index Map).
- i. A map of project area and 10 km area from boundary of the proposed/existing project area, / international boundaries.
- j. A map covering aerial distance of 25 km from the project site boundary delineating environmental sensitive areas.
- k. Boundaries of the proposed site will be shown therein with latitude and longitude.
- l. Area drainage contour map of the project area.
- m. Land use map of the study area to 1: 50,000 scale based on recent satellite imagery of the project area and 10 km from the proposed project boundary delineating the cropping pattern, wastelands, forest area, built-up areas, water bodies, human habitation and other surface features such as roads/National Highways, railway lines, ports, airports, major industries etc.

3. Legal & Legislative Frameworks

- a. The proponent should confirm that the project meets the environmental regulations and standards applicable for the project.
- b. Any litigation pending against the proposed project and/ or any direction/ order passed by any court of law against the project, if so, details thereof.

4. Project Proponent

- a. Profile of the project proponent, name and contact address with e-mail.
- b. Implementing organization, organizational chart, project consultants, etc.
- c. Project Concept & Components.
- d. Project Schedule.

5. Project Description

This chapter should cover the broader details of the basic activities, layout and implementation schedule of the project.

- a. Type of the project- new/expansion, etc.
- b. Relevance of the project in the light of the existing development plans of the region.
- c. Project coverage, master plan, phasing and scope.
- d. Description of a project site, geology, topography, transport and connectivity, demographic aspects, socio, cultural and economic aspects.
- e. Capacity of the port project, equipments, ancillary operations, housing, parking details, etc.
- f. Technologies involved for design, construction, equipment and operation.
- g. Use of existing infrastructures – road, railway and inland waterway net works, water supply, electrical power, etc.
- h. Water budget - during construction/ operation stages.
- i. Estimated cost of development of the project, environmental cost, rehabilitation of communities / villages present status of such activities.
- j. Resources, manpower and time frame, etc. –required for project implementation.
- k. Environmental Management/Pollution Mitigation Measures.

Essential Maps to be Provided

- a. Site lay out plan of the proposed development will be submitted by clearly marking the layout, workshops, container freight station, cargo handling systems, covered and uncovered storage yards, warehouses, roads, effluent disposal point, administrative and operational buildings, utilities, town ships, greenbelt, dredged material disposal sites, etc.
- b. Hydrographic charts, if applicable, giving general morphology of the river/coastal stretch to a scale of 1:50,000 will be submitted covering water depth up to 10 m beyond the maximum.
- c. proposed dredging depths of the project and covering a distance of 10 km along the river/coast from the project limits on both sides.

6. Analysis of Alternatives (Technology & Sites)

- a. Description of various alternatives like locations or layouts or technologies studied.
- b. Description of each alternative.
- c. Selection of alternative.

7. Description of the Environment

Study Area: As a primary requirement of EIA process, the proponent should collect primary baseline data in the project area as well as in the area falling 10 km aerial distance from the project boundary (25 km towards Sundarbans).

The study areas for different environmental attributes (water, air, noise, soil, etc.) is to be submitted considering the proposed activities and location, along with proper reasoning, for review and approval.

8. Land Environment

- a. Availability of land for earmarking for the Project without causing a due hardship to local habitat and their socio cultural and economic aspects.
- b. Justification for the proposed quantum of the area is to be given.
- c. Study of land use pattern, habitation, cropping pattern, forest cover, environmentally sensitive places, etc, by employing remote sensing techniques and also through secondary data sources.

9. Topography

Baseline data to be given on description of existing status of the land at the project area including description of slopes/ inland topography, features (lowland, littoral areas, shoal areas), terrain features, slope and elevation.

10. Geology

- a. Baseline data to be provided on rock types, regional tectonic setting (reported fractures/faulting, folding, warping), history of any volcanic activity, seismicity and associated hazards, mainly in the Project & Coastal area.
- b. Information on quarry yields, strengths of rock, distance of quarries from habitat, restrictions for quarrying, environmental controls, statutory permissions, etc., should be provided.

11. Soil

- a. Soil data including type, classification, characteristics, soil properties, etc., are important from engineering considerations for design of structures, loading capacities of cargo stockpiles, green belt development etc.
- b. Changes in parameters of soil also may affect plantation and vegetative growth which in turn may endanger the health of local habitat.
- c. Baseline data of the soil, results of investigations carried out to be provided.

12. Water Environment

a. Ground Water

Baseline data of ground water including data of pH, dissolved solids, suspended solids, BOD, DO, coli-form bacteria, oil, heavy metals, etc. is to be collected at least for one season. Usage purpose of the ground water, if any, is to be indicated.

b. Surface Water

Baseline data on location of surface water like lagoons, lakes, tidal inlets, streams, rivers, their details, present quality and their utility, if any, is to be provided. Details of water bodies in the project area will be described specifically. Water quality is to be monitored for one season.

13. Marine Environment

a. Hydrology/Geomorphology

Hydrology requires collection of oceanographic data during the study period, covering the following parameters:

- Tides
- Waves (wind waves and swells)
- Storm surges
- Currents
- Salinity
- Sea water temperature
- Suspended load, and
- Seabed bathymetry.

Baseline oceanographic data should extend at least to depths more than 10 m of proposed deepening of the harbor approach and basin as proposed. A study on likely changes in the sediment transport and littoral drift due to the construction of port should be taken up.

Details of mangroves, marshes and other vegetation, sand dunes, stability, seismic characteristics, history of any endangered species, river bank/coastal erosion and shoreline changes should be furnished.

b. Bed Sediment Contamination

Baseline data on bottom sediments and the associated bottom biota and other physical habitat, at the project area and the neighborhood areas has to be collected and analyzed.

14. Biological Environment

a. Ecology

- Baseline data of both terrestrial and aquatic flora and fauna at the project area is to be ascertained by proper surveys including mangroves and marshes and other inland waters & coastal vegetation, etc.
- Data on rare & endangered species are also necessary.
- Flora-Fauna Status & Species Diversity of Sundarbans will be included.

a. Flora and Fauna

Details on secondary data on the existing flora and fauna in the study area will be carried out along with classification as per Schedule.

15. Air Environment

a. Meteorological Data

Meteorological data for at least a 10 year period should be presented from the nearest meteorological station, except for the history of cyclones and tidal surges for which 100 year data is required. The list of Meteorological parameters includes: Wind speed and direction (Wind Rose diagrams), Temperature, Relative humidity, Barometric pressures, Rainfall, History of cyclones, etc.

b. Ambient Air Quality

Baseline data of ambient air parameters namely particulate matter size less than 2.5 μm or PM_{2.5} (in $\mu\text{g}/\text{m}^3$), particulate matter size less than 10 μm or PM₁₀ ($\mu\text{g}/\text{m}^3$), sulphur dioxide ($\mu\text{g}/\text{m}^3$), nitrogen dioxide ($\mu\text{g}/\text{m}^3$), ozone ($\mu\text{g}/\text{m}^3$), carbon monoxide (mg/m^3), heavy metals (Particulate Lead; $\mu\text{g}/\text{m}^3$) will be monitored. One season data should be monitored other than monsoon. At least one station should be in the up-wind/ non-impact/ non-polluting area as a control station and one station in down wind direction.

16. Noise Levels

Baseline data on noise levels at the project area and the neighbourhood areas is to be monitored and reported with applicable Noise Standards.

17. Socio-Economics and Occupational Health Environment

Baseline data at the project area will include the demography, particularly on human settlements, health status of the communities, existing infrastructure facilities in the Project area and area of impact due to the proposed activity. Present employment and livelihood of these populations, awareness of the population about the proposed activity will also be included.

18. Public Utilities

Base line data of existing public utility infrastructure will be ascertained and reported to assess the impact of the project on these public utilities in order to incorporate desired methods in the EMP and monitor the same during the construction as well as operational phases of the Project.

19. Anticipated Environmental Impact and Mitigation Measures

This Chapter should describe the likely impact of the project on each of the environmental parameters, methods adopted for assessing the impact such as model studies, empirical methods, reference to existing similar situations, reference to previous studies, details of mitigation methods proposed to reduce adverse effects of the project, best environmental practices and conservation of natural resources.

The identification of specific impact followed with mitigation measures should be done for different stages i.e., Construction Phase (including dredging) and Operation Phase (ship traffic including discharges from vessels and cargo operations).

a. Land Environment

Anticipated Impacts:

- Impact of project construction/operation on the land requirement / land use pattern should be assessed.
- Effect of future growth of the port facility and/or of the ancillaries should be carefully assessed. Impact on the public utilities arising out of the utilities for the project activities and impact on the natural drainage system will be assessed.
- Prediction of impact should include impact on the existing infrastructures like road networks, housing, ground water/surface waters, etc. and loss of productive soil (if any) and impact on the natural drainage pattern.

Mitigation Measures:

- Mitigation measures to reduce adverse effects like adopting soil improvement techniques and adopting suitable design methods to reduce land requirement.
- Strengthening of road and rail network infrastructure to handle the increase in traffic and truck parking arrangements, integration of Port development with the local land use plan should be planned.

b. Topography, Geology and Soil

Anticipated Impacts:

- Impact of construction/operation on the topography due to activities like large scale quarrying, filling of low lying area with dredged spoils and borrowed material, damage to existing vegetation/green belt and plantation, changes in land use patterns, disturbance to existing protected areas like mangroves, forests and environmentally sensitive areas/zones should be assessed.
- Flooding due to filling up of low-lying areas should be assessed.
- Impact on the surrounding land use pattern, on infrastructure like housing, ground water, etc should be assessed.
- Impact of the project construction on the geology and vice-versa should to be studied in detail.
- Impact of project construction/operation on the soil parameters, probability of settlement, subsidence, slides, surface drainage, leachets etc., are to be assessed and addressed.

Mitigation Measures:

- Mitigation measures to reduce adverse effects include improving green belt, obtaining construction materials from other sources, usage of alternative construction materials like fly ash, where possible, storm water management, etc.
- Adopting soil improvement techniques and suitable designs, ground covering, etc.

c. Water Environment

Anticipated Impact:

- Impact of operations on surface waters, contamination due to cargo operations, impact on utility of surface water resources by the neighboring colonies, impact on surface water flow (ex. flooding) due to anticipated alterations/obstructions, etc.
- Discharge of trade effluent/sewage and its impact.
- Impact of project construction/operation on the ground water on account of leachets, run off from material and cargo storages and toxic or harmful substances, percolation, etc.

Mitigation Measures:

- Protection measures to surface water resources to prevent reduction in their quality due to construction and operational activities.
- Proposals to treat effluents confirming to standards notified under EC Act should be submitted.
- Mitigation measures to reduce adverse effects like impervious paving the cargo areas, impervious roads, lined drains, routing surface drainage to settlement tanks/pits, etc.
- Treatment of effluent, recycle/ reuse and disposal should be planned.
- Groundwater study on leaches should be carried out periodically and should be correlated with baseline data.
- Remedial measures should be taken in case of any deviation.
- Based on the total water budget of the project, the use of ground water should to be reviewed and alternatives to be presented.

d. Hydrology

Anticipated impact:

Impact of the project construction/operation on the hydrology on account of port construction should be assessed by suitable model studies.

Mitigation measures:

- Careful port design should be planned to minimize impact due to changes in current patterns and other hydrology.
- Model experiments or computer simulations of these changes are useful in developing an appropriate design.
- Shore protection works like construction of sea walls, sand by passing or bank nourishment should be studied.

e. Sediment Contamination

Anticipated Impact:

Impact of the project construction/operation on the bed sediment contamination on account of port construction/operations is to be assessed by suitable empirical/model studies.

Mitigation Measures:

A survey of contamination of bottom sediments should be undertaken before dredging.

f. Biological Environment

Anticipated impact:

- Impact of the project construction/operation on the river/marine/coastal ecology on account of port construction should be assessed by suitable empirical/model studies.
- Impact due to floodlights on the species should be studied.

Mitigation Measures:

Mitigation measures to reduce adverse effects should be provided.

g. Air Environment

Anticipated Impact:

- Impact of project construction/operation on the ambient air quality on account of emissions of dust during construction and cargo handling as well as emission of gases from equipment deployed for construction and cargo handling should be assessed.
- Prediction due to emissions during cargo handling/ emissions from the ships in the port area/ emissions due to increased traffic, emission inventory for critical pollutants with and without mitigation measures, prediction of the impact due to the existing activity on the proposed project, prediction of impact due to on going projects in the surrounding area on the proposed project and the ambient environment will be carried out.

Mitigation Measures:

- Mitigation measures proposed during the construction stage should include dust suppression measures by suitable techniques.
- Mitigation measures proposed during the operation stage should include alternative solutions such as closed conveyor system, closed silos, closed vehicles to transport dusty cargo, etc, mitigation measures to lower the emissions from the ships and green belt development.

h. Noise Pollution

Anticipated Impact:

Impact of project construction/operation on the noise and vibration on account of construction equipment, cargo handling equipment and road traffic.

Mitigation Measures:

Mitigation measures to comply the norms should be planned.

i. Solid Waste Management

Anticipated Impact:

- Details of municipal solid waste facilities, biomedical treatment facilities and hazardous waste disposal facilities in the area should be inventorised, in case if it is proposed to utilize the same.
- Impact due to non-hazardous and hazardous solid waste generated during the construction and operational stages should be assessed.

Mitigation Measures:

- Mitigation measures to comply the norms should be planned.
- Options for minimization of solid waste and environmentally compactable disposal/ recycling of waste to
- Conservation of natural resources should be planned.
- Management and disposal of temporary structures made during construction phase should be planned.

j. Socio-Economic and Occupational Health Environment

Anticipated Impact:

- Predicted impact on the communities of the proposed activity.
- Present status of health, housing, public utilities, commercial structures and transportation should be collected.
- Impact of the project on socio cultural aspects should be assessed.
- Socio-economic impact due to displacement of fishing settlements (if any) and population influx due to increased activities should be assessed.

Mitigation Measures:

- Mitigation measures to reduce adverse effects including satisfactory R&R methods should be planned.

20. Environmental Monitoring Programme

This Chapter will include details of environmental monitoring programme. It should include the technical aspects of monitoring the effectiveness of mitigation measures (including measurement methodologies, data analysis, reporting schedules, emergency procedures, detailed budget & procurement schedules).

21. Additional Studies

Risk Assessment, Risk Mitigation Measures, Disaster Management Plan, etc. will be prepared and submitted.

22. Environmental Cost Benefit Analysis

If recommended at the scoping stage, this chapter will be included for the environmental cost benefit analysis of the project.

23. Environmental Management Plan (EMP)

- EMP for Construction Phase: Land Environment, Water Environment, Air Quality, Noise Levels, Biological Environment and Socioeconomics.
- EMP for Operation Phase: Land Environment, Water Environment, Air

Quality, Noise Levels, Biological Environment and Socioeconomics.

- Occupational Health & Safety Measures.
- Green Belt Development.
- Rain Water Harvesting Measures.
- Project Cost & EMP Budget.
- CSR Budget.
- EMP Cell for Implementation of proposed EMP Measures.

24. Project Benefits

This chapter will include benefits accruing to the locality, neighborhood, region and nation as a whole. It should bring out details of benefits by way of:

- Improvements in the physical infrastructure by way addition of project infrastructure, ancillary industries that may come up on account of the project.
- Improvements in the social infrastructure like roads, railways, townships, housing, water supply, electrical power, drainage, educational institutions, hospitals, etc.
- Employment potential –skilled; semi-skilled and unskilled, both during construction and operational phases of the project with specific attention to employment potential of local population as well as necessity for imparting any specialized skills to them to be eligible for such employment in the project on a long term basis and Other tangible benefits like improved standards of living, health, education etc.

25. Public Consultation

The Public Consultations, both at Local and National Level, have to be conducted and the Proceedings, Issues raised, Response & Action Plan to address the Public Issues are to be prepared and disclosed.

26. Conclusions and Recommendations

27. Annexures

Annex 6-19: Structure of EMP

The features of an EMP proposed for BIFFL are:

1. Technical aspects of the project

- a. Final design
- b. Equipment
- c. Construction methods
- d. Construction: Contract Document
- e. Construction and operations of pollution control measures (Air, Liquid, Solid)
- f. Green
- g. Reuse/Recycling of wastes
- h. Schedule of implementation
- i. Estimates of capital and operational costs

2. Management Organization

- a. Personnel
- b. Resources (equipment, labs etc.)
- c. Environmental Monitoring Program (for Air, Water, Soil, Terrestrial/aquatic biology, which ever is applicable)

3. Ambient environmental quality monitoring

- a. Location
- b. Parameters
- c. Frequency

4. Effluent and emissions monitoring

- a. Air Emissions (stacks, parameters and frequency)
- b. Effluent (locations, parameters and frequency)
- c. Solid wastes-(Quality and composition of each type of waste and frequency)

5. Monitoring of environmentally significant parameters of fuels and raw materials (e.g. Sulphur and ash content in fuels, metal contents in mineral ores)

6. Data presentation and submission of reports

7. Estimate of annual costs

ANNEX 6-20: EFFLUENT DISCHARGE REQUIREMENTS BY WORLD BANK

Type of project	Parameter	Maximum values
General Manufacturing	pH	6-9
	BOD ₅	50
	COD	250
	TSS	50
	Oil and grease	10
	Phenol	0.5
	Cyanide	Total: 1 ; Free: 0.1
	Nitrogen	10 as ammonia
	Phosphorus	2
	Fluorine	20
	Chlorine	0.2
	Coliform	400 MPN/100 ml
	Temperature increase	≤ 3° C
	Sulfide	1
	Silver	0.5
	Arsenic	0.1
	Cadmium	0.1
	Chromium ⁺⁶	0.1
	Chromium	0.5
	Iron	3.5
	Mercury	0.01
	Nickel	0.5
	Lead	0.1
Thermal Power	Zinc	2
	Selenium	0.1
	Total metals	10
	pH	6-9
	TSS	50
	Oil and grease	10
	Total residual chlorine	0.2
	Temperature increase	≤ 3° C
	Chromium, total	0.5
	Copper	0.5
	Iron	1
	Zinc	1

Type of project	Parameter	Maximum values
Electronic Manufacturing	pH	6-9
	BOD	50
	TSS	50 (max.); 20 (monthly average);
	Oil and grease	10
	Cyanide	0.1 (free); 1 (total)
	Nitrogen	10 as ammonia
	Phosphorus	5
	Total chlorocarbons and hydrocarbons	0.5 (total)
	Fluoride	20
	Arsenic	0.1
	Cadmium	0.1
	Chromium ⁺⁶	0.1
	Copper	0.5
	Mercury	0.01
	Nickel	0.5
	Lead	0.1
	Tin	2
	Metal, total	10

All units in mg/l, except pH and otherwise specified

Note:

- BOD₅: Biochemical oxygen demand (understood as BOD measured over 5 days)
- COD: Chemical oxygen demand
- TSS: Total suspended solids
- MPN/100 ml: Coliform count expressed as most probable number per 100 milliliters

ANNEX 6-21: EMISSION DISCHARGE REQUIREMENTS BY WORLD BANK

Type of project	Parameter	Maximum values
General manufacturing	PM	50 for ≥ 50 Mwe ; 100 for < 50 Mwe
	SOx	2,000 as sulphur dioxide
	NOx	750 (260 ng/J or 365 ppm) for coal
		460 (130 ng/J or 225 ppm) for oil
		320 (86 ng/J or 155 ppm) for gas
	Dioxins: 2,3,7,8-TCSS equivalent	1 ng/J maximum
Thermal power, new plants	PM	50 for > 50 MWe ; 100 for < 50 Mwe
	SOx	0.2 tpd/Mwe (to 500 Mwe) ; 0.1 tpd/Mwe (over 500 Mwe) ; Not to exceed 2,000 mg/Nm ³ in flue gases ; Not to exceed 500 tpd
	NOx	Thermal Power Plant - Coal : 750 (260 ng/J or 365 ppm) ; Oil : 460 (130 ng/J or 225 ppm) ; Gas : 320 (86 ng/J or 155 ppm)
		Combustion turbine Unit - Gas : 125 ; Diesel fuel (No.2 oil) : 165 ; Fuel oil (No.6 and other) : 300
		Coal with $< 10\%$ volatile matter NOx is 1,500 mg/Nm ³
Thermal power plant, rehabilitation	PM	100 ; In rare cases, 150 mg/Nm ³ is acceptable
Electronic manufacturing	VOCs	20
	Phosphine	1
	Arsine	1
	Hydrogen fluoride	5
	Hydrogen chloride	10

All units in (mg/Nm³, unless otherwise specified

Note:

PM: Particulate matter
 SOx: Sulphur oxides
 NOx: Nitrogen oxides
 Mwe: Megawatts of electricity

ng/J: Nanograms per joule
 ppm: Parts per million
 tpd: Metric tons per day
 VOCs: Volatile organic compounds

ANNEX 6-22: NOISE LEVEL REQUIREMENTS BY WORLD BANK

Type of Project	Receptor	Maximum allowable log equivalent (hourly measurements), dB (A)	
		Day (07:00-22:00)	Night (22:00-07:00)
<i>Thermal power plant</i>	Residential		
	Institutional		
	Educational	55	45
	Industrial		
	Commercial	70	70
<i>Electronic manufacturing</i>	Residential		
	Institutional		
	Educational	55	45
	Industrial		
	Commercial	70	70

Source: Pollution Prevention and Abatement Handbook, World Bank, 1998

Note 1: dB (A) Decibels measured on the A scale

Note 2: Noise abatement measures should achieve either the levels given above or a maximum increase in background levels of 3 decibels (measured on the A scale) [dB (A)]. Measurements are to be taken at noise receptors located outside the project property boundary.

ANNEX 6-23: WORLD BANK SAFEGUARDS POLICIES (SOCIAL)

World Bank's Safeguard Policies

OP 4.12 Requirements (Involuntary Resettlement)

The primary objective of the World Bank policy on is to explore all alternatives to avoid or minimize involuntary resettlement. Where resettlement is unavoidable, the resettlement activities should be conceived and executed as sustainable development programs, providing sufficient resources to enable affected persons to share in project benefits and to assist in their efforts to improve their livelihood and standard of living or at least to restore them to pre-project level. The policy also requires that affected people are meaningfully consulted and should have opportunities to participate in planning and implementing resettlement programs. Measures required to ensure that resettlement has a positive outcome are as follows:

- Providing project-affected persons with options; permitting their participation in planning and selecting these options; prompt compensation at full replacement cost for losses; choosing relocation sites that provide, at a minimum, the same benefits as the sites that are replaced;
- Providing allowances and other assistance to make a smooth transition after displacement
- Identifying vulnerable groups and providing special assistance to these groups
- Implementing an institutional structure that supports the process to a successful end.

World Bank's policy on Involuntary Resettlement requires payment of compensation and other assistance to project affected people before they are displaced from their existing locations. Further, the policy requires income rehabilitation assistance to those affected due to the loss of their productive assets or loss of incomes and livelihood.

Absence of legal title does not exclude affected individuals from the eligibility to receive compensation and or other assistance. The Policy also requires that those without a legal title should be given assistance to meet the objectives of the policy. The genesis of these rights may come from continued possession of public land where the government has not sought eviction. The Bank's policy also recognizes that stakeholders who illegally occupy project-affected areas after established cut-off-date for any components are not eligible for compensation. Institutional or project specific policies are being prepared to address these gaps and to meet the requirements of multilateral agencies and best practices.

The principles for World Bank OP 4.12, involuntary resettlement are

- Involuntary resettlement should be avoided where feasible, or minimized, exploring all viable alternative project plans
- Where it is not feasible to avoid resettlement, resettlement activities should be conceived and executed as sustainable development programs.

- Displaced persons should be assisted in their efforts to improve their livelihoods and standards or at least restore them, in real terms, to pre-displacement levels or to levels prevailing prior to the beginning of project implementation, whichever is higher.

OP 4.10 Requirements (Indigenous Peoples)

This policy contributes to the WB's mission of poverty reduction and sustainable development by ensuring that the development process fully respects the dignity, human rights, economies, and cultures of. The Bank recognizes that the identities and cultures of Indigenous Peoples are inextricably linked to the lands on which they live and the natural resources on which they depend. These distinct circumstances expose Indigenous Peoples to different types of risks and levels of impacts from development projects, including loss of identity, culture, and customary livelihoods, as well as exposure to disease. As social groups with identities that are often distinct from dominant groups in their national societies, Indigenous Peoples are frequently among the most marginalized and vulnerable segments of the population. A project proposed for Bank financing that affects Indigenous Peoples requires:

- Screening by the Bank to identify whether Indigenous Peoples are present in, or have collective attachment to, the project area
- A social assessment by the borrower
- A process of free, prior, and informed consultation with the affected Indigenous Peoples' communities at each stage of the project, and particularly during project preparation, to fully identify their views and ascertain their broad community support for the project.
- The preparation of an Indigenous Peoples Plan (IPP) or an Indigenous Peoples Planning Framework (IPPF).
- Disclosure of the draft Indigenous Peoples Plan or draft Indigenous Peoples Planning Framework

The borrower makes the social assessment report and draft IPP/IPPF available to the affected Indigenous Peoples' communities in an appropriate form, manner, and language. Before project appraisals, the borrower sends the social assessment and draft IPP/IPPF to the Bank for review. Once the Bank accepts the documents as providing an adequate basis for project appraisal, the Bank makes them available to the public in accordance with *The World Bank Policy on Disclosure of Information*, and the borrower makes them available to the affected Indigenous Peoples' communities in the same manner as the earlier draft documents.

Comparison of Requirements of GOB And WB

While Bangladesh law provides for land acquisition, the regulations and rules are not very explicit regarding the resettlement and rehabilitation of Project Affected Persons (PAP). The World Bank OPs, on the other hand, are more comprehensive, requiring the following:

- Payment of adequate compensation for various losses at replacement value.
- Rehabilitation to ensure improvement/or at least restoration of lost economic activities, income and standard of living.
- Public consultation during the entire process of social assessment and disclosure of information (where ensuring that the information available to stakeholders is adequate and accessible is key)
- A grievance procedure that is accessible and adequately responsive (time for processing claims, etc) A Resettlement Action Plan, and where appropriate, an Indigenous Peoples' Development Plan

The table below lists some key comparisons between Bangladesh Law and WB policy regarding land acquisition and compensation.

Table 6-3: Comparison: Bangladesh Law and World Bank OP 4.12

Issue	1982 ORDINANCE	OP 4.12
Coverage	Legal owners Share-croppers Tenants	All affected parties, including squatters and illegal occupants
Compensation	Based on market values over previous 12 months	Replacement cost at current market price Requires livelihood restoration component.
Uses of material from dismantled structures	Material is to be auctioned after being compensated for it	Material can be taken and re-used by affected party
Minimization of impacts	Discourages unnecessary acquisition but no mechanism to monitor	Alternative analysis required to justify avoidance and/or mitigation of impacts
Cut-off dates	Not addressed	Important to ensure that squatters are included in compensation and to prevent rent-seeking behavior of additional squatters settling onto project land
Consultation	No consultation required	Consultation as core issue in RAP preparation and implementation
Relocation assistance	No assistance provided	Relocation assistance required
Livelihood restoration	Not addressed	Livelihood restoration component and attention to post-resettlement required

Addressing World Bank Safeguard Policies

The table below lists the relevant WB policies regarding social aspects such as land acquisition and compensation

Table 6-4: Addressing World Bank Safeguard Policies

World Bank Policy	Reasons of Applicability	Addressed by
Physical Cultural Resources OP 4.11	Project may pass through culturally significant areas – including areas that are an integral part of a people’s cultural identity and practices	Usually avoided by Govt. authorities while selecting the site
Involuntary Resettlement OP 4.12	Project may require physical relocation of people or entire communities and/or private property acquisition (land, commercial, residential and other structures) leading to resettlement.	A process of consultation, including adequate and accessible disclosure of any resettlement information. Preparation of a Resettlement Framework and compensation plan and entitlement policy and updating them in accordance with changes in the Project involving land acquisition. This includes an accessible grievance procedure.
Indigenous Peoples OP 4.10	Project affecting the dignity, human rights, economic and cultures of Indigenous Peoples because of acquisition of land and common resources on which they depend for their social, economic and cultural needs and also their homesteads requiring relocation.	Preparation of Indigenous Peoples’ Development Plan (IPDP) based on Social Impact Assessment that involved free, prior and informed consultations and updating it in accordance with changes in the Project that involves Indigenous Peoples.

Social Compliance Matrix

The table below has been designed considering OP 4.12 and OP 4.10 of World Bank. This will assist the project sponsor to comply with the requirement for social aspects for BIFFL funding.

Table 6-5: Social Compliance Matrix

Project Stage	Action required	Applicable standards and institutional responsibility
Pre-feasibility studies	Social screening For projects not requiring specific resettlement and land acquisition studies social performance frameworks will be designed	OP 4.12 para 27 For FIL projects requiring involuntary resettlement, the financial intermediary (FI) screens sub-projects to ensure consistency with OP 4.12. If the initial screening identifies indigenous peoples present, screening as stated in OP 4.10 will be undertaken.
	Public information campaign/ communication, consultation and participation	The project sponsor will carry out a public information campaign (which may be monitored by the relevant implementing agency) that will comply with OP 4.12
Project Design	Scoping: Identify significant potential impacts and project alternatives and propose terms of reference for SIA	WB OP 4.12 The project sponsor will identify the presence of significant potential impacts and assess against project alternatives and then propose terms of reference for the SIA
	Baseline Data Collection (SIA): Identify current socio-economic conditions without the project and update it during the project with changes	The project sponsor will carry out a sample socio-economic survey of both proposed project area as well as surrounding areas of the project that would have impacts. The socio-economic survey would have to comply to OP 4.12 Annex A
	Estimate social impacts	The project sponsor will estimate impacts based on OP 4.12 For IPDP, OP 4.10 Annex A will be considered.
	Preparation of SMP	The project sponsor to design plan for specific actions during land acquisition, engineering design and construction stages to conform to GoB requirement (Acquisition & Requisition of Property Ordinance 1982) as well as World Bank's requirements OP 4.12
	Preparation of RAP on the basis of SIA	The project sponsor will prepare a RAP following OP 4.12 Annex A
	Preparation of IPDP on the basis of SIA	The project sponsor will prepare IPDP (if indicated by the SIA) following OP 4.10 Annex B
	Design mitigation measures	The project sponsor will design mitigation measure in line with OP 4.12. For Indigenous peoples OP 4.10 will be consulted.
	Review and Approval of RAP	BIFFL is to review report to assess compliance with WB 4 OP.12 and if relevant OP 4.10.
Project Appraisal,		

Project Stage	Action required	Applicable standards and institutional responsibility
Approval		Also to check for compliance with GoB's requirements for compensation for LA.
Implementation Plan	Implement SMP	Appointed Agency (Consulting company or NGO) to implement
Engineering Design & Construction Works	Social Monitoring of SMP	Appointed Agency (Consulting company or NGO) to implement
Post-Construction	Social Audit and regular monitoring arrangements and evaluation	Selected Agency (Consulting company or NGO) to audit

ANNEX 6-24: RESETTLEMENT ACTION PLAN (RAP) PREPARATION

According to WB OP 4.12, the Resettlement Action Plan (RAP) for any project will cover the following elements.

1. Description of the project:

General description of the project and identification of the project area

2. Potential impacts:

Identification of:

- The project component or activities that give rise to resettlement
- The zone of impact of such component or activities
- The alternatives considered to avoid or minimize resettlement
- The mechanisms established to minimize resettlement, to the extent possible, during project implementation

3. Objectives:

The main objectives of the resettlement program

4. Socio-economic studies:

The findings of socio-economic studies to be conducted in the early stages of project preparation and with the involvement of potentially displaced people, to include:

- (a) The results of a census survey covering:
 - (i) Current components of the affected area to establish a basis for the design of the resettlement program and to exclude subsequent inflows of people from eligibility for compensation and resettlement assistance
 - (ii) Standard characteristics of displaced households, including a description of production systems, labor, and household organization; and baseline information on livelihoods (including, as relevant, production levels and income derived from both formal and informal economic activities) and standards of living (including health status) of the displaced population. Annex 9.1 is an illustration of the typical format of data collection for operation.
 - (iii) the magnitude of the expected loss – total or partial – of assets, and the extent of displacement, physical or economic. Annex 9.2 is an illustration of the typical format of data collection for such an operation
 - (iv) information on vulnerable groups or persons as provided for in OP 4.12, para. 8, for whom special provisions may have to be made
 - (v) provisions to update information on the displaced people's livelihoods and standards of living at regular intervals so that the latest information is available at the time of their displacement.

- (b) Other studies describing the following
 - (i) land tenure and transfer systems, including an inventory of common property natural resources from which people derive their livelihoods and sustenance, non-title-based usufruct systems (including fishing, grazing, or use of forest areas) governed by local recognized land allocation mechanisms, and any issues raised by different tenure systems in the project area;
 - (ii) the patterns of social interaction in the affected communities, including social networks and social support systems, and how they will be affected by the project;
 - (iii) public infrastructure and social services that will be affected; and
 - (iv) social and cultural characteristics of displaced communities, including a description of formal and informal institutions (e.g., community organizations, ritual groups, nongovernmental organizations (NGOs)) that may be relevant to the consultation strategy and to designing and implementing the resettlement activities.

5. Legal Framework:

The findings of an analysis of the legal framework will cover the following:

- (a) the scope of the power of eminent domain and the nature of compensation associated with it, in terms of both the valuation methodology and the timing of payment
- (b) the applicable legal and administrative procedures, including a description of the remedies available to displaced persons in the judicial process and the normal timeframe for such procedures, and any available alternative dispute resolution mechanisms that may be relevant to resettlement under the project;
- (c) relevant law (including customary and traditional law) governing land tenure, valuation of assets and losses, compensation, and natural resource usage rights; customary personal law related to displacement; and environmental laws and social welfare legislation;
- (d) laws and regulations relating to the agencies responsible for implementing resettlement activities;
- (e) gaps, if any, between local laws covering eminent domain and resettlement and the Bank's resettlement policy, and the mechanisms to bridge such gaps; and
- (f) any legal steps necessary to ensure the effective implementation of resettlement activities under the project, including, as appropriate, a process for recognizing claims to legal rights to land--including claims that derive from customary law and traditional usage (see OP 4.12, para.15 b).

6. Institutional Framework:

The findings of an analysis of the institutional framework will cover:

- (a) the identification of agencies responsible for resettlement activities and NGOs that may have a role in project implementation;
- (b) an assessment of the institutional capacity of such agencies and NGOs;
- (c) any steps that are proposed to enhance the institutional capacity of agencies and NGOs responsible for resettlement implementation.

7. Eligibility:

The definition of displaced persons and criteria for determining their eligibility for compensation and other resettlement assistance, including relevant cut-off dates.

8. Valuation of and compensation for losses:

The methodology to be used in valuing losses to determine their replacement cost; and a description of the proposed types and levels of compensation under local law and such supplementary measures as are necessary to achieve replacement cost for lost assets.

9. Resettlement measures:

A description of the packages of compensation and other resettlement measures that will assist each category of eligible displaced persons to achieve the objectives of the policy (see OP 4.12, para. 6). In addition to being technically and economically feasible, the resettlement packages should be compatible with the cultural preferences of the displaced persons, and prepared in consultation with them.

10. Site selection, site preparation and relocation:

Alternative relocation sites considered and explanation of those selected, covering

- (a) institutional and technical arrangements for identifying and preparing relocation sites, whether rural or urban, for which a combination of productive potential, locational advantages, and other factors is at least comparable to the advantages of the old sites, with an estimate of the time needed to acquire and transfer land and ancillary resources;
- (b) any measures necessary to prevent land speculation or influx of ineligible persons at the selected sites;
- (c) procedures for physical relocation under the project, including timetables for site preparation and transfer; and
- (d) legal arrangements for regularizing tenure and transferring titles to resettlers.

11. Housing, infrastructure, and social services:

Plans to provide (or to finance resettlers provision of) housing, infrastructure (e.g., water supply, feeder roads), and social services (e.g., schools, health services); plans to ensure comparable services to host populations; any necessary site development, engineering, and architectural designs for these facilities.

12. Environmental protection and management:

A description of the boundaries of the relocation area; and an assessment of the environmental impacts of the proposed resettlement and measures to mitigate and manage these impacts (coordinated as appropriate with the environmental assessment of the main investment requiring the resettlement).

13. Community participation, Involvement of resettlers and host communities:

This includes the following components:

- (a) A description of the strategy for consultation with and participation of resettlers and hosts in the design and implementation of the resettlement activities;
- (b) a summary of the views expressed and how these views were taken into account in preparing the resettlement plan;
- (c) a review of the resettlement alternatives presented and the choices made by displaced persons regarding options available to them, including choices related to forms of compensation and resettlement assistance, to relocating as individuals families or as parts of preexisting communities or kinship groups, to sustaining existing patterns of group organization, and to retaining access to cultural property (e.g. places of worship, pilgrimage centers, cemeteries)
- (d) institutionalized arrangements by which displaced people can communicate their concerns to project authorities throughout planning and implementation, and measures to ensure that such vulnerable groups as indigenous people, ethnic minorities, the landless, and women are adequately represented.

14. Integration with host populations:

Measures to mitigate the impact of resettlement on any host communities, including

- (a) consultations with host communities and local governments;
- (b) arrangements for prompt tendering of any payment due the hosts for land or other assets provided to resettlers ;
- (c) arrangements for addressing any conflict that may arise between resettlers and host communities; and
- (d) any measures necessary to augment services (e.g., education, water, health, and production services) in host communities to make them at least comparable to services available to resettlers.

15. Grievance procedures:

Affordable and accessible procedures for third-party settlement of disputes arising from resettlement; such grievance mechanisms should take into account the availability of judicial recourse and community and traditional dispute settlement mechanisms.

16. Organizational responsibilities:

The organizational framework for implementing resettlement, including identification of agencies responsible for delivery of resettlement measures and provision of services; arrangements to ensure appropriate coordination between agencies and jurisdictions involved in implementation; and any measures (including technical assistance) needed to strengthen the implementing agencies' capacity to design and carry out resettlement activities; provisions for the transfer to local authorities or resettlers themselves of responsibility for managing facilities and services provided under the project and for transferring other such responsibilities from the resettlement implementing agencies, when appropriate.

17. Implementation schedule:

An implementation schedule covering all resettlement activities from preparation through implementation, including target dates for the achievement of expected benefits to resettlers and hosts and terminating the various forms of assistance. The schedule should indicate how the resettlement activities are linked to the implementation of the overall project.

18. Costs and budget:

Tables showing itemized cost estimates for all resettlement activities, including allowances for inflation, population growth, and other contingencies; timetables for expenditures; sources of funds; and arrangements for timely flow of funds, and funding for resettlement, if any, in areas outside the jurisdiction of the implementing agencies.

19. Monitoring and evaluation:

Arrangements for monitoring of resettlement activities by the implementing agency, supplemented by independent monitors as considered appropriate by the Bank, to ensure complete and objective information; performance monitoring indicators to measure inputs, outputs, and outcomes for resettlement activities; involvement of the displaced persons in the monitoring process; evaluation of the impact of resettlement for a reasonable period after all resettlement and related development activities have been completed; using the results of resettlement monitoring to guide subsequent implementation.

Abbreviated Resettlement Plan

Where the impacts on the entire displaced population are minor, or fewer than 200 people are displaced, an abbreviated resettlement plan may be agreed with the project sponsor. This is in line with OP 4.12 para. 25. Information disclosure procedures set forth in para. 22 are applicable.

An abbreviated plan covers the following minimum elements:

- I. a census survey of displaced persons and valuation of assets;
- II. description of compensation and other resettlement assistance to be provided
- III. consultations with displaced people about acceptable alternatives;

- IV. institutional responsibility for implementation and procedures for grievance redress
- V. arrangements for monitoring and implementation
- VI. a timetable and budget.

Indigenous Peoples Development Plan (IPDP)

If the SIA indicates the need for an Indigenous Peoples Development Plan IPDP, then an IPDP is will be prepared by the project sponsor through social experts. The level of detail will vary depending on the nature of the sub-project and the nature of the effects to be addressed. The IPDP will include the following elements, as mentioned in the WB OP 4.10.

- I. A summary of the information referred to in the OP.
- II. A summary of the social assessment.
- III. A summary of results of the free, prior, and informed consultation with the affected Indigenous Peoples' communities that was carried out during project preparation and that led to broad community support for the project.
- IV. A framework for ensuring free, prior, and informed consultation with the affected Indigenous Peoples' communities during project implementation (see paragraph 10 of this policy).
- V. An action plan of measures to ensure that the Indigenous Peoples receive social and economic benefits that are culturally appropriate, including, if necessary, measures to enhance the capacity of the project implementing agencies.
- VI. When potential adverse effects on Indigenous Peoples are identified, an appropriate action plan of measures to avoid, minimize, mitigate, or compensate for these adverse effects.
- VII. the cost estimates and financing plan for the IPP.
- VIII. Accessible procedures appropriate to the project to address grievances by the affected Indigenous Peoples' communities arising from project implementation. When designing the grievance procedures, the borrower takes into account the availability of judicial recourse and customary dispute settlement mechanisms among the Indigenous Peoples.
- IX. Mechanisms and benchmarks appropriate to the project for monitoring, evaluating, and reporting on the implementation of the IPP. The monitoring and evaluation mechanisms should include arrangements for the free, prior, and informed consultation with the affected Indigenous Peoples' communities.

Grievance Redress Mechanism

Despite best efforts to arrive at fair rewards in project involving involuntary resettlement, there will always be a few unsatisfied citizens. The relevant authority will make efforts at project level to resolve grievances through negotiations involving community leaders and PAP's representatives. Grievance Resolution Committee: In case the dispute is not resolved at local level, the matter through submission of a relevant formal reference would be placed before a Grievance Resolution Committee. Grievance Resolution Committee (GRC) will be constituted consisting of a panel of five (5) Members, one of whom will be community relation officer, to be appointed by the concerned authority. Other members would include two (2) representatives from the PAPs community of the project area, those who are publicly known to be persons of integrity, good judgment and command respect.

ANNEX 6-25: WORLD BANK ENVIRONMENTAL CLEARANCE PROCESS

1. Environmental Screening

The environmental screening of each proposed project is undertaken to determine the appropriate extent and type of EA. A preliminary EA, also called EIA (Environmental Impact Assessment) is conducted for each project to screen and set the project into Environmental Assessment category.

World Bank classifies the proposed project into one of following four categories, depending on the type, location, sensitivity, and scale of the project and the nature and magnitude of its potential environmental impacts.

(a) Category A: A proposed project is classified as Category A if it is likely to have significant adverse environmental impacts that are sensitive, diverse or unprecedented. These impacts may affect an area broader than the sites or facilities subject to physical works.

EA for a Category A project examines the project's potential negative and positive environmental impacts, compares them with those of feasible alternatives (including the "without project" situation), and recommends any measures needed to prevent, minimize, mitigate, or compensate for adverse impacts and improve environmental performance. For a Category A project, the borrower is responsible for preparing a report, normally an EA.

(b) Category B: A proposed project is classified as Category B if its potential adverse environmental impacts on human populations or environmentally important areas—including wetlands, forests, grasslands, and other natural habitats—are less adverse than those of Category A projects. These impacts are site-specific; few if any of them are irreversible; and in most cases mitigation measures can be designed more readily than for Category A projects. The scope of EA for a Category B project may vary from project to project, but it is narrower than that of Category A EA. Like Category A EA, it examines the project's potential negative and positive environmental impacts and recommends any measures needed to prevent, minimize, mitigate, or compensate for adverse impacts and improve environmental performance. The findings and results of Category B EA are described in the project documentation (Project Appraisal Document and Project Information Document).

(c) Category C: A proposed project is classified as Category C if it is likely to have minimal or no adverse environmental impacts.

Beyond screening, no further EA action is required for a Category C project.

(d) Category F: A proposed project is classified as Category FI if it involves investment of Bank funds through a financial intermediary, in subprojects that may result in adverse environmental impacts.

2. Environmental Impact Assessment (EA or EIA)

Breadth, depth, and type of analysis in the EA process depend on the nature, scale, and potential environmental impact of the proposed project. EA evaluates a project's potential environmental risks and impacts in its area of influence; examines project alternatives; identifies ways of improving project selection, planning, design, and implementation by preventing, minimizing, mitigating, or compensating for adverse environmental impacts and enhancing positive impacts; and includes the process of mitigating and managing adverse environmental impacts throughout project implementation. World Bank favors preventive measures over mitigation or compensatory measures, whenever feasible.

1. EA takes into account the natural environment (air, water, and land); human health and safety; social aspects (involuntary resettlement, indigenous peoples, and physical cultural resources); and trans-boundary and global environmental aspects. EA considers natural and social aspects in an integrated way. It also takes into account the variations in project and country conditions; the findings of country environmental studies; national environmental action plans; the country's overall policy framework, national legislation, and institutional capabilities related to the environment and social aspects; and obligations of the country, pertaining to project activities, under relevant international environmental treaties and agreements. World Bank does not finance project activities that would contravene such country obligations, as identified during the EA. EA is initiated as early as possible in project processing and is integrated closely with the economic, financial, institutional, social, and technical analyses of a proposed project.
2. The borrower is responsible for carrying out the EA. For Category A projects, the borrower retains independent EA experts not affiliated with the project to carry out the EA. For Category A projects that are highly risky or contentious or that involve serious and multidimensional environmental concerns, the borrower should normally also engage an advisory panel of independent, internationally recognized environmental specialists to advise on all aspects of the project relevant to the EA. The role of the advisory panel depends on the degree to which project preparation has progressed, and on the extent and quality of any EA work completed, at the time the Bank begins to consider the project.
3. World Bank advises the borrower on the Bank's EA requirements. World Bank reviews the findings and recommendations of the EA to determine whether they provide an adequate basis for processing the project for Bank financing. When the borrower has completed or partially completed EA work prior to the Bank's involvement in a project, World Bank reviews the EA to ensure its consistency with this policy. World Bank may, if appropriate, require additional EA work, including public consultation and disclosure.

Impacts as found, assessed and recommended in the EIA report are taken up in the Environment Management and Monitoring Plan (EMP) for addressing or mitigation. The EMP, if not already a part of the EA; and the EA, need prior approval of World Bank for accepting a loan proposal for the project. The EMP should be integrated in the bidding documents.

3. Contents of an EA report

Contents of an EA report **for a category –A project** is given in OP 4.01, Annex B outlines of which in brief are as under:

- 1) Executive summary: findings & recommendations.
- 2) Policy, legal and administrative framework: frame work in which EA is carried out, environmental requirements of co-financiers, country's international environmental agreements.
- 3) Project description: the project and its geographic, ecological, social and temporal context-off site investments for pipeline, access roads, power plants, water supply, housing, raw material and storage facilities-RAP,IPP, site and influence area map.
- 4) Baseline data: biological, physical, socioeconomic-changes anticipated before commencement,-current and proposed development activities- relevancy and indicate accuracy, reliability and sources of data.
- 5) Environmental Impacts: project's positive and negative impacts (quantitative)-mitigation measures and residual immitigable impacts-opportunities for environmental enhancement-extent and quality of available data, key data gaps and uncertainties.
- 6) Analysis of alternatives: compare feasible alternatives to project's site, technology, design and operation- without project- feasibility of migrating these impacts, their capital and current costs, suitability to local conditions-institutional, training and monitoring needs –basis for selecting a particular project design, justify recommended emission levels and approaches to pollution prevention and abatement.
- 7) Environmental Management plan: mitigation measures, monitoring and institutional strengthening
- 8) Appendixes
- 9) List of EA report papers- individuals and organizations.
- 10) References- published and unpublished materials
- 11) Records of interagency and consultation meetings-views of affected people and NGOs/ surveys.
- 12) Data Tables
- 13) List of associated reports-RAP,IPP

4. Contents of EMP

EMP consists of a set of mitigation, monitoring and institutional measures to be taken during implementation and operation to eliminate adverse environmental and social impacts, offset them or reduce them to acceptable levels. The plan includes actions needed to implement these measures. The Management Plans are essential elements of EA reports for category-A projects. For many category-B projects the EA results in an EMP only. The contents of an EMP are given in OP 4.01, Annex-C and in brief areas under:

a) Mitigation:

Feasible, cost effective and sufficient measures-else; compensatory measures.

- EMP identifies and summarises all anticipated significant adverse environmental impacts including those involving indigenous people or involuntary resettlement.
- describes each measure with technical details- type of impact and condition (required continuously or in contingencies) designs, equipments, operating procedure
- potential impacts of these measures
- linkage with other mitigation plans

b) Monitoring:

During implementation- impacts and effectiveness of mitigation measures-evaluate success, corrective actions needed. This section provides (a) a specific description and technical details of monitoring measures, parameters to be measures, methods, sampling locations, frequency of measurements, detection limits thresholds for corrective actions and (b) monitoring and reporting procedures for early detection of conditions and reporting progress and mitigation results.

c) Capacity Development and Training:

To support timely and effective implementation- existence, role and capability of environmental units on site, at the agency, and ministry level (as assessed in EA).- need for establishment or expansion for carrying out the mitigatory and monitoring measures. Most EMPs cover (a) technical assistance programs (b) procurement of equipment and supplies (c) organizational changes

d) Implementation Schedule and cost estimate:

For all the 3 aspects (mitigation, monitoring and capacity development) provide (a) implementation schedule-phasing, coordination with implementation plans (b) capital and current cost and sources of fund.

e) Integration of EMP with project:

Decision of both the borrower and the Bank to proceed with the project rest on the expectation that the EMP will be executed effectively. So the plan to describe specifically individual mitigation and monitoring measures and its assignment of

institutional responsibilities and be integrated into project's over all planning, design, budget, and implementation.

5. Environment Management System (EMS)

Environment Management System (EMS) of an organization refers to the tool the organization employs toward management of its environmental programs in a comprehensive, systematic, planned and documented manner.

Such a tool enables an organization to achieve and demonstrate sound environmental performance by controlling the impacts of its activities and services on the environment in the context of the available legislation, development of economic policies and other measures that foster environmental protection, and increased concern expressed by interested parties about environmental matters and sustainable development. The EMS of an organization, as such, embodies the organizational structure, planning and resources for developing, implementing and maintaining policy for environmental protection.

Environment Management System (EMS) of an organization, hence, of necessity, has the following objectives:

- To serve as a **tool to improve the environmental performance** of the organization;
- To provide a **systematic way of managing** the organization's environmental affairs;
- To ensure that the overall **management structure** of the organization **adequately addresses immediate and long-term impacts** of its projects, products, activities, services and processes on the environment;
- To provide **order and consistency** for the organization to address its environmental objectives and concerns through the allocation of resources, assignment of responsibility and ongoing evaluation of practices, procedures and processes;
- To provide for **regular Environmental Audits**.
- Focuses on **continual improvement** of the system.

ANNEX 6-26: ENVIRONMENTAL IMPACT AND MITIGATION MEASURES

1. Overview

The analysis of the environmental impacts is an important tool in determining the appropriate mitigation measures to be employed. Identifying the related impacts is an exhaustive process that always has to be updated. The most important impact for the projects is the activities and its associated effects on the environment.

This section gives a broad overview of the probable impacts that might occur during the project implementation and execution, including a general indication to the seriousness or significance of potential impacts. It also discusses the possible mitigation measures that can be put in place to mitigate any negative impacts and enhance the positive ones.

2. Potential Impacts

Considering the interaction between engineering operations and activities and the environment, the potential impacts occurring in the following three phases are considered:

- i) Pre – Construction (planning and design)
- ii) Construction
- iii) Operation & Maintenance

It is recommended that the impact assessment measures the performance during all the three phases against the baseline position.

Pre-Construction (Planning and Design) Phase

The location of industry in an environmentally sensitive area such as a wildlife sanctuary, game reserve, protected wetland, forest or area of cultural importance has the potential to cause adverse impacts. The table below represents the framework for assessing the sensitivity of a sub-project depending upon its location.

Table 6-6: Site Location and Sensitivity

	Environmental Issues	Low Sensitivity	Medium Sensitivity	High Sensitivity
1	Natural Habitats	No critical natural habitats; no other natural habitats	No critical natural habitats; other non-forest natural habitats present	Critical natural habitats ³ along with unprotected natural habitats of known high importance for biodiversity conservation.
2	Induced Development	Area or region is well consolidated; dense road network	Medium density road network; open lands are still available	Area or region is undeveloped; road network is incipient or non-existent

³ Critical natural habitats are defined as existing and proposed protected areas

	Environmental Issues	Low Sensitivity	Medium Sensitivity	High Sensitivity
3	Soil Stability / Erosion	Flat terrain; no potential erosion problems.	Medium slopes; some erosion potential	Mountainous terrain; high slopes; unstable soil formations; high erosion potential

Construction Phase

Generally, construction impacts are expected to affect the environment for a relatively short time period and these are expected to cease soon after the completion of construction. The potential major impacts are described herein:

i. Impact on Flora and Fauna

The construction of buildings and other infrastructure could result in clearing and depletion of vegetation that will result in loss of plant cover, disturbance and loss of fauna habitats, weakening and degradation of soils, disturbance of the natural landscape and disfiguring of the natural morphology.

ii. Impact on Soil and Land Degradation

Earthmoving equipments such as excavators are likely to be used in cutting/excavation works. These equipments may expose the soil to erosion and also compact it and may break down the soil structure, which may potentially decrease the drainage of the areas. Furthermore, the risks of accidental discharge of hazardous products, leakage of hydrocarbons, oils or grease from construction machinery also constitute potential sources of soil and water pollution.

iii. Impact on Vehicular Traffic

Construction work may result in a high traffic volume within the area. Wastes generated from project activities such as cement bags, paint drums and debris may result in pollution and constitute obstructions to vehicular traffic.

iv. Impact on Waste Management

Activities at construction sites may produce construction wastes such as excavated soils and debris. In addition, unmanaged waste can have a significant indirect economic impact by reducing the price of properties nearby and in general, by affecting the aesthetics, child- friendliness and overall condition of the environment.

v. Impact on Slope, Erosion and Drainage

If the topography of the project area is hilly, erosion problems during construction are likely to be more severe, as compared to a flat area. However, if the area is flat, water may not drain away easily, in general, and stagnant pools of water may get created. These pools, if not drained regularly may provide favorable breeding grounds for mosquitoes.

vi. Impact on Air Quality

Air quality will be impacted by emissions from vehicles, fossil-fuel fired power plants, combustion devices, earthmoving equipment and released particulate matter.

vii. Impact on Water Quality

Water quality may be impacted by wastewater discharge from construction activities. These will include discharge from onsite sewage system and rainwater run-off. The discharge of this wastewater into surface water may impact water quality by causing changes to its physical, chemical and biological properties.

Given the high anticipated volume of waste/spoil that may be generated, it is likely that the waste may get stockpiled on road sides. If it is not properly contained, rains could carry it along with runoff into other surface waters, leading to increased turbidity and siltation. This could affect aquatic resources such as fisheries etc.

Operation and Maintenance Phase

Operational impacts continue during the life of the project after the completion of construction phase and may last for a longer time. Operation of industries emits a number of pollutants that need to be carefully mitigated and monitored. Main pollutants during the operation phase are as follows:

a) Liquid Wastes

- Process Water from the industries
- Domestic Sewage
- Oils & Grease
- Contaminated Water
- Solvents

b) Air Emission

- Smoke
- Volatile organic compounds
- Particulates
- Oxides of Nitrogen
- Sulphur dioxide
- Carbon Dioxide

c) Smell

- Effluent discharge
- Use of chemicals

d) Noise

- Use of machineries

e) Solid Wastes

- Packaging materials
- Raw material residues
- Container residues
- Scrap metals
- Sludge containing various toxic metals

f) Occupational Health & Safety

- Water pollution hazards
- Potential Fire Hazards
- Handling of Chemicals
- Heat Generation
- Air pollution hazards

g) Visual impact

- Construction of industries and buildings
- Smoke

3. Major Environmental Impacts

Following table (Major Environmental Impacts) gives an overview of the major environmental impacts resulting from the pre-construction, construction and operation of BIFFL Projects.

Table 6-7: Major Environmental Impacts

SL	Parameters	Impacts
1	Soil	▪ Possible increase in soil erosion as a result of the construction activities (clearing of vegetation and soil excavation) ▪ Loss of productive topsoil resulting from site clearance ▪ Soil contamination resulting from the release of chemicals (lubricant, fuel, paint) from the machinery
2	Water	▪ Run-off erosion may occur from unprotected excavated areas during heavy rain resulting to sedimentation of the nearby water-bodies ▪ Potential pollution of the water bodies through run-off of hazardous construction waste (lubricants, cements, paint and fuels) ▪ Contamination of surrounding water bodies through improper disposal of industrial effluent and sewage during operation ▪ Possible run off from the temporary solid waste storage site into the drainage system (during the project operational phase)
3	Ecosystem	▪ Destruction of vegetation, loss of habitat and loss of biodiversity ▪ Possible loss of endangered and rare species ▪ Change in land use and disturbance of ecosystem ▪ Contamination of biota
4	Air	▪ Particulate matters emission from construction activities ▪ Emission of pollutants (NO_x, SO_x, CO) from the construction ▪ Machineries ▪ Possible release of micro organisms to the air during the operational phase of the project
5	Noise	▪ Construction equipment will generate noise above the usual level
6	Landscape	▪ Change in natural drainage pattern ▪ Destruction of vegetation and trees ▪ Deforestation and desertification ▪ Visual impact

4. Assessment of Impacts

This section contains a discussion of the possible environmental effects of the BIFFL sub-projects for the specific issue areas that were identified as having the potential to experience significant impacts. “Significant effect” can be defined as ‘a substantial, or potentially substantial, adverse change in any of the physical conditions within the area affected by the project including land, air, water, minerals, flora, fauna, ambient noise, and objects of historic or aesthetic significance.’

An economic or social change by itself will not be considered a significant effect on the environment, but may be considered in determining whether the physical change is significant.” In the impact analysis, significance thresholds are those criteria which are universally recognized, or are developed specifically for a single project analysis to determine whether potential effects are significant.

Environmental impacts of the sub-projects can broadly be classified as those taking place during pre-construction, construction and operational phases. The overall strategy has the following sequence:

Types of Impacts are grouped into four classes. These are:

i. **Class I: Significant and Unavoidable**

An impact that cannot be reduced to below the threshold level, given reasonably available and feasible mitigation measures. Significance thresholds are those criteria which are developed specifically to determine whether potential effects are significant.

ii. **Class II: Significant but Mitigable**

An impact that can be reduced to below the threshold level, given reasonably available and feasible mitigation measures.

iii. **Class III: Not Significant**

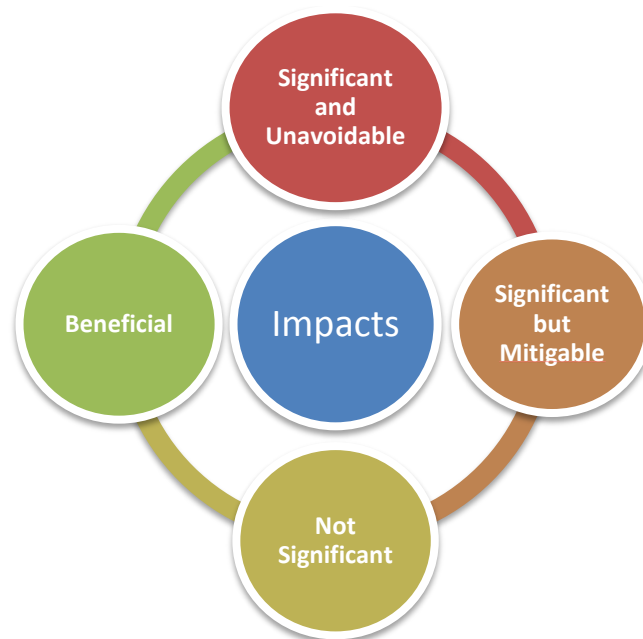
An impact that may be adverse, but does not exceed the threshold levels and does not require mitigation measures. However, mitigation measures that could further lessen the environmental effect may be suggested if readily available and easily achievable.

iv. **Class IV: Beneficial**

An effect that would reduce existing environmental problems or hazards.

In those cases where the mitigation measure for an impact could have a significant environmental impact in another issue area, this impact is discussed as a residual effect. The impact analysis concludes with a discussion of cumulative impacts, which evaluates the impacts associated with the proposed project in conjunction with other future development in the area.

Figure 6-1: Impact Structure



5. Cumulative Impacts

Cumulative impacts can be defined as impacts, which potentially develop from the combined impacts of more than one project or large scale program occurring within the same area of influence and time span. In such cases, cumulative impacts will have to be assessed based on the combined effects of potential impacts from the various program inputs. The cumulative impacts may occur due to the following environmental parameters:

- Change in land use
- Traffic
- Industrial wastewater effluent
- Solid waste disposal
- Noise

6. Mitigation Measures

This section includes the principles, procedures and mitigation measures that are relevant and suitable to the project for ensuring the most appropriate environmental mitigation and enhancement plans applicable during different stages of project implementation. To avoid and minimize the impacts resulting from the activities of different sub-projects, measures/management plans, which are essential to mitigate the impacts as discussed, are based upon appropriate technological design, improvements or adjustments, policy including good site operational practices etc.

Therefore, the mitigation plan has been adopted to highlight the action procedures to avoid/minimize/control the resultant impacts arising out of the different project phases i.e. pre-construction, construction and operation which are described in the following table.

Table 6-8: Mitigation Measures

Issue	Mitigation Measure	Time Frame	Responsibility	
			Implement ation	Supervision /Monitoring
A. CONSTRUCTION PHASE				
a) Soil				
Disposal from excavated earthworks	- Earthwork cuts will be used where possible - Residual spoil will be disposed of according to the erosion prevention plan	During Construction	Project Sponsor	BIFFL
Loss of Top Soil	- Topsoil from all areas to be permanently covered will be stripped, stored and used for re-vegetation works; - Construction vehicles, machinery, and equipment will move or be stationed in designated areas. Access to adjacent agricultural land will be minimized	During Construction	Project Sponsor	BIFFL
Soil erosion and Water Pollution	- Measures as per design or as directed by the Consultants to control soil erosion. Earth materials and stone will be properly disposed of so as not to block rivers, resulting in adverse impact on water quality. - Measures will be taken to prevent earth works - Mitigation measures will be taken to prevent the untreated wastewater produced in construction from entering into creek and streams	During Construction	Project Sponsor	BIFFL
b) Water Resources				
Siltation of water bodies	Measures to prevent erosion will be taken in accordance with the erosion prevention plan	During Construction	Project Sponsor	BIFFL
Contaminated runoff from inappropriately stored hazardous materials	Solid hazardous and non hazardous wastes will be appropriately stored to prevent contaminated runoff to adjacent waterways	During Construction	Project Sponsor	BIFFL
c) Air Quality				
Generation of Dust	- Water will be sprayed on construction sites, exposed sites and earth roads to minimize dust; - Concrete batching plants and crushing plants will be sited at least 500 m from the nearest habitation and fitted with dust extraction units in compliance with national	During Construction	Project Sponsor	BIFFL

Issue	Mitigation Measure	Time Frame	Responsibility	
			Implement ation	Supervision /Monitoring
	air quality standards			
Emissions	- All emissions will meet standards. - Developer will submit dust suppression program before construction	During Construction	Project Sponsor	BIFFL
d) Noise Quality				
Noise from vehicles, plant, and earth moving equipment	- Within 200 m of the nearest habitation, construction work will be restricted to between 0600 to 2100 hours. - Maintenance of machinery and vehicles should be enhanced to keep their noise within acceptable level	During Construction	Project Sponsor	BIFFL
e) Impact on Flora				
Loss of or damage to Vegetation	Minimum damage or disruption to the flora. Trees or shrubs will be removed that impinge directly on the permanent works or necessary temporary works.	During Construction	Project Sponsor	BIFFL
f) Impact on Fauna				
Loss or Damage or Disruption of Fauna	All works are to be carried out in such a fashion that the damage or disruption of the fauna is minimum.	During Construction	Project Sponsor	BIFFL
g) Waste Management				
Generation of Solid wastes	Solid waste generated to be accumulated at one place and treated.	During Construction	Project Sponsor	BIFFL
B. OPERATION & MAINTENANCE PHASE				
Industrial Effluent	- All recommendations on ESMP will be implemented. - All industrial effluent will be directed to central effluent treatment plant before disposal - Industrial Effluent will be treated to applicable standards	Throughout Operation Period	Project Operator	BIFFL
Air Emission	- All recommendations of ESMP will be implemented. - Suitable dust collectors (bag filters) will be provided at all major emission source. - For dispersion of the hot air from boiler stack, it is recommended that the height of the stack chimney should be at least 2.5 times of the nearest height of the building - The effluents will be treated to applicable standards		Project Operator	BIFFL
Noise	All recommendations of ESMP will be implemented.	Throughout Operation	Project Operator	BIFFL

Issue	Mitigation Measure	Time Frame	Responsibility	
			Implement ation	Supervision /Monitoring
	• Proper design and sound management practice will be adopted • The noise emission will conform to applicable standards • Green belts as sound barriers will be established. • Monitoring will be conducted during the operational phase to confirm noise-modeling results.	Period		
Solid Waste	• All materials will be disposed through central solid waste management system	Throughout Operation Period	Project Operator	BIFFL
Occupational Health	• Occupational, Health and Safety Program will be adopted. • Review and evaluate to improve the effectiveness of Environmental Health and Safety Program	Throughout Operation Period	Project Operator	BIFFL

ANNEX 6-27: GOB SOCIAL MANAGEMENT PROCESS

1. Legal and Institutional Framework

The table below lists the relevant National laws relevant for the project sponsor to comply with and the government agencies responsible for different aspects of the social management for sub-project to be funded by BIFFL.

Table 6-9: Relevant National Legislation

Sl. No	Laws/Regulations	Enforcing Agencies Ministry/Authorities
1.	Antiquities Act 1968	Ministry of Cultural Affairs
	<i>Regulated/ Enforced Items</i> This legislation governs preservation of the national cultural heritage, protects and controls ancient monuments, regulates antiquities as well as the maintenance, conservation and restoration of protected sites and monuments, controls planning, exploration and excavation of archaeological sites.	
2.	The Building Construction Act 1952 (with amendments)	Ministry of Works
	<i>Regulated/ Enforced Items</i> An Act to provide for the prevention of haphazard construction of building and excavation of tanks which are likely to interfere with the planning of certain areas in Bangladesh.	
3.	The Land Acquisition Act, 1894; The Acquisition and Requisition of Immovable Property Ordinance 1982 and subsequent amendments in 1994, 1995 and 2004	Revenue Department
	<i>Regulated/ Enforced Items</i> Current GoB Act & guidelines, relating to acquisition of land	
4.	The Factories Act, 1965 and Bangladesh Labor Law, 2006	Ministry of Labor
	<i>Regulated/ Enforced Items</i> This Act pertains to the occupational rights and safety of factory workers and the provision of a comfortable work environment and reasonable working conditions.	

2. Acquisition and Requisition Ordinance, 1982

In 1982, the Acquisition and Requisition of Immovable Property Ordinance came in force. This law replaced the earlier law, The Land Acquisition Act, 1894 (I of 1894) and others that have been in force between 1947 and 1982, and is the major basis for all the present actions regarding acquisition, resettlement and rehabilitation. The relevant and salient points of the 1982 law are as follows:

"Property" means only immovable property (land and building) and includes any right in such property and "Owner" includes the occupiers.

Matters to be considered in determining compensation include

- the market value of property
- damage to standing crops or trees due to acquisition
- damage due to severance of acquired property from other property at the time of actual taking of permission by concerned authorities
- damage to other properties or earnings
- expenses for relocation of residence
- damage due to lowering of profit of the property to be acquired between the serving of acquisition notice and actual acquisition.

While the rules under the Act 'appear' fairly generous, problems often arise in implementing the provisions of the Act. Determining market value and damage to property or income are difficult to establish in normal circumstances. It becomes more so in an emotion charged situation such as acquisition of land in a land-scarce and agricultural-dependent society. Arbitrariness is almost certainly likely to result. The situation is not helped by assigning by the government to itself sweeping powers in matters of acquisition. Read with the Article 42, sub clause (2) of the Constitution referred to above, there is every chance that the acquisition may not be entirely necessary, that compensation may be rather inadequate or paid such later than actual acquisition.

To sum up the legal context, while there are constitutional and legal provisions for acquisition and payment of compensation for acquisition of land and other immovable property for state-determined purposes, there are also uncertainties in the actual practice because of lack of clear guidelines, sweeping power given to Deputy Commissioners who actually serve notice and take possession of acquired property and the denial of legal instruments to citizens against such actions. The present laws, acts, regulations and rules are not very explicit regarding resettlement and rehabilitation of project-affected people (PAP). One can, however, as indicated earlier interpret the spirit and letter of these laws (including constitutional provisions) and rules to imply that resettlement and rehabilitation of people adversely affected due to the project is the responsibility of the project authorities.

As a result, in recent years there have been several instances where attempts have been made and are being made to take affirmative action. In all such cases the core idea is to prepare an entitlement matrix based upon an entitlement policy. Entitlement here means the rights of the persons adversely affected by the project to receive certain benefits from the project authorities to compensate for their losses that may include land and other immovable property, income, standing crops, occupation etc. The compensation is often in terms of cash grants but also includes training and credit facilities and other necessary facilities in resettlement and rehabilitation.

3. Land Acquisition for Projects

The Acquisition and Requisition of Immovable Property Ordinance 1982 (Ordinance II of 1982), now constitutes the sole instrument of land acquisition in Bangladesh (except in the 3 hill districts). This law provides for acquisition and requisition of immovable properties for a public purpose or in the public interest. Places of worship, graveyard or cremation ground, however, were exempted. Detailed procedures have been prescribed to ensure that a deputy commissioner proceeds systematically and on sound principles in such cases, leaving room for owners to raise objections which must be disposed of after due hearing.

This delegation of decision-making power was dictated by the delay in the acquisition process under the previous law. Delay in acquisition of land caused delay in the implementation of development projects. Moreover, foreign financing agencies were insistent upon expeditious land acquisition. Accordingly, a time limit was fixed according to the following scheme:

- a) if the quantity of land proposed to be acquired was up to 50 bighas⁴ and there was no objection - 45 days; and if objection was raised - 75 days; and
- b) if the quantity was over 50 bighas and no objection was raised - 105 days; and if objection was raised - 135 days. An additional maximum time limit of 112 days was allowed for finalization of the award. Thus the maximum time of finalization of an acquisition case is now 187 days if the quantity of land is up to 50 bighas and 274 days if it is over 50 bighas.

Laws, rules and procedures for land acquisition in the plain land districts of Bangladesh do not apply in the Chittagong Hill Tracts, which now comprise the three districts of Rangamati, Khagrachhari and Bandarban.

4. Steps in Land Acquisition

The following are the major steps of Bangladesh Government's Immovable and Movable Properties / Land Acquisition (LA) after the finalization of the engineering designs:

- Preparation of the land acquisition proposals (LAPs)
- LAPs submitted to the Ministry for administrative approval
- Administrative approval received from the Ministry
- LAPs submitted to the Deputy Commissioners (of respective Districts)
- LAPs approved by the District Land Acquisition Committees (DLACs)
- LAPs approval by the Ministry of Land
- Notice-3 issued by the DCs
- Notice-6 issued by the DCs
- Notice-7 issued by the DCs

⁴ 1 acre = 3 bighas

- Compensation Budgets submitted by the DCs to project authority
- Compensation Budgets forwarded to the concerned Line Ministry for approval
- Compensation Budgets approved by the Ministry
- Authority to place the Compensation funds with the DCs
- DCs start the payment of the 'compensation under the law' (CUL)

5. Land Acquisition Plan (LAP)

For any land acquisition, a Land Acquisition Plan (LAP) is needed. Preparation of the land acquisition plans will include locating each of the components; sorting out the ownership status of the lands in which project will be established; and preparing the schedules of plots (with 'dag' or Plot numbers and other information required by the acquiring body) from which lands are to be acquired. These are the documents the relevant project authority will submit to DC along with a request to start the acquisition process. LAPs will be prepared for proposed site and will ensure that the total amount of land in the individual acquisition packages does not exceed 50 standard bighas, which the Divisional Commissioners can approve without referring to the Ministry of Land.

ANNEX 6-28: WB SOCIAL MANAGEMENT PROCESS

1. World Bank's Safeguard Policies

BIFFL will evaluate the approach of the project company in responding to social compliances with respect to WB policies in context of lending to WB funded projects. Social issues are usually common to all sectors with the degree of effect of any issue varying on an individual basis. The major social issues related to infrastructure projects include:

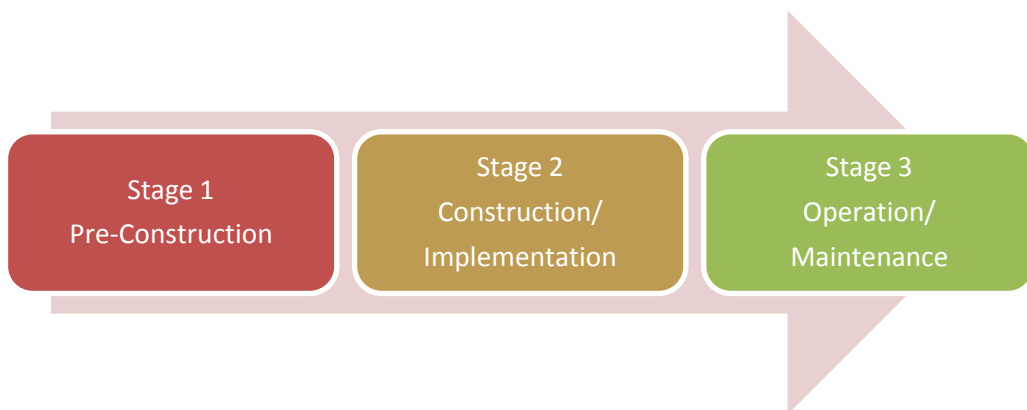
- (i) site planning
- (ii) involuntary resettlement;
- (iii) protection of marginalized and vulnerable groups; and
- (iv) public consultation, participation, and consensus building.

The World Bank Safeguard policies which cover the above issues are described in **Annex 6-23**.

2. Stages In Project Development

All projects and policies go through a series of steps or stages, starting with initial planning, then implementation and construction, carrying through to operation and implementation. It is imperative for the project sponsor to understand that Social impacts will be different for each stage; therefore it is essential to define the stages in project development:

Figure 6-2: Basic Stages in Project Development



a) Pre-Construction

This refers to all activity that takes place from the time a project or policy is conceived to the point of construction activity or policy implementation. Examples include project design, revision, public comment, licensing, the evaluating of alternatives, and the decision to go ahead.

b) Construction/Implementation

This stage begins when a decision is made to proceed, a permit is issued or a law or regulation takes place. This involves clearing land, building access roads, developing utilities, displacement and relocation of people. The build-up of a migration work force may occur creating social stresses.

c) Operation/Maintenance

This stage occurs after the construction is complete. It is during this stage that communities seeking industrial development will focus their attention on the long-term economic benefits that accrue from a development.

3. Flowchart for SMP

The flow chart is prepared to illustrate the social compliance activities for preparation of an infrastructure project. The flow chart () provides a guideline to the project sponsor in carrying out the necessary steps to address social requirements of a sub-project that utilizes BIFFL funding.

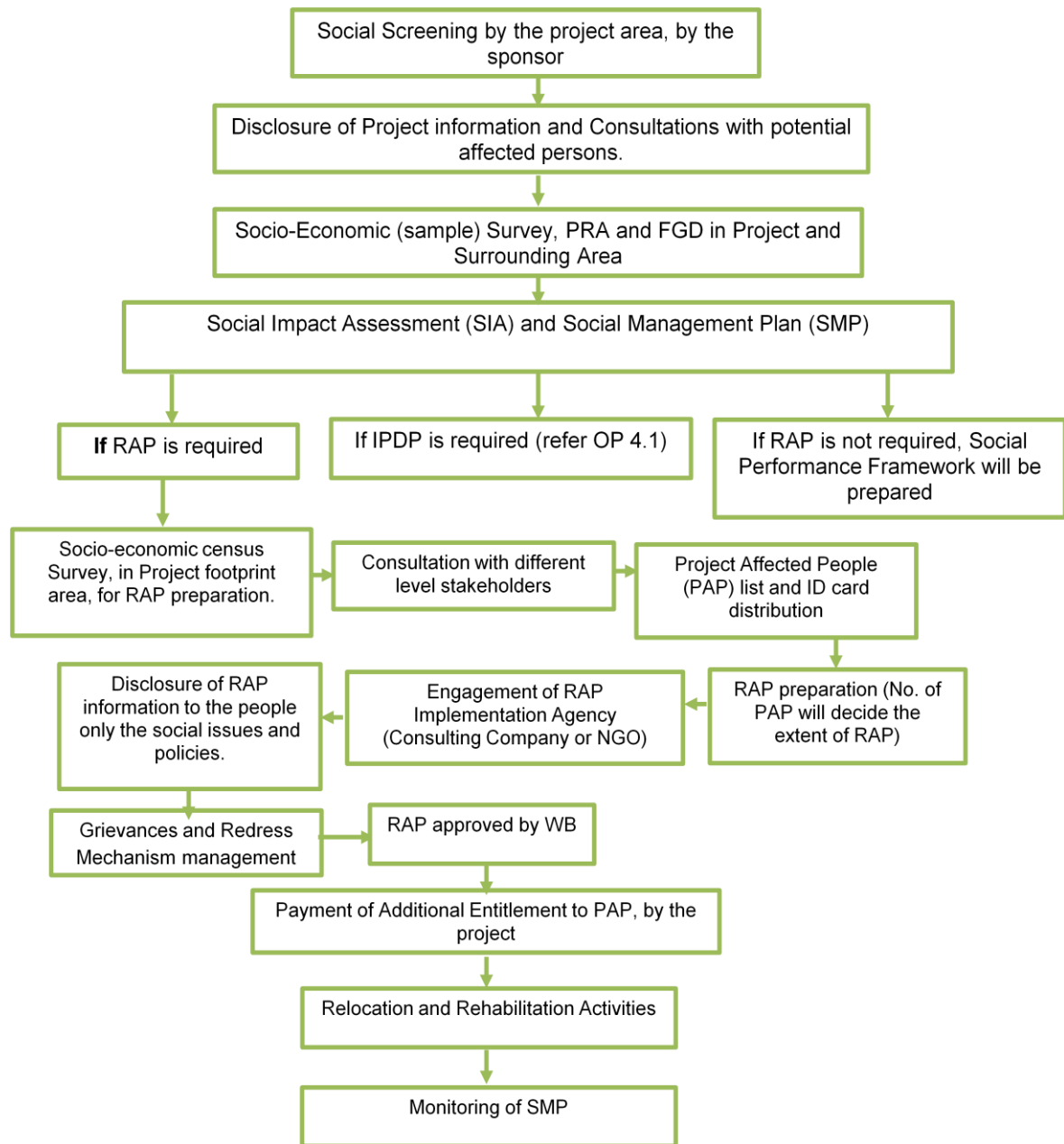
A Social Impact Assessment (SIA) of the project, with a socio-economic sample survey of the project and influence area is a must for assessment of the project. This SIA will determine if any RAP or IPDP will be required to be done. If the land is procured by the project sponsor through voluntary negotiation or the land has been acquired by the government beforehand, then no RAP etc, will be required.

4. Social Screening

To assess the precise nature and magnitude of social impacts, social screening may be conducted as part of the feasibility study for each subproject. To ensure consistency in the application of social screening criteria, a standard field level social screening checklist needs to be used.

To the extent possible, social and environment screening should be carried out together with technical and economic screening. Social screening will identify the probability for loss of land assets/structures, livelihoods, willingness of the community to donate land to the project, and other significant social impacts. Social screening also enables the categorization of subprojects based on the level of social impacts. Where the extent of adverse social impacts is minor and no displacement or loss of assets or livelihoods is expected, no further action is required. However, where the social screening indicates that land acquisition and/or loss of assets are unavoidable, appropriate resettlement plans are of paramount importance.

Figure 6-3: Flowchart for SMF



5. Public Information Campaign and Consultation

The World Bank requires participation by interested parties and PAPs in the assessment of social effects of a project. The public Involvement takes two forms: a formal and an informal process. The primary purposes of the consultation and participation process are: dissemination of information to the participants; solicitation of information relative to the interests of the participants; and solicitation of comments, concerns and suggestions for inclusion in the development of the project.

The first step is the identification of who will actually be involved in the social assessment process. In general, the participants will include all governmental agencies and ministries that have statutory responsibility for management of resources affected by the project. A second group of potential participants are the NGOs

including national organizations and CBOs. The third group includes the local population that will be directly or indirectly affected by project implementation.

a) Involvement of the Host Population

The population living in the areas identified as hosts for a relocated population also needs to be integrally involved in the planning and implementation of a resettlement plan.

b) Formal Consultation Process

Frequently, the initial contact with participants by the project sponsor is achieved through distribution of the Scoping Document. Included in the scoping document is an invitation to participate in a workshop to discuss the project and its objectives and to discuss the particular social concerns of the agencies and the local population.

A second workshop is advised, especially for larger, more complex projects to review the status of the studies being conducted are obtaining the information necessary. Upon completion of a draft RAP and IPDP, a final workshop is recommended to obtain oral comments and suggestions from the various stakeholders. The formal consultation process should also include the solicitation and receipt of written comments and suggestions on all social documents.

c) Focus Group Discussion

Formal consultation like Focus Group Discussion (FGD) with various stakeholders throughout the Social Assessment process is strongly encouraged. This consultation includes discussions with the various governmental organizations involved and interactions with NGOs and local populations to discuss various social issues pertinent to the planning and implementation of the project.

A record of the informal discussions with agencies and/or other participants should be included in the SMP.

6. Record of Consultation

A record of the formal and informal consultation with participants is an important component of the SMP. The record will verify that PI requirements of WB have been met satisfactorily. The PI record should include for workshops: an agenda of workshops, minutes of the workshop proceedings, issues discussed, responses to issues, a list of attendees, for written comments: actual letter containing comments and suggestions, the project sponsors responses to the comments and suggestions with rationale; for informal discussions with Agencies: brief memoranda describing topics discussed, information obtained from agencies, issues and suggestions raised and responses; for informal discussions: brief memoranda describing persons and situations of the discussions, issues discussed, information obtained and responses of the Project sponsor to issues and concerns raised.

7. Social Impact Assessment (SIA)

The SIA is one of the primary requirements of the project sponsor in order to comply with social requirements. The project sponsor should use social impact assessment specialists for the SIA.

a. Identifying Social Impact Assessment Variables

This is another important consideration before conducting the SIA, as this incorporates the local characteristics of the project affected areas and establishes a baseline for variables that would be used to collect data. A general list of social variables is given below:

i. Population Characteristics

Present population and expected change, ethnic and racial diversity, and influxes and outflows of temporary residents as well as the arrival of seasonal or leisure residents.

ii. Community and Institutional Structures

The size, structure, and level of organization of local government including linkages to the larger political system.

iii. Political and Social Resources

Distribution of power authority, the interested and affected publics, and the leadership capability and capacity within the community.

iv. Individual and Family changes

Factors, which influence the daily life of the individuals and families, including attitudes, perceptions, family characteristics and friendship networks.

v. Community Resources

Patterns of natural resource and land use; the availability of housing and community services.

The table below gives a matrix relating project stage to Social Impact Assessment Variables.

Table 6-10: Variables Matrix

Social Impact Assessment Variable	Pre-Construction	Implementation/Construction	Operation/Maintenance
Population Characteristics			
Population change			
Ethnic and racial distribution			
Relocated populations			
Influx or outflows or temporary workers			
Seasonal residents			

Social Impact Assessment Variable	Pre-Construction	Implementation/Construction	Operation/Maintenance
Community and Institutional Structures			
Voluntary association			
Interest group activity			
Size and structure of local government			
Historical experience with change			
Employment/income characteristics			
Employment equity of minority groups			
Local/regional/national linkages			
Political and Social Resources			
Distribution of power and authority			
Identifications of stakeholders			
Interested and affected publics			
Leadership capability and characteristics			
Individual and Family Changes			
Perceptions of risk, health, and safety			
Displacement/relocation concerns			
Trust in political and social institutions			
Residential stability			
Community Resources			
Change in community infrastructure			
Land use patterns			
Effects on cultural, historical, and archaeological sites			

b. Combining Social Impact Assessment Variables, Project Stage, and Setting

Social impact specialists engaged by the project sponsor will construct a matrix to that will serve as a guide for their assessing of significant social impacts. For each project stage, the assessor should identify potential impacts on each social variable identified in the matrix. This approach ensures that no critical areas are overlooked. The analytical procedures would be repeated for *each of the SIA variable* for *each stage of the project*. The procedures for accomplishing these tasks are outlined below:

c. Steps in the Social Impact Assessment Process

The process in the social assessment is described below in 10 steps.

- i) Public Involvement

- ii) Identification of Alternatives
- iii) Baseline Conditions
- iv) Scoping
- v) Projection of Estimated Effects-Investigate the probable impacts
- vi) Predicting responses to impacts
- vii) Indirect and Cumulative Impacts
- viii) Changes in Alternatives
- ix) Mitigation
- x) Monitoring

i) Public Involvement

Develop an effective public Involvement plan to involve all potentially affected publics. Requires identifying and working with all potentially affected groups starting at the very beginning of planning for the proposed action.

ii) Identification of Alternatives

Describe the proposed action of policy change and reasonable alternatives. The proposed action is described in detail to begin to identify the data requirements needed from a project proponent (World Bank) to frame the SIA

iii) Baseline Conditions

Describe the relevant impact zone/area of influence and baseline conditions. These are the existing conditions and past trends associated with human environment in which the proposed activity is to take place.

The Base conditions would include the following set aspects of human environment:

- Relationship with the biophysical environment
- Historical background: including initial settlement and subsequent shifts in population
- Political and social resources
- Culture, attitudes and socio-psychological conditions
- Population characteristics including the demographics of relevant groups (including all significant stakeholders and groups)

iv) Scoping

Identify the full range of probable social impacts that will be addressed based on discussion or interviews with numbers of al potentially affected. After initial scoping, the social impact assessor (social impact specialist engaged by the project sponsor) selects the SIA variables for further assessment situations. Considerations need to be given to both the impacts perceived by the project sponsor/executing agency (as the case may be) and to those perceived by the affected groups and communities through consultations.

The principal methodology employed here is reviews of the existing social science literature, public scoping, public surveys, and public participation techniques. It is a requirement of OP 4.12 that views of affected people are taken into consideration.

Relevant criteria for selecting impacts include:

- Probability of the event occurring
- Number of people including indigenous populations that will be affected
- Duration of impacts (long-term vs. short-term)
- Value of benefits and costs to impacted groups (intensity of impacts)
- Extent that the impact is reversible or can be mitigated
- Likelihood of causing subsequent impacts
- Relevance to present and future policy decisions
- Uncertainty over possible effects
- Presence or absence of controversy over the issue.

v) *Projection of Estimated Effects-Investigate the probable impacts*

The probable social impacts will be formulated in terms of predicted conditions without the actions (baseline projection); predicted conditions with the actions; and predicted impacts which can be interpreted as the differences between the future with and without the proposed action. Investigation of the probable impacts involves five major sources of information:

- Data from project proponents
- Records of previous experience with similar actions as represented in reference literature as well as other SIA's.
- Census and vital statistics
- Documents and secondary sources
- Field research, including informant interviews, hearing, group meetings, and surveys of the general population.

The record of previous experience is very important to the estimation of future impacts. Expert knowledge of the social impact specialists is used to enlarge this knowledge base and to judge how the study case is likely to deviate from the typical patterns.

vi) *Predicting responses to impacts*

This determines the significance of the identified social impacts. After direct impacts have been estimated the Social Impact Specialist must next estimate how the affected people will respond in terms of attitude and actions. The actions of affected groups are to be estimated using comparable cases and interviews with affected people about what they expect to do.

vii) Indirect and Cumulative Impacts

This estimates subsequent impacts and cumulative impacts. Indirect impacts are those caused by the direct impacts; they often occur later than the direct impacts, or further away. Cumulative impacts are those impacts which result from the incremental impacts of an action added to other past, present, and reasonably foreseeable future actions regardless of which agency or person undertakes them.

viii) Changes in Alternatives

Recommend new or changed alternatives and estimate or project their consequences. Each new alternative or recommended change should be assessed separately. Here more innovative alternatives and changes probably should be presented in an experimental structure.

ix) Mitigation

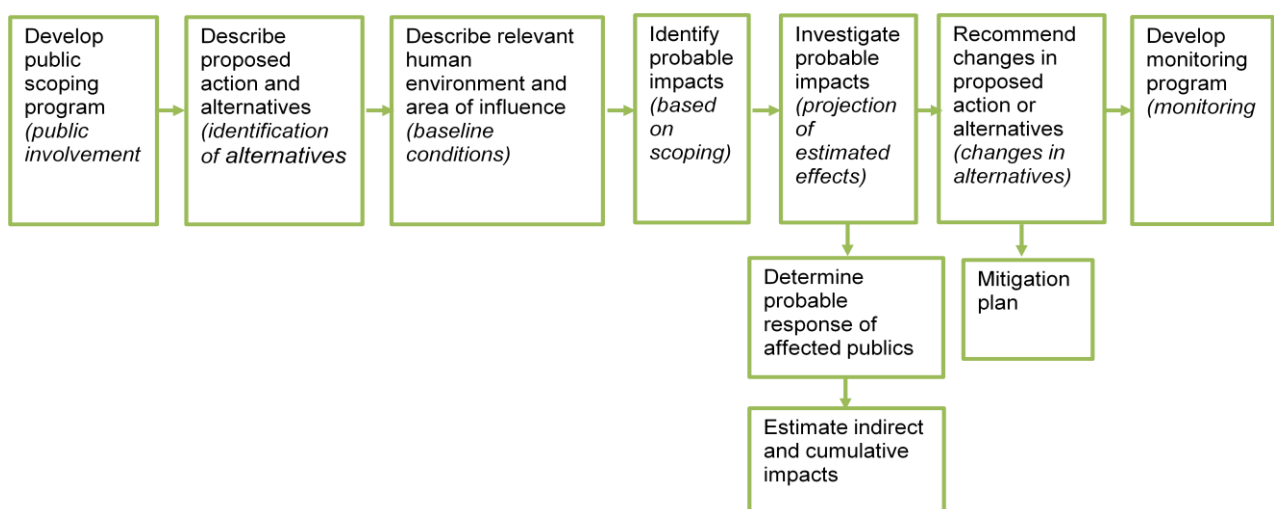
Develop a mitigation plan. A SIA should identify means to mitigate adverse impacts. These measures include avoiding the impact by not taking or modifying an action; minimizing, rectifying, or reducing the impacts through the design or operation of the project; or compensation for the impact by providing substitute facilities, resources, or opportunities

x) Monitoring and Impact Evaluation

Develop a monitoring and impact evaluation plan. A monitoring and impact evaluation program should be developed for identifying deviations from the proposed action and any important unanticipated impacts. A monitoring plan should be developed to track project and program development and compare real impacts with projected areas.

A flowchart of the Steps in the Social Impact Assessment Process is provided in the following.

Figure 6-4: Steps in the Social Impact Assessment Process



8. Identification of Impact

Basic categories of issues/impacts those needed to be addressed under the entitlement framework are the following:

- Loss of land
- Loss of water source
- Loss of structure
- Loss of source of livelihood
- Loss of access to common resources and facilities
- Loss of standing crops, trees and perennial trees
- Loss of public infrastructure
- Loss of services (electricity, water, transport, etc)
- Loss of or imperilled access to any of the above

Table 6-11: List of Probable Impacts during Various Phases of the Project

Pre-Construction	Construction	Operation & Maintenance
Site Surveys & Investigations • <i>Inducement of uncertainties relating to land acquisition</i> • <i>Stimulation of new construction to obtain higher compensation</i>	Contractors' Camps. Employment • Employment opportunities for local population • Conflicts between the imported workers and local population • Risk of HIV/AIDS and other transmittable diseases • Inducement of traffic congestion & related road safety hazards	• Increased employment opportunities • Increased economic activity
Land & Property Acquisition • <i>Loss of and displacement from homestead land</i> • <i>Loss of agricultural land and other productive assets</i> • <i>Loss of business/commercial land or premises</i> • <i>Loss of public buildings and facilities</i> • <i>Loss of cultural heritage/Resources</i> • <i>Loss of sensitive habitats</i>	Clearance of Site • Interference with utility services Construction Materials • Loss of and displacement from productive land • Waterborne disease risks/Safety hazard on abandonment • Visual alteration in landscape quality Construction Traffic • Damage to road pavement & structures • Increased traffic congestion & related road safety hazards.	

9. Impact Mitigation

Impact mitigation includes the principles, procedures and mitigation measures that are relevant and suitable to the project for ensuring the most appropriate environmental mitigation and enhancement plans applicable during different stages of project implementation. To avoid and minimize the impacts resulting from the activities of the different sub-projects, measures/management plans, which are essential to mitigate the impacts as discussed, are based upon appropriate technological design, improvements or adjustments, policy including good site operational practices etc.

Environmental and social impacts of the subprojects can broadly be classified as those taking place during pre-construction, construction and operational phases. The overall strategy has the following sequence:

- Impact avoidance: Changing project location, design and construction methods to avoid impacts.
- Impact minimization: Where impacts cannot be avoided, implementing mitigation measures to reduce the impact to acceptable levels.
- Compensation: Arranging compensation where impacts cannot be avoided or sufficiently mitigated.
- Enhancement: measures, which, at insignificant cost to the project, give appreciable social or developmental benefits.

The mitigation plan is adopted to highlight the action procedures to avoid/ minimize/ control the resultant impacts arising out of the different project phases i.e. pre-construction, construction and operation.

10. Resettlement Action Plan (RAP)

a. Resettlement Types

For acquisition of land, if the SIA indicates that a Resettlement Action Plan (RAP) is required, then the project sponsor will engage a Social Specialist who will prepare a RAP considering all World Bank OPs.

i) **Short RAP:** As per WB OP 4.12, where impacts on the entire displaced population are minor or fewer than 200 people are displaced, a short RAP is required.

ii) **Full RAP** In other cases, a full RAP is to be prepared.

Resettlement effects result from a wide range of project types envisaged under the BIFFL. These range from small plots of land required for schools or health centers may create limited resettlement effects. Long alignments required for power lines, roads, railways, or canals may cause resettlement along a narrow right of way, or disrupt community networks, dividing roads, paths, irrigation systems, and landholdings. Most types of projects have the resettlement effects as set out in the table below:

Table 6-12: Resettlement in Different Project Types

Sector	Project Components	Type of Resettlement Effect
Power and Energy	- Transmission alignment - Power generation plants, transmission stations, substations, and access roads - Hydroelectric power reservoirs	- Minor resettlement effects from construction of pylons. These might be severe if landholdings are small. Right-of-way restrictions, without land acquisition, might affect people's land use along the transmission alignment. May require temporary land borrow during construction - May cause severe localized effects, and temporary land borrow during construction. Power plants may cause resettlement effects through pollution of land, air, or water. - Reservoir construction can have severe and often widespread effects, displacing whole communities from construction and inundation areas, and disrupting patterns of communication, landholdings, social and economic systems, and resource use. Temporary land borrow for construction.
Transport	- Road or rail alignment - Associated stations, terminals, bridges - Airports, seaports, river ports	- Resettlement effects over alignment. Disruption can usually be addressed within existing community units because alignment is narrow. However, if the alignment is long, cutting across administrative boundaries, the distribution of responsibilities may be unclear and entitlements may vary between sections. Alignments might divide landholdings, local roads and paths, irrigation systems, economic and social networks, or access to resources. May require temporary land borrow for construction - May cause localized resettlement effects, and necessitate temporary land borrow for construction - Severe resettlement effects for communities currently occupying land are possible. Can displace whole communities, or disrupt patterns of communication, landholdings, social and economic systems and resource use. Temporary land borrow for construction
Water supply and sanitation	- Reticulation systems - Pumping stations, treatment sites - Reservoirs for water supply	- Temporary land borrow. Use of existing rights-of-way (for example, roads) can minimize disruption. Narrow land corridors might be acquired permanently with minor disruption. - May cause more severe localized effects. Temporary land borrow for construction. - Reservoir construction can have severe and often widespread effects
Solid waste	Transfer stations, landfill sites	May cause severe localized effects
Urban	Urban	May cause severe localized effects

Sector	Project Components	Type of Resettlement Effect
renewal	infrastructure sites	
Health	Sites for hospitals, clinics, teaching facilities	May cause severe localized effects. Communities might be prepared to volunteer small sites for community services.
Education	Sites for schools, training institutions, etc	May cause severe localized effects. Communities might be prepared to volunteer small sites for community services.
Irrigation and Flood control	- Canal alignments, protective embankments, and associated works - Dams	- Resettlement effects over a narrow alignment. - Dam construction can have severe and widespread effects
Mining operations	Strip mining	May cause severe localized effects, or resettlement effects due to severe loss of environmental quality (ex. Polluted land or water)
Forestry developments	Reforestation, industrial plantations, forest closure	May cause loss of access to forest products for cash and subsistence. Loss of grazing rights. Displacements of communities.
Parks, conservation sites	National parks or biodiversity areas	May cause loss of grazing rights, or disruption of grazing routes. May displace communities from park.

The preparation for RAP according to WB OP 4.12 is described in **Annex 6-24**.

Comparison of Bangladesh Requirements with OD 4.01

The EIA process under both the legislation in Bangladesh and OD 4.01 have their own requirements, which do not necessary match. A comparison is made to identify similarities and differences, so that a process meeting more stringent of the requirements can be applied to BIFFL projects. **Table 6.13** lists some key comparisons, and discusses the requirements for the Bangladesh law with OD 4.01. In general, OD 4.01 requirements are more comprehensive when compared with the requirements of Bangladesh legal system.

Table 6-13: Comparison of Bangladesh Law with World Bank OD 4.01

Criteria	Requirements as per Law in Bangladesh	Requirements as per OD 4.01
1. Type of Environmental Analysis	project specific	project specific, regional and sectoral
2. Basis for categorization	Currently screening criteria available only for industrial projects, where they are done based on. • level of pollution emission • type of project and location • scale of project • operational activities Non-industrial projects are reviewed on a case-by-case basis by DOE.	Detailed screening criteria for all projects based on • Sensitivity • nature and magnitude of potential impacts • social assessment component • resettlement and rehabilitation
3. EA Outputs	Since detailed rules and regulations for EA have not been prescribed, EA outputs are not specified. However, the industrial sector guide the water sector guidelines, anti the infrastructure development guidelines have specific EA output requirements, such as • baseline survey • EIA/IEE report • site clearance • risk analysis and management • analysis of alternatives	• EA Report • Analysis of alternatives • Acceptable Environmental Management Plan • Acceptable Resettlement Action Plan • Plan • Acceptable Indigenous Peoples • Development Plan
4. Public Consultation	No special mention is made for public consultation in BECA Sectoral guidelines mentioned above have prescribed consultation	Mandatory at • preparation of EA • project appraisal plus • project design • project implementation and monitoring

Criteria	Requirements as per Law in Bangladesh	Requirements as per OD 4.01
5. Disclosure of Information	BECA makes no reference to disclosure. The Sectoral guidelines prescribe some provisions for disclosure.	• Mandatory • Summary of project description and potential adverse impacts • Summary of EA report and conclusions • EA report

Table 6-14: Bangladesh Law Vs WB Policy on Land Acquisition and Resettlement

ISSUE	1982 Ordinance	OP 4.12
Coverage	Legal owners Share-croppers Tenants	All affected parties, including squatters and illegal occupants
Compensation	Based on market values over previous 12 months No provision for restoration of income streams	Replacement cost at current market price Requires livelihood restoration component.
Uses of material from dismantled structures	Material is to be auctioned after being compensated for it	Material can be taken and re-used by affected party
Minimization of impacts	Discourages unnecessary acquisition but no mechanisms to monitor	Alternative analysis required to justify avoidance and/or mitigation of impacts
Cut-off dates	Not addressed	Important to ensure that squatters are included in compensation and to prevent rent-seeking behavior of additional squatters settling onto project land
Consultation	No consultation required	Consultation as core issue in RAP preparation and implementation
Relocation assistance	No assistance provided	Relocation assistance required
Livelihood restoration	Not addressed	Livelihood restoration component and attention to post-resettlement required

ANNEX 6-29: ADB SAFEGUARD POLICY STATEMENTS

1. *Environmental Safeguards*

Objectives: To ensure the environmental soundness and sustainability of projects and to support the integration of environmental considerations into the project decision-making process.

Scope and Triggers: Environmental safeguards are triggered if a project is likely to have potential environmental risks and impacts.

Policy Principles:

- 1) Use a screening process for each proposed project, as early as possible, to determine the appropriate extent and type of environmental assessment so that appropriate studies are undertaken commensurate with the significance of potential impacts and risks.
- 2) Conduct an environmental assessment for each proposed project to identify potential direct, indirect, cumulative, and induced impacts and risks to physical, biological, socioeconomic (including impacts on livelihood through environmental media, health and safety, vulnerable groups, and gender issues), and physical cultural resources in the context of the project's area of influence. Assess potential trans-boundary and global impacts, including climate change. Use strategic environmental assessment where appropriate.
- 3) Examine alternatives to the project's location, design, technology, and components and their potential environmental and social impacts and document the rationale for selecting the particular alternative proposed. Also consider the no project alternative.
- 4) Avoid, and where avoidance is not possible, minimize, mitigate, and/or offset adverse impacts and enhance positive impacts by means of environmental planning and management. Prepare an environmental management plan (EMP) that includes the proposed mitigation measures, environmental monitoring and reporting requirements, related institutional or organizational arrangements, capacity development and training measures, implementation schedule, cost estimates, and performance indicators. Key considerations for EMP preparation include mitigation of potential adverse impacts to the level of no significant harm to third parties, and the polluter pays principle.
- 5) Carry out meaningful consultation with affected people and facilitate their informed participation. Ensure women's participation in consultation. Involve stakeholders, including affected people and concerned nongovernment organizations, early in the project preparation process and ensure that their views and concerns are made known to and understood by decision makers and taken into account. Continue consultations with stakeholders throughout project implementation as necessary to address issues related to environmental assessment. Establish a grievance redress mechanism to receive and facilitate

resolution of the affected people's concerns and grievances regarding the project's environmental performance.

- 6) Disclose a draft environmental assessment (including the EMP) in a timely manner, before project appraisal, in an accessible place and in a form and language(s) understandable to affected people and other stakeholders. Disclose the final environmental assessment, and its updates if any, to affected people and other stakeholders.
- 7) Implement the EMP and monitor its effectiveness. Document monitoring results, including the development and implementation of corrective actions, and disclose monitoring reports.
- 8) Do not implement project activities in areas of critical habitats, unless
 - i. there are no measurable adverse impacts on the critical habitat that could impair its ability to function,
 - ii. there is no reduction in the population of any recognized endangered or critically endangered species, and
 - iii. any lesser impacts are mitigated.

If a project is located within a legally protected area, implement additional programs to promote and enhance the conservation aims of the protected area. In an area of natural habitats, there must be no significant conversion or degradation, unless

- i. alternatives are not available,
- ii. the overall benefits from the project substantially outweigh the environmental costs, and
- iii. any conversion or degradation is appropriately mitigated.

Use a precautionary approach to the use, development, and management of renewable natural resources.

- 9) Apply pollution prevention and control technologies and practices consistent with international good practices as reflected in internationally recognized standards such as the World Bank Group's Environmental, Health and Safety Guidelines. Adopt cleaner production processes and good energy efficiency practices. Avoid pollution, or, when avoidance is not possible, minimize or control the intensity or load of pollutant emissions and discharges, including direct and indirect greenhouse gases emissions, waste generation, and release of hazardous materials from their production, transportation, handling, and storage. Avoid the use of hazardous materials subject to international bans or phase outs. Purchase, use, and manage pesticides based on integrated pest management approaches and reduce reliance on synthetic chemical pesticides.
- 10) Provide workers with safe and healthy working conditions and prevent accidents, injuries, and disease. Establish preventive and emergency preparedness and response measures to avoid, and where avoidance is not possible, to minimize, adverse impacts and risks to the health and safety of local communities.

- 11) Conserve physical cultural resources and avoid destroying or damaging them by using field based surveys that employ qualified and experienced experts during environmental assessment. Provide for the use of —chance find|| procedures that include a pre-approved management and conservation approach for materials that may be discovered during project implementation.

2. Involuntary Resettlement Safeguards

- 1) The objectives of ADB's Safeguard Policy Statement with regard to involuntary resettlement are: (i) to avoid involuntary resettlement wherever possible; (ii) to minimize involuntary resettlement by exploring project and design alternatives; (iii) to enhance, or at least restore, the livelihoods of all displaced persons in real terms relative to pre-project levels; and (iv) to improve the standards of living of the displaced poor and other vulnerable groups⁵.
- 2) ADB's SPS covers physical displacement (relocation, loss of residential land, or loss of shelter) and economic displacement (loss of land, assets, access to assets, income sources, or means of livelihoods) as a result of (i) involuntary acquisition of land, or (ii) involuntary restrictions on land use or on access to legally designated parks and protected areas. It covers displaced persons whether such losses and involuntary restrictions are full or partial, permanent or temporary.
- 3) For any ADB operation requiring involuntary resettlement, resettlement planning is an integral part of the project design, to be dealt with from the earliest stages of the project cycle, taking into account the following basic principles⁶
 - i. Screen the project early on to identify past, present, and future involuntary resettlement impacts and risks. Determine the scope of resettlement planning through a census and socioeconomic surveys of displaced persons, including a gender analysis, specifically related to resettlement impacts and risks.
 - ii. Carry out meaningful consultations⁷ with affected persons, host communities, and concerned non-government organizations. Inform all displaced persons of their entitlements and resettlement options. Ensure their participation in planning, implementation, and monitoring and evaluation of resettlement programs. Pay particular attention to the

⁵ Vulnerable groups include (i) those below the poverty line (ii) the landless (iii) the elderly (iv) female headed households (v) women and children (vi) Indigenous People (vii) those without legal title to land (Source ADB Safeguard Policy Statement, 2009, Appendix 2 para 28).

⁶ Source: ADB Safeguard Policy Statement, 2009, page 17

⁷ Meaningful consultation is a process that should : (i) begin early in the preparation stage and is carried out on an ongoing basis throughout the project life cycle; (ii) provide timely disclosure of relevant and adequate information that is understandable and readily accessible to the affected people; (iii) be undertaken in an atmosphere free of intimidation or coercion; (iv) be gender inclusive and responsive, and tailored to the needs of disadvantaged and vulnerable groups; and (v) incorporate all relevant views of the affected people and other stakeholders into decision making, such as project design, mitigation measures, the sharing of development benefits and opportunities and implementation issues.

needs of vulnerable groups, especially those below the poverty line, the landless, the elderly, women and children, and Indigenous Peoples, and those without legal title to land, and ensure their participation in consultations. Establish a grievance redress mechanism to receive and facilitate resolution of the affected persons' concerns. Support the social and cultural institutions of displaced persons and their host population. Where involuntary resettlement impacts and risks are highly complex and sensitive, compensation and resettlement decisions should be preceded by a social preparation phase.

- iii. Improve, or at least restore, the livelihoods of all displaced persons through (i) land-based resettlement strategies when affected livelihoods are land based where possible or cash compensation at replacement value for land when the loss of land does not undermine livelihoods, (ii) prompt replacement of assets with access to assets of equal or higher value, (iii) prompt compensation at full replacement cost for assets that cannot be restored, and (iv) additional revenues and services through benefit sharing schemes where possible.
- iv. Provide physically and economically displaced persons with needed assistance, including the following: (i) if there is relocation, secured tenure to relocation land, better housing at resettlement sites with comparable access to employment and production opportunities, integration of resettled persons economically and socially into their host communities, and extension of project benefits to host communities; (ii) transitional support and development assistance, such as land development, credit facilities, training, or employment opportunities; and (iii) civic infrastructure and community services, as required.
- v. Improve the standards of living of the displaced poor and other vulnerable groups, including women, to at least national minimum standards. In rural areas provide them with legal and affordable access to land and resources, and in urban areas provide them with appropriate income sources and legal and affordable access to adequate housing.
- vi. Develop procedures in a transparent, consistent, and equitable manner if land acquisition is through negotiated settlement to ensure that those people who enter into negotiated settlements will maintain the same or better income and livelihood status.
- vii. Ensure that displaced persons without titles to land or any recognizable legal rights to land are eligible for resettlement assistance and compensation for loss of non-land assets.
- viii. Prepare a resettlement plan elaborating on displaced persons' entitlements, the income and livelihood restoration strategy, institutional arrangements, monitoring and reporting framework, budget, and time-bound implementation schedule.
- ix. Disclose a draft resettlement plan, including documentation of the consultation process in a timely manner, before project appraisal, in an

accessible place and a form and language(s) understandable to affected persons and other stakeholders. Disclose the final resettlement plan and its updates to affected persons and other stakeholders.

- x. Conceive and execute involuntary resettlement as part of a development project or program. Include the full costs of resettlement in the presentation of project's costs and benefits. For a project with significant involuntary resettlement impacts, consider implementing the involuntary resettlement component of the project as a stand-alone operation.
- xi. Pay compensation and provide other resettlement entitlements before physical or economic displacement. Implement the resettlement plan under close supervision throughout project implementation.
- xii. Monitor and assess resettlement outcomes, their impacts on the standards of living of displaced persons, and whether the objectives of the resettlement plan have been achieved by taking into account the baseline conditions and the results of resettlement monitoring. Disclose monitoring reports.

3. Indigenous Peoples Safeguards

- 1) The important elements of the indigenous peoples policy are: to design and implement projects in a way that fosters full respect for Indigenous Peoples' identity, dignity, human rights, livelihood systems, and cultural uniqueness as defined by the Indigenous Peoples themselves so that they (i) receive culturally appropriate social and economic benefits, (ii) do not suffer adverse impacts as a result of projects, and (iii) can participate actively in projects that affect them.
- 2) The Indigenous Peoples safeguards are triggered if a project directly or indirectly affects the dignity, human rights, livelihood systems, or culture of Indigenous Peoples or affects the territories or natural or cultural resources that Indigenous Peoples own, use, occupy, or claim as an ancestral domain or asset.
- 3) The term Indigenous Peoples is used in a generic sense to refer to a distinct, vulnerable, social and cultural group possessing the following characteristics in varying degrees: (i) self identification as members of a distinct indigenous cultural group and recognition of this identity by others; (ii) collective attachment to geographically distinct habitats or ancestral territories in the project area and to the natural resources in these habitats and territories; (iii) customary cultural, economic, social, or political institutions that are separate from those of the dominant society and culture; and (iv) a distinct language, often different from the official language of the country or region. In considering these characteristics, national legislation, customary law, and any international conventions to which the country is a party will be taken into account. A group that has lost collective attachment to geographically distinct habitats or ancestral territories in the project area because of forced severance remains eligible for coverage under this policy.

- i. For any ADB operation involving impact to indigenous peoples, indigenous peoples planning is an integral part of project design, to be dealt with from the earliest stages of the project cycle, taking into account the following basic principles i) Screen early on to determine
- ii. whether Indigenous Peoples are present in, or have collective attachment to, the project area; and
- iii. whether project impacts on Indigenous Peoples are likely. ii) Undertake a culturally appropriate and gender-sensitive social impact assessment or use similar methods to assess potential project impacts, both positive and adverse, on Indigenous Peoples. Give full consideration to options the affected Indigenous Peoples prefer in relation to the provision of project benefits and the design of mitigation measures. Identify social and economic benefits for affected Indigenous Peoples that are culturally appropriate and gender and intergenerationally inclusive and develop measures to avoid, minimize, and/or mitigate adverse impacts on Indigenous Peoples.
- iv. Undertake meaningful consultations with affected Indigenous Peoples communities and concerned Indigenous Peoples organizations to solicit their participation (i) in designing, implementing, and monitoring measures to avoid adverse impacts or, when avoidance is not possible, to minimize, mitigate, or compensate for such effects; and (ii) in tailoring project benefits for affected Indigenous Peoples communities in a culturally appropriate manner. To enhance Indigenous People's active participation, projects affecting them will provide for culturally appropriate and gender inclusive capacity development. Establish a culturally appropriate and gender inclusive grievance mechanism to receive and facilitate resolution of the Indigenous Peoples' concerns.
- v. Ascertain the consent of affected Indigenous Peoples communities to the following project activities: (i) commercial development of the cultural resources and knowledge of Indigenous Peoples; (ii) physical displacement from traditional or customary lands; and (iii) commercial development of natural resources within customary lands under use that would impact the livelihoods or the cultural, ceremonial, or spiritual uses that define the identity and community of Indigenous Peoples. For the purposes of policy application, the consent of affected Indigenous Peoples communities refers to a collective expression by the affected Indigenous Peoples communities, through individuals and/or their recognized representatives, of broad community support for such project activities. Broad community support may exist even if some individuals or groups object to the project activities.
- vi. Avoid, to the maximum extent possible, any restricted access to and physical displacement from protected areas and natural resources. Where avoidance is not possible, ensure that the affected Indigenous Peoples communities participate in the design, implementation, and monitoring

and evaluation of management arrangements for such areas and natural resources and that their benefits are equitably shared.

- vii. Prepare an Indigenous Peoples plan (IPP) that is based on the social impact assessment with the assistance of qualified and experienced experts and that draw on indigenous knowledge and participation by the affected Indigenous Peoples communities. The IPP includes a framework for continued consultation with the affected Indigenous Peoples communities during project implementation; specifies measures to ensure that Indigenous Peoples receive culturally appropriate benefits; identifies measures to avoid, minimize, mitigate, or compensate for any adverse project impacts; and includes culturally appropriate grievance procedures, monitoring and evaluation arrangements, and a budget and time bound actions for implementing the planned measures.
- viii. Disclose a draft IPP, including documentation of the consultation process and the results of the social impact assessment in a timely manner, before project appraisal, in an accessible place and in a form and language(s) understandable to affected Indigenous Peoples communities and other stakeholders. The final IPP and its updates will also be disclosed to the affected Indigenous Peoples communities and other stakeholders.
- ix. Prepare an action plan for legal recognition of customary rights to lands and territories or ancestral domains when the project involves (i) activities that are contingent on establishing legally recognized rights to lands and territories that Indigenous Peoples have traditionally owned or customarily used or occupied, or (ii) involuntary acquisition of such lands.
- x. Monitor implementation of the IPP using qualified and experienced experts; adopt a participatory monitoring approach, wherever possible; and assess whether the IPP's objective and desired outcome have been achieved, taking into account the baseline conditions and the results of IPP monitoring. Disclose monitoring reports.

ANNEX 6-30: GENERIC OUTLINE FOR AN INDIGENOUS PEOPLES DEVELOPMENT PLAN

During the initial evaluation of the proposed project a determination will be made as to whether an indigenous group of people will be affected by the proposed infrastructure project. The determination will be made on the basis of:

- Government recognition of the group as an indigenous, and
- Preliminary evaluation by a recognized professional anthropologist or social scientist.

If it is determined that the project will affect one or more indigenous groups of people, the Project Sponsor will be responsible for preparing an Indigenous Peoples Development Plan (IPDP) for each affected group in addition to the Resettlement Action Plan. Much of the information acquired for presentation in the RAP may also be used for the IPDP, as appropriate. As appropriate, information presented in the ESAR and RAP that is pertinent to the Indigenous Peoples Group should be incorporated into the IPDP for completeness. A generic outline for an IPDP is presented below:

1. Executive Summary

2. Introduction and Background Information

- a. Brief description of the main objectives of the project, its components and development schedule.
- b. Brief description of the effects of the project on the Indigenous Group
- c. Assessment of the legal status of the group
 1. Bangladesh constitution
 2. Bangladesh legislation and subsidiary legislation
 3. Description of rights, entitlements and vulnerability afforded the group as a consequence of legal status.

3. Description of the indigenous group

- a. Distribution of group relative to project (include appropriate maps)
- b. Brief history of the group
- c. Demographics of the group
- d. Description of social and cultural framework of the group
 1. Social status hierarchy
 2. Description of general life of group (eg diet, social interactions, family structure, etc.)
 3. Religious context of group
 4. Relationship to general population
- a. Economic foundation for the group
 1. Main sources of income/occupation
 2. Markets, cottage industry, etc.

3. Technical information regarding systems of production
4. Land tenure (customary rule and continued use) and land use by group

4. Effects of Project on Indigenous Group

- a. Inventory of assets of group (subset of information used for Resettlement Action Plan)
- b. Land Acquisition
- c. Effects on social and economic structure
- d. Effects on cultural relationships within group (including potential for fragmentation of the group)

5. Special Plans for Mitigating Project Effects to Indigenous group

- a. Description of Relocation Site (if necessary)
 1. Land area (agricultural, residential, common area, etc.)
 2. Availability of water for drinking, irrigation, livestock, etc.
 3. Waste management (solid, water, sanitary)
- b. Description of infrastructure services to be provided to group
 1. Education
 2. Health (potentially including traditional medical programs)
 3. Financial (including incentive programs for investments)
 4. Legal assistance
 5. Access to markets, cultural sites, religious sites, etc.
- c. Description of special programs
 1. Programs to maintain cultural integrity
 2. Programs to promote investments by indigenous group
 3. Programs to train indigenous groups to improve production systems, etc
- d. Entitlements and compensation
 1. Immovable assets (Use entitlements described in Annex 6-VI).
 2. Religious sites (temples, cemeteries, and other culturally significant structures and facilities.
 3. Procedure for distribution of entitlements
- e. Programs specifically directed to vulnerable groups within indigenous groups
 1. Women
 2. Aged
 3. Orphans
 4. Disabled
 5. Other vulnerable groups within the society

6. Institutional Capacity for management of Indigenous Peoples Development Plan

- a. Responsible Governmental institutions
 - 1. Number of Staff (Administrative and field)
 - 2. Training of Staff (Administrative and field)
 - 3. Experience and expertise of relevant staff members (Administrative and Field)
 - 4. Relationship of responsible institution to other governmental agencies
- b. Financial Organization
 - 1. Funding for investment in development plan
 - 2. Funding for administrative and field operations
 - 3. Prior experience in budgetary aspects for indigenous peoples programs.

7. Record of Consultation and Participation

- a. Strategy for involving indigenous group in planning and implementation process
- b. Record of Consultation and Participation during planning process (attach minutes of meetings, workshops, letters, etc. as appropriate)
 - 1. Schedule of activities
 - 2. Issues raised by Indigenous group and discussion of how those issues were incorporated into the IPDP
- c. Strategy for continued participation of indigenous group during implementation and rehabilitation period.
- d. Grievance and Appeal Protocol

8. Implementation Schedule

- a. Schedule for implementation of plan
 - 1. Construction and site preparation (if any)
 - 2. Relocation (if necessary)
 - 3. Compensation
 - 4. Social and cultural programs
- b. Monitoring and Evaluation of IPDP
 - 1. Benchmarks in each aspect to enable evaluation of progress
 - 2. Details of monitoring plan (i.e. data collection, parameters, frequency, field staff etc.)
 - 3. Evaluation and analysis protocol
 - 4. Reporting (protocol, schedule, etc)
 - 5. Audits of IPDP

- Responsible organization
- Audit schedule
- Reporting procedure

9. Detailed Cost Estimate and Financing Plan

- a. Entitlements and Compensation
- b. Administration and Management
- c. Rehabilitation Costs
 - 1. Services rehabilitation
 - 2. Subsidies
 - 3. Monitoring and Evaluation
- d. Revolving Credit Programs
- e. Sources of funds
 - 1. Entitlements and Compensation
 - 2. Administration and Management
 - 3. Rehabilitation
 - 4. Revolving credit

ANNEX 6-31: GENERAL CHECK LIST TO EVALUATE IEE/ ESIA REPORTS

Purpose of the Checklist: This checklist is to be used for various projects in order to assess that a quality IEE/ EIA report has been prepared by the project company (IE), when applying for a BIFFL loan.

(A) Project Background

1.	Name of Proposed Project				
2.	Location				
3.	Project objective				
4.	Capacity or size of the Project				
5.	Proposed date of commencement of work				
6.	Proposed construction period				
7.	Sector				
8.	Executing Agency				
9.	Category of the project	Green	Orange A	Orange B	Red

(B) Status of Consents and Approvals

	Items	Yes	No	Comments
1.	No Objection Certificate (NOC) from local authorities			
2.	Initial Environment Examination (IEE)			
3.	Site Clearance Certificate (SCC)			
4.	EIA approval from DOE (in case of red category project)			
5.	Environmental Clearance Certificate (ECC)			

(C) Assessment of Quality of the IEE Report⁸

	Items	Yes	No	Comments
1.	IEE conducted based on a good quality pre-feasibility study			
2.	Name and experience of the firm conducting IEE in similar projects			
3.	Scope of works and team composition conducting IEE			
4.	Assessment of quality of the IEE report			
	a) Executive Summary			
	b) Methodology for IEE			
	c) Project description including project location, site map, project rationale, project development plan, manpower requirement, resource utilization etc.			
	d) Legal and Administrative Framework			
	e) Alternative Site and Technology assessment and modification			
	f) Baseline Information			
	g) Acceptability to local community			
	h) Impact assessment, mitigation measure, Environmental Management Plan (EMP)			
	i) Conclusion: IEE report is complete and of good quality			

(D) Assessment of Quality of the ESIA Report

	Items	Yes	No	Comments
	Project Description			
1.	ESIA conducted based on techno-economic feasibility study			
2.	Name and experience of the firm conducting ESIA in similar projects			
3.	Scope of works and team composition conducting ESIA			

⁸ Projects in Orange B and Red categories need to conduct IEE

	Items	Yes	No	Comments
4.	Assessment of quality of the ESIA report			
	a) Executive Summary			
	i. Significant findings and recommended actions			
	ii. Alternatives, management and monitoring plan			
	iii. Costs related to mitigation, operation and monitoring			
	iv. Public consultation etc.			
	b) Methodology for ESIA			
	c) Policy, legal and administrative framework			
	i. Legislation governing the project			
	ii. Regulation and policies relevant to the project			
	iii. Relevant International agreements			
	d) Project Description			
	i. Project Background			
	ii. Need of the Project			
	iii. Importance of the project to the region and country			
	iv. Project coverage, master plan, phasing and scope			
	v. Project location, site map, topography and access ways (protected area and national monument to be taken into account)			
	vi. Land area and ownership			
	vii. Project cost			
	viii. Project layout and development plan			
	ix. Technical description of plant and machineries with plant efficiency			
	x. Materials to be utilized as input and output produced			
	xi. Requirement of land, water, energy, fuels and other resources and their management/storage facility			
	xii. Project processes and operations including process flow diagram			
	xiii. Cooling System of the plant (if required)			

	Items	Yes	No	Comments
	xiv. Type, quantity, and technology/methods/location to reduce, manage, treat and dispose of all project-related waste products: • Air emissions; • Solid wastes; • Liquid wastes; • Hazardous wastes; • Noise, heat, light, or radiation.			
	xv. Offsite investment (linked projects)			
	xvi. Transport requirements and their modes for Incoming and outgoing materials			
	xvii. Utilities and facilities and their capacities			
	xviii. Project implementation process (i.e., estimated duration, and start and finish dates for design, construction, and operation phases)			
	e) Baseline Environmental Conditions ⁹			
	i. Land Use Pattern (agriculture, forest, wet land, residence etc)			
	ii. Natural Physical Resources			
		<i>Climate</i>		
		<i>Geology</i>		
		<i>Air Quality</i>		
		<i>Noise</i>		
		<i>Soil</i>		
		<i>Seismicity</i>		
		<i>Water/ Hydrology</i>		
		<i>Climate</i>		
		<i>Humidity</i>		
	iii. Natural Biological Resources (including Forests)			
		<i>Flora</i>		
		<i>Fauna</i>		
		<i>Rare and endangered species</i>		
		<i>Ecosystem /habitats</i>		
		<i>Biodiversity</i>		

⁹ As per DOE Guidelines Environmental base maps to cover a minimum of 10 km radius

	Items	Yes	No	Comments
	<i>Ecosystem services and ecosystem function</i>			
	i. Socio-economic environment including demographic aspects, household, Literacy, Utility Services (Drinking Water, Electricity, Health & Hygiene), Occupation, Sources of Income, Population, Ethnic and Religious composition of population, indigenous people, health and others			
	ii. Environmentally sensitive areas of special or unique			
	iii. Scientific, historic sites, socio-economic or cultural Significance			
	f) Environmental Impacts and Mitigations			
	Environmental Impacts			
	i. Pre-construction Phase			
	<i>Change in landscape</i>			
	<i>Disruption of earth surface</i>			
	<i>Loss of and displacement from homestead land</i>			
	<i>Damage to nearby activities</i>			
	<i>Disruption to drainage pattern</i>			
	<i>Inadequacy of buffer zone</i>			
	<i>Encroachment into precious ecological aspects</i>			
	ii. Construction Phase			
	<i>Run off erosion</i>			
	<i>Worker accident</i>			
	<i>Air quality</i>			
	<i>Solid waste</i>			
	<i>Sanitation diseases hazard</i>			
	<i>Noise/vibration hazard</i>			
	<i>Traffic congestion</i>			
	<i>Blockage of wildlife passage</i>			
	<i>Employment</i>			
	<i>Others</i>			
	iii. Post-construction Phase			
	<i>Pollution from liquid discharge</i>			
	<i>Pollution from solid waste</i>			
	<i>Carbon and other Emissions</i>			

	Items	Yes	No	Comments
	<i>Air quality</i>			
	<i>Occupational health and safety</i>			
	<i>Odor hazard</i>			
	<i>Traffic congestion</i>			
	<i>Noise hazard</i>			
	<i>Employment</i>			
	<i>Others</i>			
	iv. Quantitative assessment of impacts			
	v. Impact identification method			
	Mitigation Measures			
	i. Mitigation measures taken to prevent, reduce and compensate all identified significant impacts			
	ii. Mitigation measures taken are appropriate/ adequate			
	g) Social Impacts and Mitigations i. Land Acquisition and Displacement ii. Economic environment iii. Vulnerable Groups iv. Employment and Labor v. Health and Safety vi. Social Conflict vii. Life Style and Culture viii. Gender Issues ix. Other Associated Impacts			
	h) Environment Management Plan (EMP)			
	i. Adequate and appropriate monitoring schedule, including a. Tasks and frequency b. Operational budget c. Clear institutional responsibilities for each identified task			
	ii. Sampling protocol including a. Sample collection method including timing, location, volume etc. b. Sample testing method			
	iii. Testing for environmental compliance a. Performance Acceptance Test after COD b. Periodic Performance Test during operational period			
	iv. Integration of ESMP with the project			
	v. Safety Management System			
	vi. Capacity Building and Training			
	i) Emergency Response and Disaster			

	Items	Yes	No	Comments
	Management Plan i. Emergency Response • Steps in Emergency Response • Reporting Incidents and accidents • Approaches to Emergency Response ii. Disaster Management Plan iii. Environment, Health and Safety			
	j) Project Alternatives			
	i. Realistic and feasible project alternatives, site alternatives in terms of environment consideration and economic value			
	ii. Without project situation			
	iii. Major environmental effects of the alternatives compared with those of proposed project			
	iv. Reasons for choosing of the proposed project explained			
	v. Suitability of the proposed project under local conditions, institutional training and monitoring requirement			
	k) Public Consultation			
	i. Objective of Stakeholders Consultation			
	ii. Consultation Process			
	iii. Stakeholder Consultation Technique			
	iv. Views of project-affected people			
	v. Views of local authority and non-governmental organizations (NGOs)			
	vi. Consideration of remarks of stakeholders			
	l) Grievance Redress Mechanism			
	i. Grievance Redress Mechanism			
	ii. Disclosure			
	m) Conclusion: EIA report is complete and of good quality			
	n) Annexure i. Air Modeling Report (if required) ii. Noise Modeling Report (if required)			

	Items	Yes	No	Comments
	iii. NO _x Calculation (if required) iv. Others			
	o) Essential Maps to be Provided v. Project Site Location Map vi. Project Layout Plan vii. Drainage Layout viii. others			

ANNEX 6-32: SECTOR SPECIFIC CHECKLISTS TO EVALUATE ESIA REPORTS

This Review may be used for the following purposes

- Assigning of the environmental category for the project
- Finalizing Terms of Reference for Environmental Due Diligence (EDD), Initial Environmental Examination (IEE), Environmental Impact Assessment (EIA), Corrective Action Plan (CAP)
- Reviewing reports of Environmental Due Diligence (EDD), Initial Environmental Examination (IEE), Environmental Impact Assessment (EIA), Corrective Action Plan (CAP)

Sector: Water Supply

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Water Withdrawal	Have the potential adverse effects of withdrawal of surface water on downstream ecosystems been evaluated?			
		Has appropriate environmental flow assessment been conducted to determine acceptable withdrawal rates?			
		Does the raw water supply have a source of pollution at the upstream like industries, agriculture, and sewage and soil erosion? Are adequate measures taken to treat these?			
		Does the project pose a hazard of land subsidence due to excessive ground water pumping?			
2.	Water Treatment	Does the treated water meet national and sponsors standards?			
		Have adequate measures been taken to reduce the solid waste residual generated through the treatment process?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		Does the effluent from the treatment process like filter backwash, reject streams from membrane filtration processes, and brine streams from ion exchange or demineralization processes meet country and sponsors standards?			
		Are adequate measures taken to prevent and control hazards during storage and use of hazardous chemicals like chlorine gas?			
		Will the emissions from the treatment process adversely affect the ambient air quality? Have measures been taken to mitigate these impacts?			
		Is the water treatment facility located in designated protected areas? Will the project activities cause any damage to these?			
		Is the water treatment facility located in a densely populated area or an area with heavy developmental activities?			
		Is sufficient buffer maintained around the pumping and treatment plants to alleviate noise and other possible nuisances and project facilities?			
3.	Water Distribution	Does the design of the distribution system include checks and measures to minimize leaks and loss of pressure?			
		Is the water used for flushing the water pipes disposed in accordance to country and sponsors requirements?			
4.	Supporting facilities	Is the increase in water supply supported by adequate sewerage network and sewage treatment facility?			
		Are the impairments associated with transmission lines and access roads adequately addressed?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
5.	Health and safety	Are adequate measures taken to prevent, minimize and mitigate health and safety hazards to workers from management of chlorine used for disinfection and other contaminants?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Sewerage and Sanitation

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	On-site systems	Are suitable facilities for storage, handling and treatment of fecal sludge from septic systems provided?			
2.	Sewerage system	Does the design of the sewerage system meet country and sponsors standards?			
		Are adequate measures taken to prevent and minimize leaks from the sewerage system into the ground?			
3.	Sewage Treatment	Is the sewage adequately treated before release into aquatic systems? Do they meet country and sponsors standards?			
		Is the sludge generated from the treatment plant disposed as per country and sponsors specifications?			
		Are adequate measures taken to minimize odor from the treatment facility?			
		Does the effluent from the treatment facility meet country and sponsors standards for disposal into aquatic systems?			
		Are adequate measures taken to prevent and control hazards during storage and use of hazardous chemicals?			
		Are adequate measures taken to prevent overflows and flooding of neighboring properties with raw sewage?			
		Is the water treatment facility located in designated protected areas? Will the project activities cause any damage to these?			
		Is the water treatment facility located in a densely populated area or an area			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		with heavy developmental activities?			
		Is sufficient buffer maintained around the pumping and treatment plants to alleviate noise and other possible nuisances and project facilities?			
4.	Industrial Waste water	Is the wastewater adequately treated before release into sewerage system or aquatic systems? Do the effluents meet country and sponsors standards?			
5.	Occupational Health and Safety	Are adequate measures taken to prevent accidents and injuries to workers while working in open water, trenches, slippery walkways, working at heights, energized circuits, and heavy equipment?			
		Are adequate measures taken to prevent chemical hazards during handling and storage of hazardous chemicals?			
		Are protective equipments and training provided to workers against exposure to bioaerosols, endotoxins?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Municipal Solid Waste

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Collection and Transport	Are adequate Litter bins and refuse collection services available to prevent littering and clandestine dumping?			
		Are adequate measures taken to mitigate air emissions like dust, bio-aerosols, odors, and vehicle emissions?			
2.	Waste Receipt, Unloading, Processing, and Storage	Are adequate measures taken to prevent migration of leachate into soil, surface water and groundwater?			
		Are adequate measures taken to prevent, minimize, and control litter?			
		Are adequate measures taken to mitigate noise and vibration from truck traffic, loading equipment (e.g., cranes, wheeled loaders), stationary compactors, balers, grinders, and other treatment and conveyance systems?			
3.	Biological treatment	Are adequate measures taken to control leachate and runoff from waste storage and processing areas?			
		Will the ambient air quality be adversely affected by direct stack emissions, fugitive emissions and emissions from burning of biogas? Do these emissions meet country and sponsors standards for ambient air emissions?			
		Are adequate measures taken to prevent combustion of waste during aerobic degradation or anaerobic digestion?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
4.	MSW Incineration Facilities	Do emissions from incinerators meet country and sponsors standards? Is the ambient air quality adversely affected by emissions from incinerators? What are the measures taken to mitigate these?			
		Are adequate measures taken to prevent, minimize, and control solid waste from incineration?			
		Are ash and other residuals disposed in accordance to country and sponsors specifications?			
		Is the flue gas from incineration treated and disposed in accordance to country and sponsors specifications?			
		Is the ambient noise levels adversely affected by exhaust fans, cooling system and turbine generators? Are measures taken to mitigate these?			
5.	Land filling	Is the landfill site located in accordance to country and sponsors specifications?			
		Are adequate measures taken for collection, treatment and disposal of leachate from landfill?			
		Is the quantity of leachate generated regularly monitored?			
		Is a landfill gas collection system designed and operated in accordance with applicable national requirements and sponsors standards?			
		Are adequate measures taken to prevent, minimize, and control dispersal of litter?			
		Is adequate buffer zone maintained around the landfill to alleviate nuisances?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
6.	Health and Safety	Are adequate measures taken to minimize and mitigate health and safety hazards to workers from toxic gases and hazardous materials on site?			
		Are adequate measures taken to protect the public and neighborhood from odor, smoke from fire, diseases transmitted by flies, rodents, insects and birds etc?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Roads

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Ecology	Does the project lie in proximity to or cut across any designated Protected Area?			
		Does the project or its activities cause disruption of terrestrial and aquatic habitats? If yes which of the following impacts does it cause - fragmentation of forested habitat; - loss of nesting sites of listed rare, threatened, or endangered species and / or high biodiversity / sensitive habitat; disruption of watercourses; - creation of barriers to wildlife movement; - visual and auditory disturbance due to the presence of machinery, construction workers, and associated equipment			
		Does the project involve regular maintenance of vegetation within road right-of-way causing likelihood of the establishment of invasive species?			
2.	Hydrology	Would the alteration of topography and installation of project components like tunnels adversely affect surface and ground water flows?			
3.	Topography and Geology	Does the project involve activities that have a likelihood of causing slope failures and landslides?			
4.	Storm water	Does the project or its activities cause increase in rate of surface water runoff?			
5.	Waste	Does the project or its activities lead to generation of the following types of waste - solid waste may be generated during construction and maintenance ☐ animal carcasses - vegetation waste from right-of-way maintenance - sediment and sludge from storm water drainage system maintenance (including sediment traps and oil /water separation			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		systems) • paint waste generated from road and bridge maintenance			
6.	Noise and Vibration	Do the ambient noise levels generated due to vehicle movement meet host country standards?			
		Will the noise and vibration due to blasting and other civil works affect the neighborhood areas?			
7.	Air Quality	Is the ambient air quality within the host country's standards for most of the times?			
		Are air polluting industries located in the vicinity of the project area. Will the project add to the air pollution load in the area?			
		Will the project cause increased local air pollution due to rock crushing, cutting and filling works and chemicals from asphalt processing?			
8.	Water Quality	Water Quality Would the surface runoff from cut and fill areas cause degradation of water quality in the downstream water bodies?			
		Would the surface runoff from roads contaminate surrounding water sources and / or ground water?			
9.	Public Health	Will the project lead to creation of temporary breeding habitats for mosquito vectors of disease?			
		Are there accident risks associated with increased vehicular traffic, leading to loss of life?			
10.	Occupational Health and Safety	Are adequate measures taken to prevent physical hazards while operating machinery, from moving vehicles and working at elevation on bridges and overpasses			
		Are adequate measures taken to prevent chemical hazards from construction and paving activities, exhaust emissions from heavy equipment and motor vehicles during all construction and maintenance activities?			
		Are adequate measures taken to prevent exposure to extremely high levels of noise from			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		heavy equipment operation and from working in proximity to vehicular traffic?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Housing

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Ecology	Will the project alter the micro-climate of the area due to building design?			
		does the project require large scale felling of trees? Have measures been planned to transplant these trees?			
2.	Energy	Are passive solar architecture features being used in the building to optimize energy consumption?			
3.	Land	Have adequate measures been planned to prevent land degradation due to dumping of construction waste and excavation spoil? Have designated locations been identified by the local body for dumping of such waste?			
		Is material for construction like sand and aggregate sourced from authorized quarries?			
		Does the project require conversion of landuse from industrial to residential? Has the soil been tested for contamination before commencement of excavation?			
		Will the housing facility impact the existing facilities adjacent to the site? (Such as open spaces, community facilities, details of the existing landuse, disturbance to the local ecology).			
		Have measures been planned for solid waste management from the housing facility after occupation?			
		Have adequate parking spaces been provided also considering the rise in social status?			
4.	Natural drainage	Has the main natural drainage of the site been retained and incorporated in the master planning to prevent future flooding?			
		Have adequate measures been planned to prevent obstruction to natural drainage due to dumping of construction waste and excavation spoil?			
5.	Air Quality	Is the ambient air quality within the host country's standards for most of the times?			
		Are air polluting industries located in the vicinity of the project area. Will the project add to the air pollution load in the area?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		Will the project cause increased local air pollution especially SPM and RSPM levels due to rock crushing, cutting and filling works and fugitive dust emissions from stockpile, construction camps, haul roads?			
		Is the exhaust from equipments used during construction like diesel generator sets, crushers etc within limits prescribed by State Pollution Control Boards?			
6.	Water Quality	Would the surface runoff from cut and fill areas cause degradation of water quality in the downstream water bodies and ground water?			
		Does the project or its activities cause increase in rate of surface water runoff?			
		Will the runoff from construction activities, stockpile areas for debris and excavation spoil affect water quality of surrounding water bodies?			
		Have measures been planned for conveyance of sewerage? Has a sewage treatment facility been installed for the housing facility?			
7.	Water Demand	How is the water demand been met for construction? Will the drawal of ground water for construction affect the water availability to existing habitations?			
		Have adequate provisions been made for water supply to the housing facility after occupancy?			
		Is ground water development of the area in critical zone?			
8.	Public Health	Will the project lead to creation of temporary breeding habitats for mosquito vectors of disease?			
		Has the housing facility and amenities provided been planned in accordance to fire hazard regulations? Has a fire fighting system been installed?			
9.	Occupational Health and Safety	Are adequate measures taken to prevent physical hazards while operating machinery, from working at elevation on scaffolding, from collapse of slabs and such structural members?			
		Are the labour provided clean water for consumption and cooking?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		Have adequate sanitation facilities been provided at labour camps?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Health Care Facilities

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Waste Management	Is a health care waste management system (HWMS) established, operated and maintained?			
		Are practices and procedures adopted to minimize waste generation through avoiding products containing hazardous materials?			
		Is waste identified and segregated at the point of generation?			
		Are the waste bags identified and labeled appropriately before removal?			
		Does the waste handling, collection, transport, storage, treatment and disposal meet national and sponsors standards?			
		Are appropriate waste disposals strategies employed for different streams of waste such as infectious waste, sharps, pharmaceutical waste, chemical waste, radioactive waste etc?			
2.	Air quality	Are adequate measures taken at treatment and disposal facilities to maintain ambient air quality within standards prescribed by national regulatory authority or sponsors?			
3.	Wastewater	Is the contaminated wastewater from medical wards and operating theaters adequately treated before discharge?			
4.	Occupational Health and safety	Are adequate safety measures adopted to prevent and control exposure to infections and diseases?			
		Are adequate measures taken to prevent, minimize and control fire hazards during storage, handling, and presence of chemicals, pressurized, gases, boards, plastics, and other flammable substrates?			
		Are adequate measures taken to prevent, minimize and control occupational radiation exposure from equipment emitting X-rays and gamma rays (e.g. CT scanners), radiotherapy machines, and equipment for nuclear medicine activities?			

(Signature of verifying officer)

Name:

(Signature of ESC In Charge, BIFFL)

Name:

Sector: Ports and Harbors

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / information
1.	Biodiversity	Would the construction and maintenance dredging activity, disposal of dredge spoil, construction of piers, wharves, breakwaters along with water-side structures, and erosion disturb important aquatic habitats such as feeding, breeding, calving, and spawning areas?			
		Is the project site located in proximity to designated protected areas? Will the project or its activities affect the protected areas?			
2.	Hazardous Waste	Is the dredged material contaminated by historical deposition and accumulation of hazardous materials, whether due to on-site or offsite activities?			
		Are measures taken to prevent discharge or dumping of hazardous waste to surrounding water areas?			
3.	Wastes	Are wastes from ships and ancillary facilities treated and disposed as per national and sponsors standards?			
		Does the off shore dumping of dredged materials meet national and sponsors specifications?			
4.	Water Quality	Are measures taken to prevent spills and discharge of oils and hazardous materials to surrounding water areas?			
		Do effluents from ships and ancillary facilities comply with national and sponsors standards for ambient water quality and effluent discharge?			
		Do general effluents from port facilities meet national and sponsors discharge standards?			
		Will any of the project components or activities cause oceanographic changes reducing sea water exchange rates?			
		For projects involving reclamation of land, are measures taken to prevent contamination of surface water, sea water			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / information
		and ground water by leaching from reclamation areas?			
5.	Air Quality	Do air pollutants like Sulfur dioxide, nitrogen oxides, soot dust, VOC's emitted from ships, vehicles and ancillary facilities meet country and sponsors emission and ambient air quality standards?			
		Are there any odor causing sources? Are adequate measures taken to prevent odors?			
6.	Noise	Do the noise levels caused due to cargo handling, vehicular traffic, loading / unloading containers and ships meet country and sponsors standards?			
7.	Topography and Geology	Will the installation of port and harbor facilities cause large scale alteration of topographic and geological features in surrounding areas or elimination of natural beaches?			
8.	Occupational Health and Safety	Are adequate measures taken to prevent physical hazards during cargo handling and use of associated machinery and vehicles?			
		Are adequate measures taken to prevent chemical hazards in cases when the workers are in contact with fuels and chemicals?			
		Are confined space entry procedures applied for workers working in cargo handling, ship cargo holds, silos, sewage tanks and water tanks?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Telecommunications

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Biodiversity	Are the towers located along important bird migratory paths?			
		Are the towers and transmission lines in proximity to critical terrestrial and aquatic habitats?			
2.	Aesthetics	Have the number of towers been optimized so as to minimize visual impacts?			
		Are the towers located against landscape views, environmental and community features? Have measures been taken to disguise the tower			
3.	Air quality	Do the emissions from operation of vehicle fleets, use of backup power generators and use of cooling and fire suppression systems meet country and sponsors standards?			
4.	Occupational health and safety	Are preventive and control measures adopted for ensuring electrical safety and optical fiber safety?			
		Have adequate measures been taken to prevent and minimize electromagnetic field exposure to workers?			
		Have adequate safety measures been incorporated for workers while working at elevated heights and for overhead work?			
		Have confined space entry procedures been adopted for underground work?			
5.	Community and public health	Are measures taken to restrict public access to antennae towers?			
		Are the towers located away from public access areas?			
		Are the towers located in proximity to airports?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Thermal Power Plant

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Air Quality	Is the plant planned and designed to control emissions of sulfur dioxide (SO ₂), nitrogen oxides (NO _x), particulate matter (PM), carbon monoxide (CO), and greenhouse gases, such as carbon dioxide (CO ₂), heavy metals, halide compounds, unburned hydrocarbons and other volatile organic compounds?			
		Does the ambient air quality near the facility meet national and sponsors standards?			
		Are adequate measures taken to avoid, minimize, and offset emissions of carbon dioxide?			
2.	Water Resource	Are adequate measures taken to conservatively use water for the facility?			
		Is wastewater recycling practiced to reduce the consumption of water?			
3.	Biodiversity	Does the facility intake water from critical aquatic habitats like spawning areas of aquatic species			
		Is the plant located in proximity to a Protected Area?			
4.	Water quality	Does the thermal discharge meet national and sponsors standards for disposal?			
		is the wastewater from various streams treated before discharge into surface water bodies?			
		Does the discharged wastewater meet national and sponsors standards for disposal?			
5.	Solid waste	Are adequate measures taken to prevent, minimize, and control the volume of solid wastes from thermal power plants?			
		Are the coal combustion wastes like fly ash, bottom ash, and boiler slag recycled in uses such as cement and other concrete products, construction fills etc?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
6.	Hazardous materials	Are adequate measures taken to prevent accidental release and minimize leaks during storage and handling of fuels and chemicals?			
7.	Noise	Is a considerable distanced maintained between the location of the plant and receptors such as residential areas?			
		Have noise control measures been employed for machines and structures?			
		Do the ambient noise levels meet national and sponsors standards?			
8.	Occupational health and safety	Are adequate measures taken to prevention and control measures to workers address heat exposure at thermal power plants?			
		Have adequate measures been taken to prevent and minimize electromagnetic field exposure to workers?			
		Are preventive and control measures adopted for ensuring electrical safety to workers?			
		Have measures been incorporated to prevent, minimize, and control occupational noise exposures?			
		Have confined space entry procedures been adopted for coal ash containers, turbines, condensers, and cooling water towers?			
		Have adequate measures been taken to prevent, minimize and control fire and explosion hazards?			
		Are adequate measures taken to prevent, minimize and control chemicals hazards?			
9.	Community health and safety	Have adequate measures been taken to ensure that the project does not compromise the availability of water for personal hygiene, agriculture, recreation, and other community needs?			
		Are practices adopted to prevent and control of increased traffic related injuries to the public?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Sector: Electric Power Transmission and Distribution

Project Title:

Date:

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
1.	Biodiversity	Are the towers located along important bird migratory paths?			
		Are the towers and transmission lines in proximity to critical terrestrial and aquatic habitats?			
		Have existing transport and utility corridors been used to prevent and minimize vegetation clearing?			
		Is the right-of way vegetation regularly monitored for fire risk?			
2.	Air quality	Do the emissions from operation of vehicle fleets, use of backup power generators and use of cooling and fire suppression systems meet country and sponsors standards?			
3.	Hazardous materials	Are adequate measures adopted for storage and handling of oils /gases (e.g. Polychlorinated Biphenyls [PCB] and sulfur hexafluoride [SF ₆], pesticides and other chemicals?			
4.	Occupational health and safety	Are adequate Prevention and control measures adopted associated with live power lines?			
		Have adequate measures been adopted to prevent and control occupational exposures to chemicals such as PCB in transformers and pesticides?			
		Have adequate measures been taken to prevent and minimize electromagnetic field exposure to workers?			
		Have adequate safety measures been incorporated for workers while working at elevated heights and for overhead work?			
5.	Community and public health	Have adequate signs and barriers been used to prevent public contact with potentially dangerous equipment?			

Sl. No	Environmental Component	Checklist	Yes	No	Comments / Information
		Have adequate measures been adopted to minimize electromagnetic interference to neighboring residential areas?			
		Are the towers located in proximity to airports?			
		Have the number of towers been optimized so as to minimize visual impacts?			
		Are the towers located against landscape views, environmental and community features? Have measures been taken to disguise the towers			
		Is the right-of-way located away from human receptors to avoid noise from transformers and emission of ozone?			

(Signature of verifying officer)

(Signature of ESC In Charge, BIFFL)

Name:

Name:

Annex 6-33: Involuntary Resettlement Screening and Categorization

Impact	Not Known	Yes	No	Indication of scope (no. of affected people, land area, land use, structures, etc.)
1. Is the prospective sub-project company (SPC) undertaking or likely to undertake any land acquisition?				
2. Is the SPC acquiring land through willing buyer to willing seller transactions?				
3. Does the SPC have any agreements or is it likely to enter into agreements with the government for provision of sites or land or rights to land?				
4. Is any of the land used by the SPC (or likely to be used by the SPC) compulsorily acquired?				
5. Will any SPC activities involve restrictions of use on adjoining land?				
6. Are the sites for land acquisition known?				
7. What is the ownership status of the land?				
8. Are non-titled persons present?				
9. Will tenants, lessees, share farmers, or other third party users be affected?				
10. Will there be loss of housing?				
11. Will there be loss of crops, trees, and other fixed assets?				
12. Will there be loss of incomes and				
13. Will access to facilities, services, or resources be lost?				
14. Will there be loss of businesses or enterprises?				
15. Will any social or economic activities be affected by land use related changes?				
If involuntary resettlement impacted are expected:				
16. Are local laws and regulations compatible with ADB's involuntary resettlement policy?				
17. Will land be acquired through the government or by the SPC?				

Impact	Not Known	Yes	No	Indication of scope (no. of affected people, land area, land use, structures, etc.)
18. Do SPC agreements with the government (if any) specify involuntary resettlement will be conducted in accordance with international standards?				
19. Does the government executing agency/SPC have sufficient skilled resources for resettlement planning Implementation?				
20. Are training and capacity-building required prior to resettlement planning and Implementation?				

ANNEX 6-34: ENVIRONMENTAL AND SOCIAL COVENANTS AND REQUIREMENTS FOR CONTRACTORS

The following guidelines pertaining to the ESMF should be included in the contractual agreements (particularly applicable to EPC contract):

- 1) Installation of the work site on areas far enough from water points, houses and sensitive areas;
- 2) Sanitary equipment and installations;
- 3) Site regulation (what is allowed and not allowed on work sites);
- 4) Compliance with laws, rules and other permits in vigor;
- 5) Inform the client if land is found to be contaminated;
- 6) Hygiene and security on work sites;
- 7) Protect neighboring properties;
- 8) Preserve existing fauna and flora;
- 9) Ensure the permanence of the traffic and access of neighboring populations during the works to avoid hindrance to traffic;
- 10) Protect and provide health and safety measures to staff working on work sites;
- 11) Soil, surface and groundwater protection: avoid any wastewater discharge, oil spill and discharge of any type of pollutants on soils, in surface or ground waters, in sewers and drainage ditches;
- 12) Protect the environment against exhaust fuels and oils;
- 13) Protect the environment against dust and other solid residues;
- 14) Waste management: install containers to collect the wastes generated next to the areas of activity;
- 15) Degradation/demolition of private properties: inform and raise the awareness of the populations before any activity causing degradation of natural vegetation and resources. Compensate beneficiaries before any work;
- 16) Use a quarry of materials according to the mining code requirements;
- 17) Compensation planting in case of deforestation or tree felling;
- 18) No waste slash and burn on site;
- 19) Speed limitation of work site vehicles and cars;
- 20) Allow the access of Public and emergency services;
- 21) Organize the storage of materials;
- 22) Organize parking and displacements of machines;
- 23) Footbridges and access of neighbors;

- 24) Signaling of works;
- 25) Respect of cultural sites;
- 26) Safe Disposal of asbestos;
- 27) Consider impacts such as noise, dust, and safety concerns on the surrounding population and schedule construction activities accordingly;
- 28) Protect soil surfaces during construction and re-vegetate or physically stabilize eligible surfaces;
- 29) Ensure proper drainage;
- 30) Prevent standing water in open construction pits, quarries or fill areas to avoid potential contamination of the water table and the development of a habitat for disease-carrying vectors and insects;
- 31) Select sustainable construction materials and construction method;
- 32) Control and clean the construction site daily;
- 33) During construction, control dust by using water or through other means;
- 34) Provide adequate waste disposal and sanitation services at the construction site;
- 35) Dispose of oil and solid waste materials appropriately;
- 36) Preserve natural habitats along streams, steep slopes, and ecologically sensitive areas;
- 37) Develop maintenance and reclamation plans and restore vegetation.
- 38) Ensure no blockage of access to households during construction and/or provide alternative access.
- 39) Camps should be located 500 m from habitations.
- 40) Ensure security and privacy of women and households in close proximity to the camps